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Electrophoresis cassettes and instrumentation

Abstract

Various implementations of electrophoretic systems and instruments are provided to improve electrophoresis and ease of use. Various electrophoretic systems and instruments utilize different electrode designs that can result in uniform electromigration of analytes. The electrophoretic systems and instrumentations optimize the size and/or location of electrodes relative to a sample, thereby increasing uniformity of electric field and reducing or minimizing drift of analyte migration. In addition, the electrophoresis systems and instruments provide a solution to conveniently check for electrical connections in the systems and instruments, and alert users of potential improper electrical connections, prior to performing electrophoresis.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4557903	12/1984	McCormick	N/A	N/A
4574729	12/1985	Wells	N/A	N/A
4683195	12/1986	Mullis	N/A	N/A
4683202	12/1986	Mullis	N/A	N/A
4800159	12/1988	Mullis	N/A	N/A
4883867	12/1988	Lee	N/A	N/A
4965188	12/1989	Mullis	N/A	N/A

5002882	12/1990	Lunnen	N/A	N/A
5061049	12/1990	Hornbeck	N/A	N/A
5130238	12/1991	Malek	N/A	N/A
5183053	12/1992	Yeh et al.	N/A	N/A
5308751	12/1993	Ohkawa	N/A	N/A
5321130	12/1993	Yue	N/A	N/A
5410030	12/1994	Yue	N/A	N/A
5436134	12/1994	Haugland	N/A	N/A
5455166	12/1994	Walker	N/A	N/A
5494810	12/1995	Barany et al.	N/A	N/A
5503980	12/1995	Cantor	N/A	N/A
5512439	12/1995	Hornes	N/A	N/A
5512462	12/1995	Cheng	N/A	N/A
5559032	12/1995	Porneroy	N/A	N/A
5582977	12/1995	Yue	N/A	N/A
5599675	12/1996	Brenner	N/A	N/A
5641658	12/1996	Adams	N/A	N/A
5648245	12/1996	Fire et al.	N/A	N/A
5658751	12/1996	Yue	N/A	N/A
5716825	12/1997	Hancock et al.	N/A	N/A
5750341	12/1997	Macevicz	N/A	N/A
5763175	12/1997	Brenner	N/A	N/A
5830711	12/1997	Barany et al.	N/A	N/A
5837832	12/1997	Chee et al.	N/A	N/A
5854033	12/1997	Lizardi	N/A	N/A
5863753	12/1998	Haugland	N/A	N/A
5871921	12/1998	Landegren et al.	N/A	N/A
5912148	12/1998	Eggerding	N/A	N/A
6013440	12/1999	Lipshutz	N/A	N/A
6027889	12/1999	Barany et al.	N/A	N/A
6060240	12/1999	Kamb et al.	N/A	N/A
6083761	12/1999	Kedar et al.	N/A	N/A
6130073	12/1999	Eggerding	N/A	N/A
6136592	12/1999	Leighton	N/A	N/A
6143496	12/1999	Brown	N/A	N/A
6153389	12/1999	Haarer	N/A	N/A
6157432	12/1999	Helbing	N/A	N/A
6165714	12/1999	Lane et al.	N/A	N/A
6210891	12/2000	Nyren	N/A	N/A
6210894	12/2000	Brennan	N/A	N/A
6214587	12/2000	Dattagupta	N/A	N/A
6221654	12/2000	Quake	N/A	N/A
6258568	12/2000	Nyren	N/A	N/A
6266459	12/2000	Walt	N/A	N/A
6274320	12/2000	Rothberg	N/A	N/A
6291180	12/2000	Chu	N/A	N/A
6300063	12/2000	Lipshutz et al.	N/A	N/A
6309824	12/2000	Drmanac	N/A	N/A
6337472	12/2001	Garner et al.	N/A	N/A
6344316	12/2001	Lockhart	N/A	N/A
6348990	12/2001	Igasaki et al.	N/A	N/A
6355431	12/2001	Chee	N/A	N/A
6368801	12/2001	Faruqi	N/A	N/A
6401267	12/2001	Drmanac	N/A	N/A
6404907	12/2001	Gilchrist	N/A	N/A
6432360	12/2001	Church et al.	N/A	N/A
6485982	12/2001	Charlton	N/A	N/A
6503713	12/2002	Rana	N/A	N/A
6506561	12/2002	Cheval et al.	N/A	N/A
6544732	12/2002	Chee	N/A	N/A
6565727	12/2002	Shenderov	N/A	N/A
6620584	12/2002	Chee	N/A	N/A
6632641	12/2002	Brennan	N/A	N/A
6673620	12/2003	Loeffler	N/A	N/A
6737236	12/2003	Pieken et al.	N/A	N/A
6770441	12/2003	Dickinson	N/A	N/A
6773566	12/2003	Shenderov	N/A	N/A
6773886	12/2003	Kaufman	N/A	N/A
6787308	12/2003	Balasubramanian	N/A	N/A
6800453	12/2003	Labaer	N/A	N/A
6812005	12/2003	Fan et al.	N/A	N/A
6828100	12/2003	Ronaghi	N/A	N/A
6833246	12/2003	Balasubramanian	N/A	N/A
6859570	12/2004	Walt	N/A	N/A
6864052	12/2004	Drmanac	N/A	N/A

6897023	12/2004	Fu	N/A	N/A
6911132	12/2004	Pamula	N/A	N/A
6911345	12/2004	Quake	N/A	N/A
6913921	12/2004	Fischer	N/A	N/A
6942968	12/2004	Dickinson et al.	N/A	N/A
6969488	12/2004	Bridgham	N/A	N/A
6977033	12/2004	Becker	N/A	N/A
7001792	12/2005	Sauer et al.	N/A	N/A
7052244	12/2005	Fouillet	N/A	N/A
7057026	12/2005	Barnes	N/A	N/A
7098041	12/2005	Kaylor et al.	N/A	N/A
7115400	12/2005	Adessi	N/A	N/A
7118883	12/2005	Inoue	N/A	N/A
7163612	12/2006	Sterling	N/A	N/A
7166431	12/2006	Chee et al.	N/A	N/A
7211414	12/2006	Hardin	N/A	N/A
7223371	12/2006	Hayenga et al.	N/A	N/A
7244559	12/2006	Rothberg	N/A	N/A
7255994	12/2006	Lao	N/A	N/A
7258976	12/2006	Mitsuhashi	N/A	N/A
7264929	12/2006	Rothberg	N/A	N/A
7297518	12/2006	Quake	N/A	N/A
7328979	12/2007	Decre	N/A	N/A
7329492	12/2007	Hardin	N/A	N/A
7361488	12/2007	Fan et al.	N/A	N/A
7378242	12/2007	Hurt	N/A	N/A
7393665	12/2007	Brenner	N/A	N/A
7405281	12/2007	Xu	N/A	N/A
7407757	12/2007	Brenner	N/A	N/A
7456012	12/2007	Ryttsen et al.	N/A	N/A
7462449	12/2007	Quake	N/A	N/A
7500637	12/2008	Marimon	244/128	B64B 1/60
7501245	12/2008	Quake	N/A	N/A
7510637	12/2008	Barlow et al.	N/A	N/A
7537897	12/2008	Brenner	N/A	N/A
7547380	12/2008	Velev	N/A	N/A
7561336	12/2008	Osaka et al.	N/A	N/A
7563576	12/2008	Chee	N/A	N/A
7582420	12/2008	Oliphant et al.	N/A	N/A
7595883	12/2008	El Gamal	N/A	N/A
7601492	12/2008	Fu et al.	N/A	N/A
7601498	12/2008	Mao	N/A	N/A
7635566	12/2008	Brenner	N/A	N/A
7641779	12/2009	Becker	N/A	N/A
7674752	12/2009	He	N/A	N/A
7709198	12/2009	Luo et al.	N/A	N/A
7741106	12/2009	Moyle	435/6.12	G01N 33/54373
7776567	12/2009	Mao	N/A	N/A
7785869	12/2009	Belgrader et al.	N/A	N/A
7803943	12/2009	Mao	N/A	N/A
7858321	12/2009	Glezer	N/A	N/A
7910304	12/2010	Drmanac	N/A	N/A
7955794	12/2010	Shen et al.	N/A	N/A
7960119	12/2010	Chee	N/A	N/A
8003354	12/2010	Shen et al.	N/A	N/A
8148068	12/2011	Brenner	N/A	N/A
8206917	12/2011	Chee	N/A	N/A
8278034	12/2011	Muraca	N/A	N/A
8288103	12/2011	Oliphant	N/A	N/A
8330087	12/2011	Domenicali	N/A	N/A
8460865	12/2012	Chee	N/A	N/A
8481257	12/2012	Van Eijk	N/A	N/A
8603743	12/2012	Liu et al.	N/A	N/A
8604182	12/2012	Luo et al.	N/A	N/A
8637242	12/2013	Shen	N/A	N/A
8778849	12/2013	Bowen	N/A	N/A
8815512	12/2013	Van Eijk	N/A	N/A
8835358	12/2013	Fodor	N/A	N/A
8900529	12/2013	Shaikh et al.	N/A	N/A
8911945	12/2013	Van Eijk	N/A	N/A
8951726	12/2014	Luo et al.	N/A	N/A
8951781	12/2014	Reed	N/A	N/A
9062348	12/2014	Van Eijk	N/A	N/A
9194001	12/2014	Brenner	N/A	N/A
9201063	12/2014	Sood et al.	N/A	N/A

9290808	12/2015	Fodor	N/A	N/A
9290809	12/2015	Fodor	N/A	N/A
9328383	12/2015	Van Eijk	N/A	N/A
9334536	12/2015	Van Eijk	N/A	N/A
9371598	12/2015	Chee	N/A	N/A
9416409	12/2015	Hayden	N/A	N/A
9506061	12/2015	Brown et al.	N/A	N/A
9557330	12/2016	Siciliano et al.	N/A	N/A
9582877	12/2016	Fu	N/A	N/A
9593365	12/2016	Frisen et al.	N/A	N/A
9644204	12/2016	Hindson et al.	N/A	N/A
9694361	12/2016	Bharadwaj	N/A	N/A
9702004	12/2016	Van Eijk	N/A	N/A
9727810	12/2016	Fodor et al.	N/A	N/A
9777324	12/2016	Van Eijk	N/A	N/A
9783841	12/2016	Nolan et al.	N/A	N/A
9799992	12/2016	Hirose et al.	N/A	N/A
9834814	12/2016	Peter et al.	N/A	N/A
9850536	12/2016	Oliphant et al.	N/A	N/A
9868979	12/2017	Chee et al.	N/A	N/A
9879313	12/2017	Chee et al.	N/A	N/A
9975122	12/2017	Masquelier et al.	N/A	N/A
10002316	12/2017	Fodor et al.	N/A	N/A
10023907	12/2017	Van Eijk	N/A	N/A
10030261	12/2017	Frisen et al.	N/A	N/A
10032064	12/2017	Hoyt	N/A	N/A
10041949	12/2017	Bendall et al.	N/A	N/A
10053723	12/2017	Hindson et al.	N/A	N/A
10059990	12/2017	Boyden et al.	N/A	N/A
10071377	12/2017	Bharadwaj et al.	N/A	N/A
10208982	12/2018	Bannish et al.	N/A	N/A
10246700	12/2018	Dunaway et al.	N/A	N/A
10266888	12/2018	Daugharthy et al.	N/A	N/A
10273541	12/2018	Hindson et al.	N/A	N/A
10308982	12/2018	Chee	N/A	N/A
10357771	12/2018	Bharadwaj	N/A	N/A
10472669	12/2018	Chee	N/A	N/A
10480022	12/2018	Chee	N/A	N/A
10480029	12/2018	Bent et al.	N/A	N/A
10494667	12/2018	Chee	N/A	N/A
10495554	12/2018	Deisseroth et al.	N/A	N/A
10550429	12/2019	Harada et al.	N/A	N/A
10590244	12/2019	Delaney et al.	N/A	N/A
10662468	12/2019	Chee	N/A	N/A
10724078	12/2019	Van Driel et al.	N/A	N/A
10725027	12/2019	Bell	N/A	N/A
10774372	12/2019	Chee et al.	N/A	N/A
10774374	12/2019	Frisen et al.	N/A	N/A
10787701	12/2019	Chee	N/A	N/A
10858702	12/2019	Lucero et al.	N/A	N/A
10913975	12/2020	So et al.	N/A	N/A
10914730	12/2020	Chee et al.	N/A	N/A
10927403	12/2020	Chee et al.	N/A	N/A
10961566	12/2020	Chee	N/A	N/A
11001879	12/2020	Chee	N/A	N/A
11008607	12/2020	Chee	N/A	N/A
11046996	12/2020	Chee et al.	N/A	N/A
11067567	12/2020	Chee	N/A	N/A
11156603	12/2020	Chee	N/A	N/A
11162132	12/2020	Frisen et al.	N/A	N/A
11208684	12/2020	Chee	N/A	N/A
11214796	12/2021	Shirai et al.	N/A	N/A
11286515	12/2021	Chee et al.	N/A	N/A
11293917	12/2021	Chee	N/A	N/A
11299774	12/2021	Frisen et al.	N/A	N/A
11313856	12/2021	Chee	N/A	N/A
11332790	12/2021	Chell et al.	N/A	N/A
11352659	12/2021	Frisen et al.	N/A	N/A
11352667	12/2021	Hauling et al.	N/A	N/A
11359228	12/2021	Chee et al.	N/A	N/A
11365442	12/2021	Chee	N/A	N/A
11371086	12/2021	Chee	N/A	N/A
11384386	12/2021	Chee	N/A	N/A
11390912	12/2021	Frisen et al.	N/A	N/A
11401545	12/2021	Chee	N/A	N/A

11407992	12/2021	Dadhwal	N/A	N/A
11408029	12/2021	Katirae et al.	N/A	N/A
11434524	12/2021	Ramachandran Iyer et al.	N/A	N/A
11459607	12/2021	Terry et al.	N/A	N/A
11479809	12/2021	Frisen et al.	N/A	N/A
11479810	12/2021	Chee	N/A	N/A
11492612	12/2021	Dadhwal	N/A	N/A
11501440	12/2021	Weisenfeld et al.	N/A	N/A
11505828	12/2021	Chell et al.	N/A	N/A
11512308	12/2021	Gallant et al.	N/A	N/A
11519022	12/2021	Chee	N/A	N/A
11519033	12/2021	Schnall-Levin et al.	N/A	N/A
11530438	12/2021	Persson et al.	N/A	N/A
11535887	12/2021	Gallant et al.	N/A	N/A
11542543	12/2022	Chee	N/A	N/A
11549138	12/2022	Chee	N/A	N/A
11560587	12/2022	Chee	N/A	N/A
11560592	12/2022	Chew et al.	N/A	N/A
11560593	12/2022	Chell et al.	N/A	N/A
11592447	12/2022	Uytingco et al.	N/A	N/A
11608498	12/2022	Gallant et al.	N/A	N/A
11608520	12/2022	Galonska et al.	N/A	N/A
11613773	12/2022	Frisen et al.	N/A	N/A
11618897	12/2022	Kim et al.	N/A	N/A
11618918	12/2022	Chee et al.	N/A	N/A
11624063	12/2022	Dadhwal	N/A	N/A
11624086	12/2022	Uytingco et al.	N/A	N/A
11634756	12/2022	Chee	N/A	N/A
11649485	12/2022	Yin et al.	N/A	N/A
11661626	12/2022	Katirae et al.	N/A	N/A
11680260	12/2022	Kim et al.	N/A	N/A
11692218	12/2022	Engblom et al.	N/A	N/A
11702693	12/2022	Bharadwaj	N/A	N/A
11702698	12/2022	Stoeckius	N/A	N/A
11713480	12/2022	Lee	N/A	N/A
11732292	12/2022	Chee	N/A	N/A
11732299	12/2022	Ramachandran Iyer	N/A	N/A
11732300	12/2022	Bava	N/A	N/A
11733238	12/2022	Chee	N/A	N/A
11739372	12/2022	Frisen et al.	N/A	N/A
11739381	12/2022	Chew et al.	N/A	N/A
11753673	12/2022	Chew et al.	N/A	N/A
11753674	12/2022	Chee et al.	N/A	N/A
11753675	12/2022	Ramachandran Iyer	N/A	N/A
11761030	12/2022	Chee	N/A	N/A
11761038	12/2022	Stoeckius	N/A	N/A
11767550	12/2022	Chee	N/A	N/A
11768175	12/2022	Kim et al.	N/A	N/A
11773433	12/2022	Gallant et al.	N/A	N/A
11781130	12/2022	Dadhwal	N/A	N/A
11788122	12/2022	Frisen et al.	N/A	N/A
11795498	12/2022	Frisen et al.	N/A	N/A
11795507	12/2022	Chell et al.	N/A	N/A
11808769	12/2022	Uytingco et al.	N/A	N/A
11821024	12/2022	Chee et al.	N/A	N/A
11821035	12/2022	Bent et al.	N/A	N/A
11827935	12/2022	Ramachandran Iyer et al.	N/A	N/A
11835462	12/2022	Bava	N/A	N/A
11840687	12/2022	Gallant et al.	N/A	N/A
11840724	12/2022	Chew et al.	N/A	N/A
11845979	12/2022	Engblom et al.	N/A	N/A
11859178	12/2023	Gallant et al.	N/A	N/A
11866767	12/2023	Uytingco et al.	N/A	N/A
11866770	12/2023	Chee	N/A	N/A
11873482	12/2023	Kim et al.	N/A	N/A
11891654	12/2023	Alvarado Martinez et al.	N/A	N/A
11898205	12/2023	Bava	N/A	N/A
11926822	12/2023	Gohil et al.	N/A	N/A
11926863	12/2023	Boutet	N/A	N/A
11926867	12/2023	Yin et al.	N/A	N/A
11933957	12/2023	Tentori et al.	N/A	N/A
11952627	12/2023	Stoeckius	N/A	N/A
11959076	12/2023	Kim et al.	N/A	N/A
11959130	12/2023	Galonska et al.	N/A	N/A
11965213	12/2023	Williams	N/A	N/A

11970739	12/2023	Chew et al.	N/A	N/A
11981958	12/2023	Galonska	N/A	N/A
11981960	12/2023	Lin et al.	N/A	N/A
11981965	12/2023	Chell et al.	N/A	N/A
RE50065	12/2023	Frisen et al.	N/A	N/A
12024741	12/2023	Tentori et al.	N/A	N/A
12031177	12/2023	Tentori et al.	N/A	N/A
12060604	12/2023	Katirae et al.	N/A	N/A
12071655	12/2023	Sukovich et al.	N/A	N/A
12076701	12/2023	Bava	N/A	N/A
12098417	12/2023	Engblom et al.	N/A	N/A
12098985	12/2023	Cox et al.	N/A	N/A
12110541	12/2023	Bava	N/A	N/A
12117439	12/2023	Delaney et al.	N/A	N/A
12128403	12/2023	Kim et al.	N/A	N/A
12129516	12/2023	Tentori et al.	N/A	N/A
12157124	12/2023	Cox et al.	N/A	N/A
12180543	12/2023	Uytingco et al.	N/A	N/A
12195790	12/2024	Sukovich et al.	N/A	N/A
12203134	12/2024	Nagendran et al.	N/A	N/A
12209280	12/2024	Mignardi et al.	N/A	N/A
12223751	12/2024	Li et al.	N/A	N/A
12228544	12/2024	Kim et al.	N/A	N/A
12234505	12/2024	Chee	N/A	N/A
12241060	12/2024	Kim et al.	N/A	N/A
12241890	12/2024	Delaney et al.	N/A	N/A
12249085	12/2024	Tentori et al.	N/A	N/A
12265079	12/2024	Bent	N/A	N/A
12270077	12/2024	Schnall-Levin et al.	N/A	N/A
12275988	12/2024	Galonska et al.	N/A	N/A
12281357	12/2024	Tentori et al.	N/A	N/A
12286673	12/2024	Bava	N/A	N/A
12287264	12/2024	Cox et al.	N/A	N/A
12297486	12/2024	Patterson et al.	N/A	N/A
12297487	12/2024	Chee	N/A	N/A
12297488	12/2024	Chee	N/A	N/A
12344892	12/2024	Schnall-Levin et al.	N/A	N/A
2002/0022261	12/2001	Anderson et al.	N/A	N/A
2002/0040275	12/2001	Cravatt	N/A	N/A
2002/0045272	12/2001	McDevitt et al.	N/A	N/A
2002/0150909	12/2001	Stuelpnagel	N/A	N/A
2002/0164611	12/2001	Bamdad	N/A	N/A
2003/0017451	12/2002	Wang et al.	N/A	N/A
2003/0022207	12/2002	Balasubramanian	N/A	N/A
2003/0113713	12/2002	Glezer	N/A	N/A
2003/0148335	12/2002	Shen et al.	N/A	N/A
2003/0162216	12/2002	Gold	N/A	N/A
2003/0175947	12/2002	Liu et al.	N/A	N/A
2003/0190744	12/2002	McGarry et al.	N/A	N/A
2003/0205632	12/2002	Kim et al.	N/A	N/A
2003/0211489	12/2002	Shen et al.	N/A	N/A
2003/0215936	12/2002	Kallioniemi et al.	N/A	N/A
2003/0224419	12/2002	Corcoran	N/A	N/A
2003/0232348	12/2002	Jones et al.	N/A	N/A
2003/0232382	12/2002	Brennan	N/A	N/A
2004/0033499	12/2003	Illesley et al.	N/A	N/A
2004/0050699	12/2003	Goncalves	N/A	N/A
2004/0067492	12/2003	Peng et al.	N/A	N/A
2004/0082058	12/2003	Schleifer et al.	N/A	N/A
2004/0096853	12/2003	Mayer	N/A	N/A
2004/0106110	12/2003	Balasubramanian	N/A	N/A
2004/0112442	12/2003	Maerkl	N/A	N/A
2004/0121456	12/2003	Fischer	N/A	N/A
2004/0219588	12/2003	Furuta	N/A	N/A
2004/0248287	12/2003	Hu et al.	N/A	N/A
2005/0037393	12/2004	Gunderson et al.	N/A	N/A
2005/0048580	12/2004	Labaer	N/A	N/A
2005/0100900	12/2004	Kawashima et al.	N/A	N/A
2005/0106617	12/2004	Besemer et al.	N/A	N/A
2005/0130173	12/2004	Leamon et al.	N/A	N/A
2005/0136414	12/2004	Gunderson et al.	N/A	N/A
2005/0170373	12/2004	Monforte	N/A	N/A
2005/0179746	12/2004	Roux et al.	N/A	N/A
2005/0191656	12/2004	Drmanac et al.	N/A	N/A
2005/0191698	12/2004	Chee et al.	N/A	N/A

2005/0202433	12/2004	Van Beuningen	N/A	N/A
2005/0226780	12/2004	Sandell et al.	N/A	N/A
2005/0227271	12/2004	Kwon	N/A	N/A
2005/0239192	12/2004	Nasarabadi et al.	N/A	N/A
2005/0260653	12/2004	LaBaer	N/A	N/A
2006/0039823	12/2005	Yamakawa et al.	N/A	N/A
2006/0041385	12/2005	Bauer et al.	N/A	N/A
2006/0063160	12/2005	West et al.	N/A	N/A
2006/0127946	12/2005	Montagu et al.	N/A	N/A
2006/0134669	12/2005	Casasanta	N/A	N/A
2006/0164490	12/2005	Kim et al.	N/A	N/A
2006/0180489	12/2005	Guiney et al.	N/A	N/A
2006/0194331	12/2005	Pamula et al.	N/A	N/A
2006/0211001	12/2005	Yu et al.	N/A	N/A
2006/0216775	12/2005	Burkart et al.	N/A	N/A
2006/0263789	12/2005	Kincaid	N/A	N/A
2007/0020640	12/2006	McCloskey et al.	N/A	N/A
2007/0023292	12/2006	Kim et al.	N/A	N/A
2007/0036511	12/2006	Lundquist et al.	N/A	N/A
2007/0054288	12/2006	Su et al.	N/A	N/A
2007/0099208	12/2006	Drmanac et al.	N/A	N/A
2007/0116612	12/2006	Williamson	N/A	N/A
2007/0128071	12/2006	Shea et al.	N/A	N/A
2007/0128624	12/2006	Gormley et al.	N/A	N/A
2007/0128656	12/2006	Agrawal	N/A	N/A
2007/0166725	12/2006	McBride et al.	N/A	N/A
2007/0172873	12/2006	Brenner et al.	N/A	N/A
2007/0184456	12/2006	Chee et al.	N/A	N/A
2007/0207482	12/2006	Church et al.	N/A	N/A
2007/0215466	12/2006	Okada	N/A	N/A
2007/0243634	12/2006	Pamula et al.	N/A	N/A
2007/0254305	12/2006	Paik et al.	N/A	N/A
2007/0269805	12/2006	Hogers	N/A	N/A
2008/0009071	12/2007	Sogard	N/A	N/A
2008/0009420	12/2007	Schroth et al.	N/A	N/A
2008/0043235	12/2007	Oldham et al.	N/A	N/A
2008/0108804	12/2007	Hayashizaki et al.	N/A	N/A
2008/0160580	12/2007	Adessi et al.	N/A	N/A
2008/0218838	12/2007	Rey-Mermet	N/A	N/A
2008/0220434	12/2007	Thomas	N/A	N/A
2008/0261204	12/2007	Lexow	N/A	N/A
2008/0286795	12/2007	Kawashima et al.	N/A	N/A
2009/0005252	12/2008	Drmanac et al.	N/A	N/A
2009/0006002	12/2008	Honisch et al.	N/A	N/A
2009/0011943	12/2008	Drmanac et al.	N/A	N/A
2009/0018024	12/2008	Church et al.	N/A	N/A
2009/0023148	12/2008	Moyle et al.	N/A	N/A
2009/0026082	12/2008	Rothberg et al.	N/A	N/A
2009/0082212	12/2008	Williams	N/A	N/A
2009/0099041	12/2008	Church et al.	N/A	N/A
2009/0105959	12/2008	Braverman et al.	N/A	N/A
2009/0117573	12/2008	Fu et al.	N/A	N/A
2009/0127589	12/2008	Rothberg et al.	N/A	N/A
2009/0155781	12/2008	Drmanac et al.	N/A	N/A
2009/0169089	12/2008	Hunt et al.	N/A	N/A
2009/0192044	12/2008	Fouillet	N/A	N/A
2009/0197326	12/2008	El Gamal	435/287.2	C12Q 1/66
2009/0233802	12/2008	Bignell et al.	N/A	N/A
2009/0239232	12/2008	Kurn	N/A	N/A
2009/0253163	12/2008	Xie et al.	N/A	N/A
2009/0253581	12/2008	van Eijk et al.	N/A	N/A
2009/0253582	12/2008	Pena et al.	N/A	N/A
2009/0270273	12/2008	Burns et al.	N/A	N/A
2009/0291854	12/2008	Weisinger-Mayr et al.	N/A	N/A
2009/0312193	12/2008	Kim et al.	N/A	N/A
2009/0321262	12/2008	Adachi et al.	N/A	N/A
2010/0009871	12/2009	Reed et al.	N/A	N/A
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2010/0035249	12/2009	Hayashizaki et al.	N/A	N/A
2010/0096266	12/2009	Kim et al.	N/A	N/A
2010/0111768	12/2009	Banerjee et al.	N/A	N/A
2010/0120097	12/2009	Matz et al.	N/A	N/A
2010/0120098	12/2009	Grunenwald et al.	N/A	N/A
2010/0126862	12/2009	Sabin et al.	N/A	N/A
2010/0145037	12/2009	Brive et al.	N/A	N/A

2010/0151511	12/2009	Gereenizer et al.	N/A	N/A
2010/0267590	12/2009	Grudzien et al.	N/A	N/A
2010/0273219	12/2009	May et al.	N/A	N/A
2010/0273679	12/2009	Cuppoletti et al.	N/A	N/A
2010/0282617	12/2009	Rothberg et al.	N/A	N/A
2011/0028685	12/2010	Purkayastha et al.	N/A	N/A
2011/0048951	12/2010	Wu	N/A	N/A
2011/0059436	12/2010	Hardin et al.	N/A	N/A
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2011/0177518	12/2010	Kartalov et al.	N/A	N/A
2011/0244448	12/2010	Shirai et al.	N/A	N/A
2011/0245111	12/2010	Chee	N/A	N/A
2012/0135871	12/2011	van Eijk et al.	N/A	N/A
2012/0160683	12/2011	Ye et al.	N/A	N/A
2012/0177543	12/2011	Battrell	N/A	N/A
2012/0195810	12/2011	Cohen et al.	N/A	N/A
2012/0202698	12/2011	van Eijk et al.	N/A	N/A
2012/0270305	12/2011	Reed et al.	N/A	N/A
2012/0279954	12/2011	Ceremony et al.	N/A	N/A
2013/0053273	12/2012	Juncker et al.	N/A	N/A
2013/0065788	12/2012	Glezer et al.	N/A	N/A
2013/0146459	12/2012	Bazant et al.	N/A	N/A
2013/0171621	12/2012	Luo et al.	N/A	N/A
2013/0203100	12/2012	Otter et al.	N/A	N/A
2013/0252847	12/2012	McKenna et al.	N/A	N/A
2013/0260372	12/2012	Buermann et al.	N/A	N/A
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2014/0011707	12/2013	Ye et al.	N/A	N/A
2014/0066318	12/2013	Frisen et al.	N/A	N/A
2014/0155295	12/2013	Hindson et al.	N/A	N/A
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2014/0323330	12/2013	Glezer et al.	N/A	N/A
2014/0378350	12/2013	Hindson et al.	N/A	N/A
2015/0000854	12/2014	Gann-Fetter et al.	N/A	N/A
2015/0292988	12/2014	Bharadwaj et al.	N/A	N/A
2015/0298091	12/2014	Weitz et al.	N/A	N/A
2015/0344942	12/2014	Frisen et al.	N/A	N/A
2016/0033496	12/2015	Chou et al.	N/A	N/A
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2016/0253584	12/2015	Fodor et al.	N/A	N/A
2016/0289669	12/2015	Fan et al.	N/A	N/A
2016/0289740	12/2015	Fu et al.	N/A	N/A
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2016/0299165	12/2015	Zhou	N/A	N/A
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2017/0067096	12/2016	Wassie et al.	N/A	N/A
2017/0089811	12/2016	Tillberg et al.	N/A	N/A
2017/0159109	12/2016	Zheng et al.	N/A	N/A
2017/0220733	12/2016	Zhuang et al.	N/A	N/A
2017/0241911	12/2016	Rockel et al.	N/A	N/A
2017/0342405	12/2016	Fu et al.	N/A	N/A
2017/0349940	12/2016	Morin et al.	N/A	N/A
2018/0051322	12/2017	Church et al.	N/A	N/A
2018/0052082	12/2017	Groll et al.	N/A	N/A
2018/0057873	12/2017	Zhou et al.	N/A	N/A
2018/0074039	12/2017	Soper et al.	N/A	N/A
2018/0080876	12/2017	Rockel et al.	N/A	N/A
2018/0088112	12/2017	Fan et al.	N/A	N/A
2018/0095067	12/2017	Huff et al.	N/A	N/A
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2018/0104964	12/2017	Uemura et al.	N/A	N/A
2018/0112248	12/2017	Lam et al.	N/A	N/A
2018/0112261	12/2017	Van Driel et al.	N/A	N/A
2018/0120255	12/2017	Wu et al.	N/A	N/A
2018/0201980	12/2017	Chee et al.	N/A	N/A
2018/0216161	12/2017	Chen et al.	N/A	N/A
2018/0216162	12/2017	Belhocine et al.	N/A	N/A
2018/0217094	12/2017	Herr et al.	N/A	N/A
2018/0245142	12/2017	So et al.	N/A	N/A
2018/0251833	12/2017	Daugharthy et al.	N/A	N/A
2018/0291439	12/2017	van Eijk et al.	N/A	N/A

2018/0305681	12/2017	Jovanovich et al.	N/A	N/A
2018/0334670	12/2017	Bharadwaj et al.	N/A	N/A
2019/0055594	12/2018	Samusik et al.	N/A	N/A
2019/0064173	12/2018	Bharadwaj et al.	N/A	N/A
2019/0085383	12/2018	Church et al.	N/A	N/A
2019/0099754	12/2018	Dupouy et al.	N/A	N/A
2019/0113532	12/2018	Tan et al.	N/A	N/A
2019/0126280	12/2018	Gach et al.	N/A	N/A
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2019/0161796	12/2018	Hauling et al.	N/A	N/A
2019/0177777	12/2018	Chee	N/A	N/A
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2019/0177789	12/2018	Hindson et al.	N/A	N/A
2019/0177800	12/2018	Boutet et al.	N/A	N/A
2019/0194709	12/2018	Church et al.	N/A	N/A
2019/0201891	12/2018	Pallas et al.	N/A	N/A
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2019/0218276	12/2018	Regev et al.	N/A	N/A
2019/0233878	12/2018	Delaney et al.	N/A	N/A
2019/0249226	12/2018	Bent et al.	N/A	N/A
2019/0262831	12/2018	West et al.	N/A	N/A
2019/0264268	12/2018	Frisen et al.	N/A	N/A
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2019/0300943	12/2018	Chee et al.	N/A	N/A
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2019/0323088	12/2018	Boutet et al.	N/A	N/A
2019/0330617	12/2018	Church et al.	N/A	N/A
2019/0338353	12/2018	Belgrader et al.	N/A	N/A
2019/0367969	12/2018	Belhocine et al.	N/A	N/A
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2019/0367997	12/2018	Bent et al.	N/A	N/A
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2020/0024641	12/2019	Nolan et al.	N/A	N/A
2020/0047010	12/2019	Lee et al.	N/A	N/A
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2020/0049599	12/2019	Alexander et al.	N/A	N/A
2020/0063191	12/2019	Kennedy-Darling et al.	N/A	N/A
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2020/0071751	12/2019	Daugharthy et al.	N/A	N/A
2020/0080136	12/2019	Zhang et al.	N/A	N/A
2020/0109443	12/2019	Chee	N/A	N/A
2020/0173985	12/2019	Dong et al.	N/A	N/A
2020/0224244	12/2019	Nilsson et al.	N/A	N/A
2020/0239946	12/2019	Dewal	N/A	N/A
2020/0256867	12/2019	Hennek et al.	N/A	N/A
2020/0277663	12/2019	Iyer	N/A	N/A
2020/0277664	12/2019	Frenz	N/A	N/A
2020/0298241	12/2019	Kabaha et al.	N/A	N/A
2020/0299757	12/2019	Chee et al.	N/A	N/A
2020/0325531	12/2019	Chee	N/A	N/A
2020/0370095	12/2019	Farmer et al.	N/A	N/A
2020/0393343	12/2019	Kennedy-Darling et al.	N/A	N/A
2020/0399687	12/2019	Frisen et al.	N/A	N/A
2020/0407781	12/2019	Schnall-Levin	N/A	N/A
2021/0010068	12/2020	Chee et al.	N/A	N/A
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2021/0130881	12/2020	Cox	N/A	N/A
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2021/0150707	12/2020	Weisenfeld et al.	N/A	N/A
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2021/0189475	12/2020	Tentori et al.	N/A	N/A
2021/0190770	12/2020	Delaney et al.	N/A	N/A
2021/0198741	12/2020	Williams	N/A	N/A
2021/0199660	12/2020	Williams et al.	N/A	N/A

2021/0207202	12/2020	Chee	N/A	N/A
2021/0214785	12/2020	Stoeckius	N/A	N/A
2021/0222235	12/2020	Chee	N/A	N/A
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2021/0222242	12/2020	Ramachandran Iyer	N/A	N/A
2021/0222253	12/2020	Uytingco	N/A	N/A
2021/0223227	12/2020	Stoeckius	N/A	N/A
2021/0230584	12/2020	Mikkelsen et al.	N/A	N/A
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2021/0237022	12/2020	Bava	N/A	N/A
2021/0238581	12/2020	Mikkelsen et al.	N/A	N/A
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2021/0238680	12/2020	Bava	N/A	N/A
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2021/0262018	12/2020	Bava et al.	N/A	N/A
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2021/0317510	12/2020	Chee et al.	N/A	N/A
2021/0317524	12/2020	Lucero et al.	N/A	N/A
2021/0324457	12/2020	Ramachandran Iyer et al.	N/A	N/A
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2021/0348221	12/2020	Chell et al.	N/A	N/A
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2022/0010367	12/2021	Ramachandran Iyer et al.	N/A	N/A
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2022/0025446	12/2021	Shah	N/A	N/A
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2022/0049293	12/2021	Frenz et al.	N/A	N/A
2022/0049294	12/2021	Uytingco et al.	N/A	N/A
2022/0064630	12/2021	Bent et al.	N/A	N/A
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2022/0106633	12/2021	Engblom et al.	N/A	N/A
2022/0112486	12/2021	Ramachandran Iyer et al.	N/A	N/A
2022/0112545	12/2021	Chee	N/A	N/A
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2022/0127666	12/2021	Katirae et al.	N/A	N/A
2022/0127672	12/2021	Stoeckius	N/A	N/A
2022/0145361	12/2021	Frenz et al.	N/A	N/A
2022/0154255	12/2021	Chee et al.	N/A	N/A
2022/0170083	12/2021	Khaled et al.	N/A	N/A
2022/0195422	12/2021	Gallant et al.	N/A	N/A
2022/0195505	12/2021	Frisen et al.	N/A	N/A
2022/0196644	12/2021	Chee	N/A	N/A
2022/0213526	12/2021	Frisen et al.	N/A	N/A
2022/0220544	12/2021	Ach et al.	N/A	N/A
2022/0241780	12/2021	Tentori et al.	N/A	N/A
2022/0267844	12/2021	Ramachandran Iyer et al.	N/A	N/A
2022/0282329	12/2021	Chell et al.	N/A	N/A
2022/0290217	12/2021	Frenz et al.	N/A	N/A
2022/0290219	12/2021	Chee	N/A	N/A
2022/0298560	12/2021	Frisen et al.	N/A	N/A
2022/0325325	12/2021	Chee et al.	N/A	N/A
2022/0326251	12/2021	Uytingco et al.	N/A	N/A
2022/0333171	12/2021	Chee	N/A	N/A
2022/0333191	12/2021	Mikkelsen et al.	N/A	N/A
2022/0333192	12/2021	Uytingco	N/A	N/A
2022/0333195	12/2021	Schnall-Levin et al.	N/A	N/A

2022/0334031	12/2021	Delaney et al.	N/A	N/A
2022/0348905	12/2021	Dadhwal	N/A	N/A
2022/0348992	12/2021	Stoeckius et al.	N/A	N/A
2022/0356464	12/2021	Kim et al.	N/A	N/A
2022/0364163	12/2021	Stahl et al.	N/A	N/A
2022/0389491	12/2021	Chee	N/A	N/A
2022/0389503	12/2021	Mikkelsen et al.	N/A	N/A
2022/0389504	12/2021	Chew et al.	N/A	N/A
2022/0403455	12/2021	Ramachandran Iyer et al.	N/A	N/A
2022/0404245	12/2021	Chell et al.	N/A	N/A
2023/0002812	12/2022	Stoeckius et al.	N/A	N/A
2023/0014008	12/2022	Shastry	N/A	N/A
2023/0017773	12/2022	Kim et al.	N/A	N/A
2023/0031305	12/2022	Hernandez Neuta et al.	N/A	N/A
2023/0033960	12/2022	Gallant et al.	N/A	N/A
2023/0034039	12/2022	Shahjamali	N/A	N/A
2023/0034216	12/2022	Bava	N/A	N/A
2023/0040363	12/2022	Chee	N/A	N/A
2023/0042088	12/2022	Chee	N/A	N/A
2023/0042817	12/2022	Mignardi	N/A	N/A
2023/0047782	12/2022	Tentori et al.	N/A	N/A
2023/0056549	12/2022	Dadhwal	N/A	N/A
2023/0064372	12/2022	Chell et al.	N/A	N/A
2023/0069046	12/2022	Chew et al.	N/A	N/A
2023/0077364	12/2022	Patterson et al.	N/A	N/A
2023/0080543	12/2022	Katiraei et al.	N/A	N/A
2023/0081381	12/2022	Chew et al.	N/A	N/A
2023/0100497	12/2022	Frisen et al.	N/A	N/A
2023/0107023	12/2022	Chee	N/A	N/A
2023/0111225	12/2022	Chew et al.	N/A	N/A
2023/0113230	12/2022	Kim et al.	N/A	N/A
2023/0126825	12/2022	Nagendran et al.	N/A	N/A
2023/0129552	12/2022	Ramachandran Iyer	N/A	N/A
2023/0135010	12/2022	Tentori et al.	N/A	N/A
2023/0143569	12/2022	Iyer et al.	N/A	N/A
2023/0145575	12/2022	Gallant et al.	N/A	N/A
2023/0147726	12/2022	Hadrup et al.	N/A	N/A
2023/0151412	12/2022	Chee	N/A	N/A
2023/0159994	12/2022	Chee	N/A	N/A
2023/0159995	12/2022	Iyer et al.	N/A	N/A
2023/0160008	12/2022	Chell et al.	N/A	N/A
2023/0175045	12/2022	Katsori et al.	N/A	N/A
2023/0183684	12/2022	Gallant et al.	N/A	N/A
2023/0183785	12/2022	Frisen et al.	N/A	N/A
2023/0194470	12/2022	Kim et al.	N/A	N/A
2023/0203478	12/2022	Kim et al.	N/A	N/A
2023/0212650	12/2022	Chew et al.	N/A	N/A
2023/0212655	12/2022	Chee	N/A	N/A
2023/0212656	12/2022	Chow et al.	N/A	N/A
2023/0220368	12/2022	Kim	N/A	N/A
2023/0220454	12/2022	Bent et al.	N/A	N/A
2023/0220455	12/2022	Galonska et al.	N/A	N/A
2023/0227811	12/2022	Dadhwal	N/A	N/A
2023/0228762	12/2022	Uytingco et al.	N/A	N/A
2023/0242973	12/2022	Frisen et al.	N/A	N/A
2023/0242976	12/2022	Tentori et al.	N/A	N/A
2023/0265488	12/2022	Gohil et al.	N/A	N/A
2023/0265489	12/2022	Uytingco et al.	N/A	N/A
2023/0265491	12/2022	Tentori et al.	N/A	N/A
2023/0267625	12/2022	Tentori et al.	N/A	N/A
2023/0279474	12/2022	Katiraei	N/A	N/A
2023/0279477	12/2022	Kvastad et al.	N/A	N/A
2023/0279481	12/2022	Marrache et al.	N/A	N/A
2023/0287399	12/2022	Gallant et al.	N/A	N/A
2023/0287475	12/2022	Chell et al.	N/A	N/A
2023/0287481	12/2022	Katsori et al.	N/A	N/A
2023/0295699	12/2022	Sukovich et al.	N/A	N/A
2023/0295722	12/2022	Bharadwaj	N/A	N/A
2023/0304072	12/2022	Gohil et al.	N/A	N/A
2023/0304074	12/2022	Chee et al.	N/A	N/A
2023/0304078	12/2022	Frisen et al.	N/A	N/A
2023/0313279	12/2022	Giacomello et al.	N/A	N/A
2023/0323340	12/2022	Dadhwal	N/A	N/A
2023/0323434	12/2022	Yin et al.	N/A	N/A
2023/0323436	12/2022	Chee	N/A	N/A

2023/0323447	12/2022	Schnall-Levin et al.	N/A	N/A
2023/0323453	12/2022	Stoeckius	N/A	N/A
2023/0332138	12/2022	Kim et al.	N/A	N/A
2023/0332211	12/2022	Chee	N/A	N/A
2023/0332212	12/2022	Chew et al.	N/A	N/A
2023/0332227	12/2022	Ramachandran Iyer	N/A	N/A
2023/0332247	12/2022	Singh et al.	N/A	N/A
2023/0351619	12/2022	Tentori et al.	N/A	N/A
2023/0358733	12/2022	Chee	N/A	N/A
2023/0366008	12/2022	Chew et al.	N/A	N/A
2023/0383285	12/2022	Kim et al.	N/A	N/A
2023/0383344	12/2022	Stoeckius	N/A	N/A
2023/0392204	12/2022	Chell et al.	N/A	N/A
2023/0393071	12/2022	Bava	N/A	N/A
2023/0407404	12/2022	Baumgartner et al.	N/A	N/A
2023/0416807	12/2022	Chee	N/A	N/A
2023/0416808	12/2022	Sukovich et al.	N/A	N/A
2023/0416850	12/2022	Singh et al.	N/A	N/A
2024/0002931	12/2023	Bava	N/A	N/A
2024/0011081	12/2023	Chee	N/A	N/A
2024/0011090	12/2023	Chew et al.	N/A	N/A
2024/0018572	12/2023	Mignardi	N/A	N/A
2024/0018575	12/2023	Gallant et al.	N/A	N/A
2024/0018589	12/2023	Schnall-Levin et al.	N/A	N/A
2024/0026445	12/2023	Ramachandran Iyer et al.	N/A	N/A
2024/0033743	12/2023	Tentori et al.	N/A	N/A
2024/0035937	12/2023	Cox et al.	N/A	N/A
2024/0043908	12/2023	Chew et al.	N/A	N/A
2024/0043925	12/2023	Bent et al.	N/A	N/A
2024/0052343	12/2023	Gallant et al.	N/A	N/A
2024/0053351	12/2023	Uytingco et al.	N/A	N/A
2024/0060115	12/2023	Chee et al.	N/A	N/A
2024/0067953	12/2023	Mikkelsen et al.	N/A	N/A
2024/0068016	12/2023	Frisen et al.	N/A	N/A
2024/0068017	12/2023	Lundeberg et al.	N/A	N/A
2024/0076723	12/2023	Mignardi	N/A	N/A
2024/0080346	12/2023	Engblom et al.	N/A	N/A
2024/0084365	12/2023	Frisen et al.	N/A	N/A
2024/0084366	12/2023	Chee	N/A	N/A
2024/0084383	12/2023	Ramachandran Iyer et al.	N/A	N/A
2024/0093274	12/2023	Frisen et al.	N/A	N/A
2024/0093290	12/2023	Stahl et al.	N/A	N/A
2024/0110228	12/2023	Uytingco et al.	N/A	N/A
2024/0124933	12/2023	Chell et al.	N/A	N/A
2024/0125772	12/2023	Delaney et al.	N/A	N/A
2024/0141327	12/2023	Kim et al.	N/A	N/A
2024/0158838	12/2023	Alvarado Martinez et al.	N/A	N/A
2024/0175080	12/2023	Galonska et al.	N/A	N/A
2024/0182968	12/2023	Bava	N/A	N/A
2024/0191286	12/2023	Boutet et al.	N/A	N/A
2024/0200121	12/2023	Boutet	N/A	N/A
2024/0209425	12/2023	Yin et al.	N/A	N/A
2024/0218427	12/2023	Sukovich et al.	N/A	N/A
2024/0218432	12/2023	Mielinis	N/A	N/A
2024/0219701	12/2023	Tentori et al.	N/A	N/A
2024/0253036	12/2023	Kim et al.	N/A	N/A
2024/0263218	12/2023	Katirae et al.	N/A	N/A
2024/0271190	12/2023	Stoeckius et al.	N/A	N/A
2024/0271195	12/2023	Mikhael et al.	N/A	N/A
2024/0279747	12/2023	Williams	N/A	N/A
2024/0287600	12/2023	Iyer et al.	N/A	N/A
2024/0294971	12/2023	Galonska	N/A	N/A
2024/0294974	12/2023	Galonska et al.	N/A	N/A
2024/0294975	12/2023	Lin et al.	N/A	N/A
2024/0301488	12/2023	Stoeckius	N/A	N/A
2024/0301489	12/2023	Chew et al.	N/A	N/A
2024/0360494	12/2023	Costa et al.	N/A	N/A
2024/0368711	12/2023	Giacomello et al.	N/A	N/A
2024/0377297	12/2023	Cox et al.	N/A	N/A
2024/0385088	12/2023	Kim et al.	N/A	N/A
2024/0392349	12/2023	Frisen et al.	N/A	N/A
2024/0392351	12/2023	Chee	N/A	N/A
2024/0392352	12/2023	Stahl et al.	N/A	N/A
2024/0392353	12/2023	Engblom et al.	N/A	N/A
2024/0401109	12/2023	Kim et al.	N/A	N/A

2024/0401117	12/2023	Bava	N/A	N/A
2024/0401118	12/2023	Tentori et al.	N/A	N/A
2024/0404301	12/2023	Li et al.	N/A	N/A
2024/0408593	12/2023	Kim et al.	N/A	N/A
2024/0416315	12/2023	Bava	N/A	N/A
2024/0417783	12/2023	Chew et al.	N/A	N/A
2024/0417784	12/2023	Sukovich et al.	N/A	N/A
2025/0002980	12/2024	Tentori et al.	N/A	N/A
2025/0002982	12/2024	Stoeckius et al.	N/A	N/A
2025/0003956	12/2024	Delaney et al.	N/A	N/A
2025/0019689	12/2024	Galonska et al.	N/A	N/A
2025/0019749	12/2024	Katiraei et al.	N/A	N/A
2025/0066762	12/2024	Man et al.	N/A	N/A
2025/0066770	12/2024	Costa	N/A	N/A
2025/0073719	12/2024	Cox et al.	N/A	N/A
2025/0075261	12/2024	Kim	N/A	N/A
2025/0101501	12/2024	Chee	N/A	N/A
2025/0101502	12/2024	Chee	N/A	N/A
2025/0101504	12/2024	Nagendran et al.	N/A	N/A
2025/0122564	12/2024	Mignardi et al.	N/A	N/A
2025/0122565	12/2024	Schnall-Levin et al.	N/A	N/A
2025/0129412	12/2024	Uytingco et al.	N/A	N/A
2025/0129421	12/2024	Schnall-Levin et al.	N/A	N/A
2025/0137043	12/2024	Tentori	N/A	N/A
2025/0145984	12/2024	Ma et al.	N/A	N/A
2025/0146057	12/2024	Schnall-Levin et al.	N/A	N/A
2025/0146058	12/2024	Tentori	N/A	N/A
2025/0146071	12/2024	Schnall-Levin et al.	N/A	N/A
2025/0146072	12/2024	Schnall-Levin et al.	N/A	N/A
2025/0154565	12/2024	Chee	N/A	N/A
2025/0154566	12/2024	Chee	N/A	N/A
2025/0154567	12/2024	Chee	N/A	N/A
2025/0154568	12/2024	Frisen et al.	N/A	N/A
2025/0154569	12/2024	Stoeckius et al.	N/A	N/A
2025/0154571	12/2024	Ramachandran Iyer et al.	N/A	N/A
2025/0154588	12/2024	Ramachandran Iyer et al.	N/A	N/A
2025/0155446	12/2024	Uytingco et al.	N/A	N/A
2025/0163501	12/2024	Singh et al.	N/A	N/A
2025/0163509	12/2024	Daugharthy et al.	N/A	N/A
2025/0171833	12/2024	Frisen et al.	N/A	N/A
2025/0171848	12/2024	Chell et al.	N/A	N/A
2025/0179475	12/2024	Borgstrom et al.	N/A	N/A
2025/0182305	12/2024	Tentori et al.	N/A	N/A
2025/0182503	12/2024	Li et al.	N/A	N/A
2025/0188526	12/2024	Sukovich et al.	N/A	N/A
2025/0189483	12/2024	Kim et al.	N/A	N/A
2025/0197847	12/2024	Kim et al.	N/A	N/A
2025/0197938	12/2024	Bjorninen	N/A	N/A
2025/0207125	12/2024	Gupta et al.	N/A	N/A
2025/0207182	12/2024	Chee	N/A	N/A
2025/0207183	12/2024	Chee	N/A	N/A
2025/0207195	12/2024	Chell et al.	N/A	N/A
2025/0208115	12/2024	Bent	N/A	N/A
2025/0215482	12/2024	Mignardi et al.	N/A	N/A
2025/0215484	12/2024	Ramachandran Iyer et al.	N/A	N/A
2025/0216300	12/2024	Delaney et al.	N/A	N/A
2025/0216303	12/2024	Cox et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS			
Patent No.	Application Date	Country	CPC
3054046	12/2019	CA	N/A
1425133	12/2002	CN	N/A
1680604	12/2004	CN	N/A
1934452	12/2006	CN	N/A
1981188	12/2006	CN	N/A
202548048	12/2011	CN	N/A
102851369	12/2012	CN	N/A
104513785	12/2014	CN	N/A
104541162	12/2014	CN	N/A
110998310	12/2019	CN	N/A
0961110	12/1998	EP	N/A
1878502	12/2007	EP	N/A
1923471	12/2007	EP	N/A
2002017	12/2007	EP	N/A
2130913	12/2008	EP	N/A

2881465	12/2014	EP	N/A
3013983	12/2015	EP	N/A
3013984	12/2015	EP	N/A
3511423	12/2018	EP	N/A
3541956	12/2018	EP	N/A
WO 1989/010977	12/1988	WO	N/A
WO 1991/006678	12/1990	WO	N/A
WO 1995/025116	12/1994	WO	N/A
WO 1995/035505	12/1994	WO	N/A
WO 1998/010277	12/1997	WO	N/A
WO 1999/063385	12/1998	WO	N/A
WO 2000/063437	12/1999	WO	N/A
WO 2002/040874	12/2001	WO	N/A
WO 2002/059355	12/2001	WO	N/A
WO 2002/077283	12/2001	WO	N/A
WO 2003/002979	12/2002	WO	N/A
WO 2003/010176	12/2002	WO	N/A
WO 2003/106973	12/2002	WO	N/A
WO 2004/028955	12/2003	WO	N/A
WO 2004/108268	12/2003	WO	N/A
WO 2005/007814	12/2004	WO	N/A
WO 2006/056861	12/2005	WO	N/A
WO 2007/073171	12/2006	WO	N/A
WO 2007/076726	12/2006	WO	N/A
WO 2007/120241	12/2006	WO	N/A
WO 2007/123744	12/2006	WO	N/A
WO 2007/145612	12/2006	WO	N/A
WO 2008/075086	12/2007	WO	N/A
WO 2008/157801	12/2007	WO	N/A
WO 2009/032167	12/2008	WO	N/A
WO 2009/152928	12/2008	WO	N/A
WO 2010/088517	12/2009	WO	N/A
WO 2010/126614	12/2009	WO	N/A
WO 2011/068088	12/2010	WO	N/A
WO 2011/102903	12/2010	WO	N/A
WO 2012/058096	12/2011	WO	N/A
WO 2012/159089	12/2011	WO	N/A
WO 2012/168003	12/2011	WO	N/A
WO 2013/123442	12/2012	WO	N/A
WO 2013/131962	12/2012	WO	N/A
WO 2013/138510	12/2012	WO	N/A
WO 2013/150082	12/2012	WO	N/A
WO 2013/150083	12/2012	WO	N/A
WO 2014/060483	12/2013	WO	N/A
WO 2014/128129	12/2013	WO	N/A
WO 2014/142841	12/2013	WO	N/A
WO 2014/210223	12/2013	WO	N/A
WO 2014/210225	12/2013	WO	N/A
WO 2014/210353	12/2013	WO	N/A
WO 2015/031691	12/2014	WO	N/A
WO 2016/138496	12/2015	WO	N/A
WO 2016/138500	12/2015	WO	N/A
WO 2016/162309	12/2015	WO	N/A
WO 2016/166128	12/2015	WO	N/A
WO 2016/168825	12/2015	WO	N/A
WO 2016/172362	12/2015	WO	N/A
WO 2017/019456	12/2016	WO	N/A
WO 2017/048871	12/2016	WO	N/A
WO 2017/075293	12/2016	WO	N/A
WO 2017/112957	12/2016	WO	N/A
WO 2017/096158	12/2016	WO	N/A
WO 2017/144338	12/2016	WO	N/A
WO 2018/064640	12/2017	WO	N/A
WO 2018/075436	12/2017	WO	N/A
WO 2018/091676	12/2017	WO	N/A
WO 2018/148471	12/2017	WO	N/A
WO 2018/175779	12/2017	WO	N/A
WO 2019/104337	12/2018	WO	N/A
WO 2019/113457	12/2018	WO	N/A
WO 2019/140334	12/2018	WO	N/A
WO 2019/213254	12/2018	WO	N/A
WO 2019/213294	12/2018	WO	N/A
WO 2020/028194	12/2019	WO	N/A
WO 2020/047002	12/2019	WO	N/A
WO 2020/047010	12/2019	WO	N/A

WO 2020/053655	12/2019	WO	N/A
WO 2020/061064	12/2019	WO	N/A
WO 2020/061066	12/2019	WO	N/A
WO 2020/061108	12/2019	WO	N/A
WO 2020/076979	12/2019	WO	N/A
WO 2020/099640	12/2019	WO	N/A
WO 2020/123301	12/2019	WO	N/A
WO 2020/123305	12/2019	WO	N/A
WO 2020/123311	12/2019	WO	N/A
WO 2020/123316	12/2019	WO	N/A
WO 2020/123317	12/2019	WO	N/A
WO 2020/123318	12/2019	WO	N/A
WO 2020/123319	12/2019	WO	N/A
WO 2020/123320	12/2019	WO	N/A
WO 2020/160044	12/2019	WO	N/A
WO 2020/167862	12/2019	WO	N/A
WO 2020/176788	12/2019	WO	N/A
WO 2020/176882	12/2019	WO	N/A
WO 2020/190509	12/2019	WO	N/A
WO 2020/198071	12/2019	WO	N/A
WO 2020/206285	12/2019	WO	N/A
WO 2020/219901	12/2019	WO	N/A
WO 2020/243579	12/2019	WO	N/A
WO 2021/041974	12/2020	WO	N/A
WO 2021/067246	12/2020	WO	N/A
WO 2021/067514	12/2020	WO	N/A
WO 2021/102003	12/2020	WO	N/A
WO 2021/102005	12/2020	WO	N/A
WO 2021/102039	12/2020	WO	N/A
WO 2021/116715	12/2020	WO	N/A
WO 2021/133842	12/2020	WO	N/A
WO 2021/133845	12/2020	WO	N/A
WO 2021/133849	12/2020	WO	N/A
WO 2021/142233	12/2020	WO	N/A
WO 2021/168261	12/2020	WO	N/A
WO 2021/168278	12/2020	WO	N/A
WO 2021/207610	12/2020	WO	N/A
WO 2021/216708	12/2020	WO	N/A
WO 2021/225900	12/2020	WO	N/A
WO 2021/236625	12/2020	WO	N/A
WO 2021/236929	12/2020	WO	N/A
WO 2021/237056	12/2020	WO	N/A
WO 2021/237087	12/2020	WO	N/A
WO 2021/242834	12/2020	WO	N/A
WO 2021/247543	12/2020	WO	N/A
WO 2021/247568	12/2020	WO	N/A
WO 2021/247593	12/2020	WO	N/A
WO 2021/252499	12/2020	WO	N/A
WO 2021/252576	12/2020	WO	N/A
WO 2021/252591	12/2020	WO	N/A
WO 2021/252747	12/2020	WO	N/A
WO 2021/263111	12/2020	WO	N/A
WO 2022/025965	12/2021	WO	N/A
WO 2022/051152	12/2021	WO	N/A
WO 2022/060798	12/2021	WO	N/A
WO 2022/060953	12/2021	WO	N/A
WO 2022/061150	12/2021	WO	N/A
WO 2022/061152	12/2021	WO	N/A
WO 2022/087273	12/2021	WO	N/A
WO 2022/098810	12/2021	WO	N/A
WO 2022/099037	12/2021	WO	N/A
WO 2022/103712	12/2021	WO	N/A
WO 2022/109181	12/2021	WO	N/A
WO 2022/132645	12/2021	WO	N/A
WO 2022/140028	12/2021	WO	N/A
WO 2022/147005	12/2021	WO	N/A
WO 2022/147296	12/2021	WO	N/A
WO 2022/164615	12/2021	WO	N/A
WO 2022/178267	12/2021	WO	N/A
WO 2022/198068	12/2021	WO	N/A
WO 2022/221425	12/2021	WO	N/A
WO 2022/226057	12/2021	WO	N/A
WO 2022/015913	12/2021	WO	N/A
WO 2022/236054	12/2021	WO	N/A
WO 2022/243303	12/2021	WO	N/A

WO 2022/226372	12/2021	WO	N/A
WO 2022/256503	12/2021	WO	N/A
WO 2022/271820	12/2021	WO	N/A
WO 2023/287765	12/2022	WO	N/A
WO 2023/018799	12/2022	WO	N/A
WO 2023/034489	12/2022	WO	N/A
WO 2023/044071	12/2022	WO	N/A
WO 2023/076345	12/2022	WO	N/A
WO 2023/086880	12/2022	WO	N/A
WO 2023/102118	12/2022	WO	N/A
WO 2023/122033	12/2022	WO	N/A
WO 2023/130019	12/2022	WO	N/A
WO 2023/147355	12/2022	WO	N/A
WO 2023/150098	12/2022	WO	N/A
WO 2023/150163	12/2022	WO	N/A
WO 2023/150171	12/2022	WO	N/A
WO 2023/215552	12/2022	WO	N/A
WO 2023/225519	12/2022	WO	N/A
WO 2023/229988	12/2022	WO	N/A
WO 2023/250077	12/2022	WO	N/A
WO 2024/015578	12/2023	WO	N/A
WO 2024/035844	12/2023	WO	N/A
WO 2024/081212	12/2023	WO	N/A
WO 2024/086167	12/2023	WO	N/A
WO 2024/086776	12/2023	WO	N/A
WO 2024/102809	12/2023	WO	N/A
WO 2024/137826	12/2023	WO	N/A
WO 2024/145224	12/2023	WO	N/A
WO 2024/145441	12/2023	WO	N/A
WO 2024/145445	12/2023	WO	N/A
WO 2024/145491	12/2023	WO	N/A
WO 2024/206603	12/2023	WO	N/A
WO 2024/220882	12/2023	WO	N/A
WO 2024/238900	12/2023	WO	N/A
WO 2024/254316	12/2023	WO	N/A
WO 2025/029605	12/2024	WO	N/A
WO 2025/029627	12/2024	WO	N/A
WO 2025/043076	12/2024	WO	N/A
WO 2025/072119	12/2024	WO	N/A
WO 2025/090912	12/2024	WO	N/A
WO 2025/096581	12/2024	WO	N/A
WO 2025/101864	12/2024	WO	N/A

OTHER PUBLICATIONS

10xGenomics.com, [online], “Visium Spatial Gene Expression Reagent Kits—User Guide,” Jan. 2022, retrieved on Jun. 27, 2024, retrieved from URL <<https://web.archive.org/web/20230326192142/https://www.10xgenomics.com/support/spatial-gene-expression-fresh-frozen/documentation/steps/library-spatial-gene-expression-reagent-kits-user-guide>>, 71 pages. cited by applicant

Kuhn et al., “A novel, high-performance random array platform for quantitative gene expression profiling,” *Genome Res*, 2004, 14:2347-2356. cited by applicant

Eastburn, “Microfluidic droplet enrichment for targeted sequencing,” *Nucleic Acids Research*, 2015, 43(13):1-8. cited by applicant

10xGenomics.com, [online], “Visium Spatial Gene Expression Reagent Kits—Tissue Optimization—User Guide,” Feb. 2022, retrieved on Mar. 29, 2024, retrieved from URL <https://cdn.10xgenomics.com/image/upload/v1660261286/support-documents/CG000238_VisiumSpatialTissueOptimizationUserGuide_RevE.pdf>, 46 pages. cited by applicant

Asp et al., “A spatiotemporal organ-wide gene expression and cell atlas of the developing human heart,” *Cell*, Dec. 12, 2019, 179(7):1647-1660. cited by applicant

U.S. Appl. No. 61/839,313, filed Jun. 25, 2013, Chee et al. cited by applicant

U.S. Appl. No. 61/839,320, filed Jun. 25, 2013, Chee et al. cited by applicant

[No Author Listed], “Chromium Next GEM Single Cell 3’ Reagent Kits v3.1—User Guide,” 10x Genomics, Document No. CG000204, Nov. 2019, 58 pages. cited by applicant

[No Author Listed], “Chromium Next GEM Single Cell 3’ Reagent Kits v3.1 (Dual Index)—User Guide,” 10x Genomics, Mar. 2021, Document No. CG000304, 43 pages. cited by applicant

[No Author Listed], “HuSNP Mapping Assay User's Manual,” Affymetrix Part No. 90094 (Affymetrix, Santa Clara, Calif.), GeneChip, 2000, 104 pages. cited by applicant

[No Author Listed], “Microarray technologies have excellent possibilities in genomics-related researches,” *Science Tools From Amersham Pharmacia Biotech* (English Translation). cited by applicant

10xGenomics.com, [online], “Visium Spatial Gene Expression Reagent Kits—Tissue Optimization—User Guide,” Jul. 2020, retrieved on May 25, 2021, retrieved from URL <https://assets.ctfassets.net/an68im79xiti/5UJrN0cH17rEk0UXwd19It/e54d99fb08a8f1500aba503005a04a56/CG000238_VisiumSpatialTissueOptimizationUserGuide_RevD.pdf>, 42 pages. cited by applicant

10xGenomics.com, [online], “Visium Spatial Gene Expression Reagent Kits—Tissue Optimization,” Nov. 2019, retrieved on Jan. 25, 2022, retrieved from URL <https://assets.ctfassets.net/an68im79xiti/4q03w6959AJFxfSw5lee9/6a2ac61cf6388a72564eeb96bc294967/CG000238_VisiumSpatialTissueOptimizationUserGuide_RevC.pdf>, 46 pages. cited by applicant

10xGenomics.com, [online], “Visium Spatial Gene Expression Reagent Kits—Tissue Optimization,” Oct. 2020, retrieved on Dec. 28, 2021, retrieved from URL <https://assets.ctfassets.net/an68im79xiti/5UJrN0cH17rEk0UXwd19It/e54d99fb08a8f1500aba503005a04a56/CG000238_VisiumSpatialTissueOptimizationUserGuide_RevE.pdf>, 43 pages. cited by applicant

10xGenomics.com, [online], “Visium Spatial Gene Expression Reagent Kits—User Guide,” Jun. 2020, retrieved on May 25, 2021, retrieved from URL <https://assets.ctfassets.net/an68im79xiti/3GGIfH3RWpd1bFVha1pexR/8baa08d9007157592b65b2cdc7130990/CG000239_VisiumSpatialGeneExpressionUserGuide_RevA.pdf>, 69 pages. cited by applicant

10xGenomics.com, [online], “Visium Spatial Gene Expression Reagent Kits—User Guide,” Oct. 2020, retrieved on Dec. 28, 2021, retrieved from URL <https://assets.ctfassets.net/an68im79xiti/3GGIfH3RWpd1bFVha1pexR/8baa08d9007157592b65b2cdc7130990/CG000239_VisiumSpatialGeneExpressionUserGuide_RevB.pdf>, 70 pages. cited by applicant

Adessi et al., "Solid phase DNA amplification: characterisation of primer attachment mechanisms," Nucl. Acids Res., 2000, 28(20):E87, 8 pages. cited by applicant

Affymetrix, "GeneChip Human Genome U133 Set," retrieved from the Internet: on the World Wide Web at affymetrix.com/support/technical/datasheets/hgu133a2.pdf, on Feb. 26, 2003, 2 pages. cited by applicant

Affymetrix, "Human Genome U95Av2," Internet Citation, retrieved from the internet: on the World Wide Web affymetrix.com, retrieved on Oct. 2, 2002, 1 page. cited by applicant

Albretsen et al., "Applications of magnetic beads with covalently attached oligonucleotides in hybridization: Isolation and detection of specific measles virus RNA," *Anal. Biochem.*, 1990, 189(1):40-50. cited by applicant

Allawi et al., "Thermodynamics and NMR of Internal GâT Mismatches in DNA," *Biochemistry*, 1996, 36(34):10581-10594. cited by applicant

Andor.com [online], "Discover new ways of seeing," Next Generation Digital Illumination, Mosaic 3, 2020, 11 pages. cited by applicant

Armani et al., "2D-PCR: a method of mapping DNA in tissue sections," *Lab Chip*, 2009, 9(24):3526-34. cited by applicant

Asp et al., "Spatially Resolved Transcriptomes-Next Generation Tools for Tissue Exploration," *Bioessays*, Oct. 2020, 42(10):e1900221, 16 pages. cited by applicant

Atkinson et al., "An Updated Protocol for High Throughput Plant Tissue Sectioning," *Front Plant Sci*, 2017, 8:1721, 8 pages. cited by applicant

Atkinson, "Overview of Translation: Lecture Manuscript," U of Texas, 2000, DD, pp. 6.1-6.8. cited by applicant

Bains et al., "A novel method for nucleic acid sequence determination," *Journal of Theoretical Biology*, 1988, 135(3), 303-7. cited by applicant

Barnes, "PCR amplification of up to 35-kb DNA with high fidelity and high yield from lambda bacteriophage templates," *Proc. Natl. Acad. Sci. USA*, 1994, 91(12):3573-3577. cited by applicant

Beattie et al., "Advances in genosensor research," *Clin Chem.*, May 1995, 41(5):700-6. cited by applicant

Beechem et al., "High-Plex Spatially Resolved RNA and Protein Detection Using Digital Spatial Profiling: A Technology Designed for Immuno-oncology Biomedical Translational Research," *Methods Mol Biol*, 2020, Chapter 25, 2055:563-583. cited by applicant

Bergenstr hle et al., "Seamless integration of image and molecular analysis for spatial transcriptomics workflows," *BMC Genomics*, Jul. 2020, 21(1):482, 7 pages. cited by applicant

Birney et al., "Identification and analysis of functional elements in 1% of the human genome by the ENCODE pilot project," *Nature*, 2007, 447(7146):799-816. cited by applicant

Blair et al., "Microarray temperature optimization using hybridization kinetics," *Methods Mol Biol.*, 2009, 529:171-96. cited by applicant

Blanchard et al., "High-density oligonucleotide arrays," *Biosensors & Bioelectronics*, 1996, 11(6-7):687-690. cited by applicant

Blokzijl et al., "Profiling protein expression and interactions: proximity ligation as a tool for personalized medicine," *J Intern. Med.*, 2010, 268(3):232-245. cited by applicant

Blow, "Tissue Issues," *Nature*, 2007, 448(7156):959-962. cited by applicant

Bolotin et al., "MiXCR: software for comprehensive adaptive immunity profiling," *Nat Methods.*, May 2015, 12(5):380-1. cited by applicant

Borm et al., "High throughput human embryo spatial transcriptome mapping by surface transfer of tissue RNA," Abstracts Selected Talks, Single Cell Genomics, 2020, 1 pages (Abstract Only). cited by applicant

Brandon et al., "Mitochondrial mutations in cancer," *Oncogene*, 2006, 25(34):4647-4662. cited by applicant

Brenner et al., "Gene expression analysis by massively parallel signature sequencing (MPSS) on microbead arrays," *Nat. Biotech.*, 2000, 18(6):630-634. cited by applicant

Brenner et al., "In vitro cloning of complex mixtures of DNA on microbeads: physical separation of differentially expressed cDNAs," *Proc. Natl. Acad. Sci. U.S.A.*, 1997, 94(12):6370-6375. cited by applicant

Brow, "35—The Cleavase I enzyme for mutation and polymorphism scanning," *PCR Applications Protocols for Functional Genomics*, 1999, pp. 537-550. cited by applicant

Brown et al., "Retroviral integration: structure of the initial covalent product and its precursor, and a role for the viral IN protein," *Proc Natl Acad Sci USA*, Apr. 1996, 93(10):5171-5176. cited by applicant

Buenrostro et al., "Transposition of native chromatin for multimodal regulatory analysis and personal epigenomics," *Nat Methods*, Dec. 2013, 10(12):1213-1221. cited by applicant

Bullard et al., "Direct comparison of nick-joining activity of the nucleic acid ligases from bacteriophage T4," *Biochem. J.* 2006, 398(1):135-144. cited by applicant

Burgess, "A space for transcriptomics," *Nature Reviews Genetics*, 2016, 17(8):436-7. cited by applicant

Burgess, "Finding structure in gene expression," *Nature Reviews Genetics*, 2018, 19(5):249, 1 page. cited by applicant

Burgess, "Spatial transcriptomics coming of age," *Nat Rev Genet.*, Jun. 2019, 20(6):317, 1 page. cited by applicant

Burton et al., "Coverslip Mounted-Immersion Cycled in Situ RT-PCR for the Localization of mRNA in Tissue Sections," *Biotechniques*, 1998, 24(1):92-100. cited by applicant

Calvert, "Materials science. Printing cells," *Science*, Oct. 2007, 318(5848):208-209. cited by applicant

Carter et al., "Stabilization of an optical microscope to 0.1 nm in three dimensions," *Applied Optics*, 2007, 46:421-427. cited by applicant

Cha et al., "Specificity, efficiency, and fidelity of PCR," *Genome Res.*, 1993, 3(3):S18-29. cited by applicant

Chandra et al., "Cell-free synthesis-based protein microarrays and their applications," *Proteomics*, 2009, 5(6):717-30. cited by applicant

Chatterjee et al., "Mitochondrial DNA mutations in human cancer. *Oncogene*," 2006, 25(34):4663-4674. cited by applicant

Chen et al., "DNA hybridization detection in a microfluidic Channel using two fluorescently labelled nucleic acid probes," *Biosensors and Bioelectronics*, 2000, 15(12):1033-1038. cited by applicant

Chen et al., "Large field of view-spatially resolved transcriptomics at nanoscale resolution," *bioRxiv*, Jan. 19, 2021, retrieved from URL <<https://www.biorxiv.org/node/1751045.abstract>>, 37 pages. cited by applicant

Chen et al., "Spatial Transcriptomics and In Situ Sequencing to Study Alzheimer's Disease," *Cell*, Aug. 2020, 182(4):976-991. cited by applicant

Chen et al., " CB-seq: microfluidic cell barcoding and sequencing for high-resolution imaging and sequencing of single cells," *Lab Chip*, Nov. 2020, 20(21):3950-3961. cited by applicant

Cho et al., "Seq-Scope: Submicrometer-resolution spatial transcriptomics for single cell and subcellular studies," *bioRxiv*, Jan. 27, 2021, retrieved from URL <<https://www.biorxiv.org/node/1754517.abstract>>, 50 pages. cited by applicant

Cockroft et al., "A single-molecule nanopore device detects DNA polymerase activity with single-nucleotide resolution," *J Am Chem Soc.*, Jan. 2008, 130(3):1009-1014. cited by applicant

Codeluppi et al., "Spatial organization of the somatosensory cortex revealed by osmFISH," *Nature Methods*, Nov. 2018, 15:932-935. cited by applicant

Constantine et al., "Use of genechip high-density oligonucleotide arrays for gene expression monitoring," *Life Science News, Amersham Life Science*, 1998, 19(1):1-6. cited by applicant

Crosetto et al., "Spatially resolved transcriptomics and beyond," *Nature Review Genetics*, 2015, 16(1):57-66. cited by applicant

Czarnik, "Encoding methods for combinatorial chemistry," *Curr Opin Chem Biol.*, Jun. 1997, 1(1):60-6. cited by applicant

Dahl et al., "Circle-to-circle amplification for precise and sensitive DNA analysis," *Proc. Natl. Acad. Sci.*, 2004, 101(13):4548-4553. cited by applicant

Dalma-Weiszhauz et al., "The affymetrix GeneChip platform: an overview," *Methods Enzymol.*, 2006, 410:3-28. cited by applicant

Daubendiek et al., "Rolling-Circle RNA Synthesis: Circular Oligonucleotides as Efficient Substrates for T7 RNA Polymerase," *J. Am. Chem. Soc.*, 1995, 117(12):3111-3116. cited by applicant

Davies et al., "How best to identify chromosomal interactions: a comparison of approaches," *Nat. Methods*, 2017, 14(2):125-134. cited by applicant

Deamer et al., "Characterization of nucleic acids by Nanopore analysis," *Acc Chem Res.*, Oct. 2002, 35(10):817-25. cited by applicant

Dean et al., "Comprehensive human genome amplification using multiple displacement amplification," *Proc Natl. Acad. Sci. USA*, 2002, 99(8):5261-66. cited by applicant

Dhindsa et al., "Virtual Electrowetting Channels: Electronic Liquid Transport with Continuous Channel Functionality," *Lab Chip*, 2010, 10:832-836. cited by applicant

Duncan et al., "Affinity chromatography of a sequence-specific DNA binding protein using Teflon-linked oligonucleotides," *Anal. Biochem.*, 1988, 169(1):103-108. cited by applicant

Eguiluz et al., "Multitissue array review: a chronological description of tissue array techniques, applications and procedures," *Pathology Research and Practice*, 2000, 196(1):1-10. cited by applicant

Eldridge et al., "An in vitro selection strategy for conferring protease resistance to ligand binding peptides," *Protein Eng Des Sel.*, 2009, 22(11):691-698. cited by applicant

Ellington et al., "Antibody-based protein multiplex platforms: technical and operational challenges," *Clin Chem*, 2010, 56(2):186-193. cited by applicant

Eng et al., "Transcriptome-scale super-resolved imaging in tissues by RNA seqFISH+," *Nature*, Apr. 2019, 568(7751):235-239, 37 pages. cited by applicant

Fire et al., "Rolling replication of short DNA circles," *Proc. Natl. Acad. Sci.*, 1995, 92(10):4641-4645. cited by applicant

Fluidigm, "Hyperion Imaging System: Visualize a new path forward," Feb. 2018, retrieved from URL <<https://www.fluidigm.com/binaries/content/documents/fluidigm/search/hippo%3Aresultset/hyperion-imaging-system-br-400326/fluidigm%3Afile>>, 27 pages. cited by applicant

Fluidigm, "Immuno-Oncology Research with the Hyperion Imaging System: A high-parameter imaging solution at subcellular resolution to characterize the tumor microenvironment," Apr. 2018, retrieved from URL <<https://www.fluidigm.com/binaries/content/documents/fluidigm/search/hippo%3Aresultset/application-immuno-oncology-research-with-the-hyperion-imaging-system-br-400332/fluidigm%3Afile>>, 6 pages. cited by applicant

Fluidigm, "Immuno-Oncology Research with the Hyperion Imaging System: A high-parameter imaging solution at subcellular resolution to characterize the tumor microenvironment," Aug. 2018, retrieved from URL <<https://www.fluidigm.com/binaries/content/documents/fluidigm/marketing/immuno-oncology-research-with-the-hyperion-imaging-system-appnote-400332/immuno-oncology-research-with-the-hyperion-imaging-system-appnote-400332/fluidigm%3Afile>>, 6 pages. cited by applicant

Fluidigm, "Maxpar Antibodies for Imaging Mass Cytometry," Mar. 2018, retrieved from URL <<https://www.fluidigm.com/binaries/content/documents/fluidigm/search/hippo%3Aresultset/hyperion-antibodies-for-imaging-mass-cytometry-br-101-7115/fluidigm%3Afile>>, 6 pages. cited by applicant

Fodor et al., "Light-directed, spatially addressable parallel chemical synthesis," *Science*, 1995, 251(4995):767-773. cited by applicant

Forcucci et al., "All-plastic miniature fluorescence microscope for point-of-care readout of bead-based bioassays," *J. Biomed Opt.*, 2015, 20(10):105010, 15 pages. cited by applicant

Forster et al., "A human gut bacterial genome and culture collection for improved metagenomic analyses," *Nature Biotechnology*, 2019, 37(2):186-192. cited by applicant

Frese et al., "Formylglycine aldehyde Tag—protein engineering through a novel post-translational modification," *ChemBioChem.*, 2009, 10(3):425-27. cited by applicant

Fu et al., "Continuous Polony Gels for Tissue Mapping with High Resolution and RNA Capture Efficiency," *bioRxiv*, 2021, 20 pages. cited by applicant

Fu et al., "Counting individual DNA molecules by the stochastic attachment of diverse labels," *PNAS*, 2011, 108(22):9026-9031. cited by applicant

Fullwood et al., "Next-generation DNA sequencing of paired-end tags (PET) for transcriptome and genome analyses," *Genome Res.*, 2009, 19(4):521-532. cited by applicant

Ganguli et al., "Pixelated spatial gene expression analysis from tissue," *Nat Commun.*, Jan. 2018, 9(1):202, 9 pages. cited by applicant

Gene@arrays[online], BeadArray Technology, available on or before Feb. 14, 2015, via Internet Archive: Wayback Machine URL <<https://web.archive.org/web/20150214084616/http://genearrays.com/services/microarrays/illumina/beadarray-technology/>>, [retrieved on Jan. 30, 2020], 3 pages. cited by applicant

Gnanaprasagam, "Unlocking the molecular archive: the emerging use of formalin-fixed paraffin-embedded tissue for biomarker research in urological cancer," *Urology*, 2010, 75(2):274-278. cited by applicant

Goh et al., "Highly Specific Multiplexed RNA Imaging in Tissues With Split-FISH," *Nat Methods*, Jun. 15, 2020, 17(7):689-693, 21 pages. cited by applicant

Goldkorn et al., "A simple and efficient enzymatic method for covalent attachment of DNA to cellulose. Application for hybridization-restriction analysis and DNA probes," *Nucleic Acids Res.*, 1986, 14(22):9171-9191. cited by applicant

Gracia Villacampa et al., "Genome-wide Spatial Expression Profiling in FFPE Tissues," *bioRxiv*, 2020, pp. 38 pages. cited by applicant

Gunderson et al., "Decoding randomly ordered DNA arrays," *Genome Research*, 2004, 14(5):870-877. cited by applicant

Guo et al., "Direct fluorescence analysis of genetic polymorphisms by hybridization with oligonucleotide arrays on glass supports," *Nucleic Acids Res.*, Dec. 1996, 24(24):5355-5360. cited by applicant

Hamaguchi et al., "Direct reverse transcription-PCR on oligo(dT)-immobilized polypropylene microplates after capturing total mRNA from crude cell lysates," *Anal Biochem.*, 2000, 282(1):44-49. cited by applicant

Hattersley et al., "Development of a microfluidic device for the maintenance and interrogation of viable tissue biopsies," *Lab Chip.*, Nov. 2008, 8(11):1842-6. cited by applicant

Hayes et al., "Electrophoresis of proteins and nucleic acids: I-Theory," *BMJ*, Sep. 1989, 299(6703):843-6. cited by applicant

He et al., "In situ synthesis of protein arrays," *Current Opinion in Biotechnology*, 2008, 19(1):4-9. cited by applicant

He, "Cell-free protein synthesis: applications in proteomics and biotechnology," *New Biotechnology*, 2008, 25(2-3):126-132. cited by applicant

Hejatko et al., "In situ hybridization technique for mRNA detection in whole mount *Arabidopsis* samples," *Nature Protocols*, 2006, 1(4):1939-1946. cited by applicant

Hiatt et al., "Parallel, tag-directed assembly of locally derived short sequence reads," *Nature Methods*, 2010, 7(2):119-25. cited by applicant

Hoyer et al., "Electrostatic spraying: a novel technique for preparation of polymer coatings on electrodes," *Anal Chem*, Nov. 1996, 68(21):3840-4. cited by applicant

Jamur et al., "Permeabilization of cell membranes," *Method Mol. Biol.*, 2010, 588:63-66. cited by applicant

Jemt et al., "An automated approach to prepare tissue-derived spatially barcoded RNA-sequencing libraries," *Scientific Reports*, 2016, 6:37137, 10 pages. cited by applicant

Kapteyn et al., "Incorporation of non-natural nucleotides into template-switching oligonucleotides reduces background and improves cDNA synthesis from vesicles," *BMC Genomics*, Jul. 2010, 11:413, 9 pages. cited by applicant

Korbel et al., "Paired-end mapping reveals extensive structural variation in the human genome," *Science*, 2007, 318(5849):420-426. cited by applicant

Kozlov et al., "A highly scalable peptide-based assay system for proteomics," *PLoS ONE*, 2012, 7(6):e37441, 10 pages. cited by applicant

Kristensen et al., "High-Throughput Methods for Detection of Genetic Variation," *BioTechniques*, Feb. 2001, 30(2):318-332. cited by applicant

Kurz et al., "cDNA—protein fusions: covalent protein—gene conjugates for the in vitro selection of peptides and proteins," *ChemBioChem.*, 2001, 2(9):666-7. cited by applicant

Kwok, "High-throughput genotyping assay approaches," *Pharmacogenomics*, Feb. 2000, 1(1):95-100. cited by applicant

Lage et al., "Whole genome analysis of genetic alterations in small DNA samples using hyperbranched strand displacement amplification and array-CGH," *Genome Res.*, 2003, 13(2):294-307. cited by applicant

Landegren et al., "Reading bits of genetic information: methods for single-nucleotide polymorphism analysis," *Genome Res.*, Aug. 1998, 8(8):769-76. cited by applicant

Langdale et al., "A rapid method of gene detection using DNA bound to Sephacryl," *Gene*, 1985, 36(3):201-210. cited by applicant

Leriche et al., "Cleavable linkers in chemical biology," *Bioorganic & Medicinal Chemistry*, 2012, 20:571-582. cited by applicant

Levene et al., "Zero-Mode Waveguides for Single-Molecule Analysis at High Concentrations," *Science*, 2003, 299:682-686. cited by applicant

Li et al., "DNA molecules and configurations in a solid-state nanopore microscope," *Nat Mater.*, Sep. 2003, 2(9):611-5. cited by applicant

Lin et al., "Microfluidic cell trap array for controlled positioning of single cells on adhesive micropatterns," *Lab Chip*, Feb. 2013, 13(4):714-721. cited by applicant

Linnarsson, "Recent advances in DNA sequencing methods—general principles of sample preparation," *Experimental Cell Research*, 2010, 316(8):1339-1343. cited by applicant

Liu et al., "High-Spatial-Resolution Multi-Omics Atlas Sequencing of Mouse Embryos via Deterministic Barcoding in Tissue," *bioRxiv*, 2019, 55 pages. cited by applicant

Liu et al., "High-Spatial-Resolution Multi-Omics Sequencing via Deterministic Barcoding in Tissue," *Cell*, Nov. 13, 2020, 183(6):1665-1681, 36 pages. cited by applicant

Liu et al., "Spatial transcriptome sequencing of FFPE tissues at cellular level," *bioRxiv* 788992, Oct. 14, 2020, 39 pages. cited by applicant

Lizardi et al., "Mutation detection and single-molecule counting using isothermal rolling-circle amplification," *Nat. Genet.*, 1998, 19(3):225-232. cited by applicant

Lu et al., "A microfluidic electroporation device for cell lysis," *Lab Chip.*, Jan. 2005, 5(1):23-29. cited by applicant

Lundberg et al., "Multiplexed homogeneous proximity ligation assays for high-throughput protein biomarker research in serological material," *Mol Cell Proteomics*, 2004, 3(10):M110.004978, 11 pages. cited by applicant

Lundquist et al., "Parallel confocal detection of single molecules in real time," *Opt. Lett.*, 2008, 33:1026-1028. cited by applicant

Macbeath et al., "Printing proteins as microarrays for high-throughput function determination," *Science*, Sep. 2000, 289(5485):1760-1763. cited by applicant

Marx, "Method of the Year: spatially resolved transcriptomics," *Nature Methods*, 2021, 18(1):9-14. cited by applicant

Merritt et al., "Multiplex digital spatial profiling of proteins and RNA in fixed tissue," *Nat Biotechnol.*, May 2020, 38(5):586-599. cited by applicant

Metzker, "Sequencing technologies—the next generation," *Nature Reviews Genetics*, 2010, 11(1):31-46. cited by applicant

Miller et al., "Basic concepts of microarrays and potential applications in clinical microbiology," *Clinical Microbiology Reviews*, 2009, 22(4):611-633. cited by applicant

Miller et al., "Chapter 11—Solid and Suspension Microarrays for Microbial Diagnostics," *Methods in Microbiology*, 2015, 42:395-431. cited by applicant

Mishra et al., "Three-dimensional genome architecture and emerging technologies: looping in disease," *Genome Medicine*, 2017, 9(1):87, 14 pages. cited by applicant

Mitra et al., "Digital genotyping and haplotyping with polymerase colonies," *Proc. Natl. Acad. Sci. USA*, May 2003, 100(10):5926-5931. cited by applicant

Mizusawa et al., "A bacteriophage lambda vector for cloning with BamHI and Sau3A," *Gene*, 1982, 20(3):317-322. cited by applicant

Ncbi.nlm.nih.gov, [online], "Molecular Inversion Probe Assay," available on or before Oct. 14, 2014, via Internet Archive: Wayback Machine <<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4141411/>>, 10 pages. cited by applicant

URL<<https://www.ncbi.nlm.nih.gov/probe/docs/techmip/>>, 2 pages. cited by applicant

Nikiforov et al., "The use of 96-well polystyrene plates for DNA hybridization-based assays: an evaluation of different approaches to oligonucleotide immobilization," *Anal Biochem*, May 1995, 227(1):201-9. cited by applicant

Nowak et al., "Entering the Postgenome Era," *Science*, 1995, 270(5235):368-71. cited by applicant

PCT International Search Report and Written Opinion in International Appln. No. PCT/US2019/065100 dated Apr. 6, 2020, 20 pages. cited by applicant

Pemov et al., "DNA analysis with multiplex microarray-enhanced PCR," *Nucl. Acids Res.*, Jan. 2005, 33(2):e11, 9 pages. cited by applicant

Perler et al., "Intervening sequences in an Archaea DNA polymerase gen," *Proc Natl Acad Sci USA*, Jun. 1992, 89(12):5577-5581. cited by applicant

Petterson et al., "Generations of sequencing technologies," *Genomics*, 2009, 93(2):105-111. cited by applicant

Polisky-Cynkin et al., "Use of DNA immobilized on plastic and agarose supports to detect DNA by sandwich hybridization," *Clin. Chem.*, 1985, 31(9):1438-1442. cited by applicant

Ranki et al., "Sandwich hybridization as a convenient method for the detection of nucleic acids in crude samples," *Gene*, 1983, 21(1-2):77-85. cited by applicant

Reijenga et al., "Buffer Capacity, Ionic Strength and Heat Dissipation in Capillary Electrophoresis," *Journal of Chromatography A*, Sep. 13, 1996, 744(1-2):1-10. cited by applicant

Reinartz et al., "Massively parallel signature sequencing (MPSS) as a tool for in-depth quantitative gene expression profiling in all organisms," *Brief Funct Genomics*, 2002, 1(1):95-104. cited by applicant

Rodrigues et al., "Slide-seq: A scalable technology for measuring genome-wide expression at high spatial resolution," *Science*, 2019, 363(6434):1463-1467. cited by applicant

Ronaghi et al., "A sequencing method based on real-time pyrophosphate," *Science*, Jul. 1998, 281(5375):363-365. cited by applicant

Ronaghi et al., "Real-time DNA sequencing using detection of pyrophosphate release," *Analytical Biochemistry*, Nov. 1996, 242(1):84-89. cited by applicant

Ronaghi, "Pyrosequencing sheds light on DNA sequencing," *Genome Res*, Jan. 2001, 11(1):3-11. cited by applicant

Rountenberg et al., "Microfluidic probe: a new tool for integrating microfluidic environments and electronic wafer-orobina," *Lab Chip*, Oct. 2009, 10: 123-128. cited by applicant

Salmén et al., "Barcoded solid-phase RNA capture for Spatial Transcriptomics profiling in mammalian tissue sections," *Nature Protocols*, Oct. 2018, 13(11):2018-2024. cited by applicant

Saxonov et al., "10x Genomics, Mastering Biology to Advance Human Health," PowerPoint, 10x, 2020, 41 pages. cited by applicant

Schena et al., "Quantitative monitoring of gene expression patterns with a complementary DNA microarray," *Science*, Oct. 1995, 270(5235):467-470. cited by applicant

Shalon et al., "A DNA microarray system for analyzing complex DNA samples using two-color fluorescent probe hybridization," *Genome Res.*, Jul. 1996, 6(7):1215-1222. cited by applicant

Soni and Meller, "Progress toward ultrafast DNA sequencing using solid-state nanopores," *Clin Chem.*, 2007, 53:1996-2001. cited by applicant

Stahl et al., "Visualization and analysis of gene expression in tissue sections by spatial transcriptomics," *Science*, Jul. 2016, 353(6294):78-82. cited by applicant

Stahl et al., "Visualization and analysis of gene expression in tissue sections by spatial transcriptomics," *Supplementary Materials*, *Science*, Jul. 2016, 353(6294):78-82. cited by applicant

Stimpson et al., "Real-time detection of DNA hybridization and melting on oligonucleotide arrays by using optical wave guides," *Proc Natl Acad Sci USA*, Jun. 2001, 98(12):13500-13505. cited by applicant

Strell et al., "Placing RNA in context and space—methods for spatially resolved transcriptomics," *The FEBS Journal*, 2019, 286(8):1468-1481. cited by applicant

Takei et al., "Integrated Spatial Genomics Reveals Global Architecture of Single Nuclei," *Nature*, Jan. 27, 2021, 590(7845):344-350, 53 pages. cited by applicant

Taylor et al., "Microfluidic local perfusion chambers for the visualization and manipulation of synapses," *Neuron.*, Apr. 2010, 66(1):57-68, 25 pages. cited by applicant

Thomas et al., "A chamber for the perfusion of in vitro tissue with multiple solutions," *J. Neurophysiol.*, Jul. 2013, 110:269-277. cited by applicant

Tijssen et al., "Overview of principles of hybridization and the strategy of nucleic acid assays" in *Techniques in Biochemistry and Molecular Biology—Hybridization*, 1993, 24(Chapter 2), 65 pages. cited by applicant

Toy et al., "A Simple Plastic Perfusion Chamber for Continuous Maintenance and Cinematography of Tissue Cultures," *Experimental Cell Research*, 1958, 14:1-10. cited by applicant

Twyman et al., "Techniques Patents for SNP Genotyping," *Pharmacogenomics*, Jan. 2003, 4(1):67-79. cited by applicant

Valley et al., "Optoelectronic tweezers as a tool for parallel single-cell manipulation and stimulation," *IEEE Trans Biomed Circuits Syst.*, Dec. 2009, 3(6):424-431. cited by applicant

Van Gelder et al., "Amplified RNA synthesized from limited quantities of heterogeneous cDNA," *Proc. Natl. Acad. Sci. USA*, 1990, 87(5):1663-1667. cited by applicant

Vasiliskov et al., "Fabrication of microarray of gel-immobilized compounds on a chip by copolymerization," *Biotechniques*, Sep. 1999, 27(3):592-606. cited by applicant

Vickovic et al., "High-definition spatial transcriptomics for in situ tissue profiling," *Nat Methods*, Oct. 2019, 16(10):987-990. cited by applicant

Vickovic et al., "SM-Omics: An automated Platform for High-Throughput Spatial Multi-Omics," *bioRxiv*, Oct. 2020, 40 pages. cited by applicant

Vogelstein et al., "Digital PCR," *Proceedings of the National Academy of Sciences*, Aug. 1999, 96(16):9236-9241. cited by applicant

Walker et al., "Strand displacement amplification—an isothermal, in vitro DNA amplification technique," *Nucleic Acids Research*, 1992, 20(7):1691-1696. cited by applicant

Wang et al., "Concentration gradient generation methods based on microfluidic systems," *RSC Adv.*, 2017, 7:29966-29984. cited by applicant

Wang et al., "High-fidelity mRNA amplification for gene profiling," *Nature Biotechnology*, Apr. 2000, 18(4):457-459. cited by applicant

Wang et al., "Imaging-based pooled CRISPR screening reveals regulators of lncRNA localization," *Proc Natl Acad Sci USA*, May 2019, 116(22):10842-10850. cited by applicant

Wheeler et al., "Microfluidic device for single-cell analysis," *Analytical Chemistry*, Jul. 2003, 75(14):3581-3586. cited by applicant

Worthington et al., "Cloning of random oligonucleotides to create single-insert plasmid libraries," *Anal Biochem*, 2001, 294(2):169-175. cited by applicant

Xia et al., "Spatial transcriptome profiling by MERFISH reveals subcellular RNA compartmentalization and cell cycle-dependent gene expression," *Proceedings of the National Academy of Sciences*, Sep. 2019, 116(39):19490-19499. cited by applicant

Yershov et al., "DNA analysis and diagnostics on oligonucleotide microchips," *Proc. Natl. Acad. Sci. USA*, May 1996, 93(10):4913-4918. cited by applicant

Yoda et al., "Site-specific gene expression analysis using an automated tissue micro-dissection punching system," *Sci Rep.*, Jun. 2017, 7(1):4325, 11 pages. cited by applicant

Zhu et al., "Reverse transcriptase template switching: a SMART approach for full-length cDNA library construction," *Biotechniques*, Apr. 2001, 30(4):892-895. cited by applicant

Kim, "Development of Microdevices for Applications to Bioanalysis," Dissertation for the degree of Doctor of Philosophy, University of Texas at Austin, Aug. 2000. cited by applicant

Sun et al., "Statistical Analysis of Spatial Expression Pattern for Spatially Resolved Transcriptomic Studies," *Nature Methods*, Jan. 27, 2020, 17(2): 193-200. cited by applicant

Svensson et al., "SpatialDE: identification of spatially variable genes," *Nature Methods*, May 2018, 15:343-346, 15 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a National Stage Application under 35 U.S.C. § 371 and claims the benefit of International Application No. PCT/US2021/032944, filed May 18, 2021, which claims the benefit of U.S. Patent Application Ser. No. 63/026,975 titled ELECTROPHORESIS CASSETTES AND INSTRUMENTATION, filed May 19, 2020, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Cells within a tissue have differences in cell morphology and/or function due to varied analyte levels (e.g., gene and/or protein expression) within

the different cells within a cell within a tissue (e.g., the cell's position relative to neighboring cells or the cell's position relative to the tissue microenvironment) can affect, e.g., the cell's morphology, differentiation, fate, viability, proliferation, behavior, signaling, and cross-talk with other cells in the tissue.

(2) Spatial heterogeneity has been previously studied using techniques that typically provide data for a handful of analytes in the context of intact tissue or a portion of a tissue (e.g., tissue section), or provide significant analyte data from individual, single cells, but fails to provide information regarding the position of the single cells from the originating biological sample (e.g., tissue).

(3) Various methods have been used to prepare a biological sample for analyzing analyte data in the sample. In some applications, a biological sample can be permeabilized to facilitate transfer of analytes out of the sample, and/or to facilitate transfer of species (such as capture probes) into the sample.

SUMMARY

(4) This document generally relates to electrophoretic devices, apparatuses, systems, instruments, and methods for analyte migration.

(5) In general, electrophoresis can be used to migrate analytes from a sample toward a substrate that includes capture probes for spatial transcriptomics applications in order to enhance sensitivity and/or resolution. The present disclosure provides various implementations of electrophoretic devices, systems and instruments to improve electrophoresis and ease of use. Electrophoretic systems and instruments described herein provide different designs for electrode geometry and/or placement, and further provide different approaches for electrical interfacing between components. For example, various configurations of electrophoretic systems and instruments described herein utilize different electrode designs that can result in uniform electromigration of analytes. Further, the electrophoretic systems described herein consider that, where a sample tissue is very thin, relatively small angles of the electric field has no or little effect on drift of analyte migration. Moreover, various implementations of electrophoretic systems and instrumentations optimize the size and/or location (x, y, z directions) of electrodes (e.g., cathode) relative to a sample, thereby increasing uniformity of electric field and reducing or minimizing drift of analyte migration. In addition, the electrophoresis systems and instruments described herein provides a solution to conveniently check for electrical connections in the systems and instruments, and alert users of potential improper electrical connections, prior to performing electrophoresis.

(6) Particular implementations of the present disclosure described herein provide an electrophoretic system for analyte migration. The system includes a substrate, a substrate cassette, a cathode assembly, and a power supply. The substrate may include a plurality of regions. Each region may be electrically conductive and further configured to comprise one or more capture probes and a biological sample comprising an analyte. The substrate may include a first electrode contact. The first electrode contact may be electrically connected to at least one of the plurality of regions. The substrate cassette may be disposed at the substrate and include a plurality of apertures corresponding to the plurality of regions of the substrate. The plurality of apertures may define a plurality of buffer chambers comprising the plurality of regions of the substrate. The substrate cassette may comprise a connection interface electrically coupled to the first electrode contact of the substrate. In some implementations, the substrate cassette may comprise the connection interface that exposes the first electrode contact of the substrate. The cathode assembly may include a plurality of electrodes positioned within the plurality of buffer chambers of the substrate cassette, respectively. The cathode assembly may include a second electrode contact. The second electrode contact may be electrically connected to at least one of the plurality of electrodes. The power supply may be electrically connected with the first electrode contact of the substrate at the connection interface of the substrate cassette, and electrically connected with the second electrode contact of the cathode assembly. The power supply may generate an electric field between the plurality of regions and the plurality of electrodes, respectively, such that the analyte in the biological sample migrates toward the capture probe on the substrate.

(7) In some implementations, the system described herein can include one or more of the following features. The capture probe may be coated, spotted or printed on each of the plurality of regions. The biological sample may be placed in contact with one or more of the plurality of regions on the substrate. The substrate cassette may include a substrate holder including a substrate mount for securing the substrate, and a gasket including a plurality of gasket apertures configured to align with the plurality of regions when the substrate is secured by the substrate holder. The plurality of apertures may include the plurality of gasket apertures. The substrate holder may include a plurality of holder apertures configured to align with the plurality of gasket apertures when the substrate is secured by the substrate holder. The plurality of apertures may include the plurality of gasket apertures and the plurality of holder apertures. The substrate may include a glass slide coated with a conductive material. The conductive material may include at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), fluorine doped tin oxide (FTO), or any combination thereof. The electrophoretic system may include a first electrical wire connecting the power supply to the first electrode contact, and a second electrical wire connecting the power supply to the second electrode contact. The plurality of buffer chambers may receive a buffer. The buffer may include a permeabilization reagent. The electrophoretic system may include a light configured to illuminate onto the plurality of regions to permeabilize the biological sample on the plurality of regions. The sample may be permeabilized using a detergent before, after, or during enzymatic treatment. The sample may be permeabilized with a lysis reagent being added to the sample. The sample may be permeabilized by exposing the sample to a protease.

(8) The substrate cassette may include a body defining a cavity configured to partially receive the substrate. The connection interface of the substrate cassette may include a slit defined at the body and configured for the substrate to partially extend out from the body such that the first electrode contact of the substrate is positioned outside the body. The body may include a main face and a lateral face extending from a periphery of the main face. The plurality of the apertures may be defined at the main face of the body. The slit may be defined at the lateral face of the body.

(9) The substrate cassette may include a body defining a cavity configured to receive the substrate. The connection interface of the substrate cassette may include a contact opening defined at the body and configured to expose the first electrode contact of the substrate there through. The body may include a main face and a lateral face extending from a periphery of the main face. The plurality of the apertures may be defined at the main face of the body. The contact opening may be defined at the main face adjacent the lateral face. The contact opening may be defined further at the lateral face of the body to extend from the main face to the lateral face.

(10) The electrophoretic system may include a first contact pin electrically attached to the substrate and providing the first electrode contact. The substrate cassette may include a body defining a cavity configured to receive the substrate. The connection interface of the substrate cassette may include a contact hole defined at the body and configured to expose the first contact pin therethrough. The body may include a main face and a lateral face extending from a periphery of the main face. The plurality of the apertures may be defined at the main face of the body. The contact hole may be defined at the main face.

(11) The substrate cassette may include a body defining a cavity configured to receive the substrate, and a contact pin extending through a wall of the body and having a first end and a second end opposite to the first end. The first end may be arranged exterior of the body, and the second end may be arranged interior of the body and configured to electrically contact with the first electrode contact of the substrate. The electrophoretic system may include a contact bracket that electrically engages with the substrate and provides the first electrode contact of the substrate. The second end of the contact pin may electrically contact with the contact bracket.

(12) The cathode assembly may include a cathode body configured to position at least partially on the substrate cassette. The plurality of electrodes may project from the cathode body. Each of the plurality of electrodes may include at least one of a conductive wire, a conductive rod, or an array of conductive wires. Each of the plurality of electrodes may include a conductive plate at a distal end of the at least one of the conductive wire, the conductive rod, or the array of conductive wires. The conductive plate may be configured as at least one of a circular plate, a square plate, or a ring. The cathode body may include the second electrode contact.

(13) The substrate cassette may include a second aperture corresponding to the first electrode contact of the substrate and configured to define a second buffer chamber on the first electrode contact of the substrate. The plurality of buffer chambers may contain a first buffer. The second chamber

may contain a second buffer different from the first buffer. The second buffer may have an electrolyte strength greater than the first buffer. The second buffer may include a sodium chloride solution. The cathode assembly may include a cathode body configured to position at least partially on the substrate cassette. The plurality of electrodes may project from the cathode body and be configured to position within the plurality of buffer chambers, respectively. The cathode assembly may include a second electrode projecting from the cathode body and configured to position within the second chamber. The cathode body may include an anode contact that is electrically connected to the second electrode. The power supply may be electrically connected to the anode contact of the cathode body so that the power supply is electrically connected with the first electrode contact of the substrate via the second electrode that positions within the second chamber containing the second buffer on the first electrode contact of the substrate.

(14) The substrate cassette may include a cassette body defining a cavity configured to receive the substrate. The cassette body may include the cathode assembly such that the plurality of electrodes are configured to position within the plurality of buffer chambers. The substrate cassette may include a plurality of cathode contact pins extending through a wall of the cassette body and electrically contacting with the second electrode contact such that the cathode contact pin is electrically connected to the at least one of the plurality of electrodes. Each of the plurality of electrodes may include at least one of a conductive wire, a conductive rod, or an array of conductive wires. Each of the plurality of electrodes may include a conductive plate at a distal end of the at least one of the conductive wire, the conductive rod, or the array of conductive wires. The cassette body may include an anode contact pin extending through the wall of the body and having a first end and a second end opposite to the first end. The first end may be arranged exterior of the cassette body and the second end may be arranged interior of the cassette body and configured to electrically contact with the first electrode contact of the substrate. The electrophoretic system may include a contact bracket that electrically engages with the substrate and provides the first electrode contact of the substrate. The second end of the anode pin may electrically contact with the contact bracket.

(15) The body may include the cathode assembly such that the plurality of electrodes are configured to position within the plurality of buffer chambers. The substrate cassette may include a cathode contact pin extending through a wall of the body and electrically contacting with the second electrode contact such that the cathode contact pin is electrically connected to the at least one of the plurality of electrodes. The substrate cassette may include an anode contact pin extending from the body, and a conductive wire having a first end and an opposite second end. The first end may be electrically connected to the anode contact pin, and the second end may be electrically connected to the first electrode contact that is positioned outside the body.

(16) The electrophoretic system may include an anode cassette body and an anode contact pin. The anode cassette body may define a cavity configured to receive the first electrode contact of the substrate that is positioned outside the body of the substrate cassette. The anode contact pin may extend through a wall of the anode cassette body and have a first end and a second end opposite to the first end. The first end may be arranged exterior of the anode cassette body and the second end may be arranged interior of the anode cassette body and configured to electrically contact with the first electrode contact of the substrate. The electrophoretic system may include an anode contact bracket that electrically engages with the substrate and provides the first electrode contact of the substrate. The second end of the anode contact pin may electrically contact with the anode contact bracket.

(17) The electrophoretic system may include a system housing including the power supply, a cassette tray that extends from the system housing and is configured to receive the substrate cassette thereon, and a cassette cover that extends from the system housing above the cassette tray.

(18) The electrophoretic system may include a system housing including the power supply, a cassette tray that extends from the system housing and is configured to receive the substrate cassette thereon, and a cassette cover that extends from the system housing above the cassette tray. The cassette cover may include a cathode connector and an anode connector. The cathode connector may electrically engage with the cathode contact pin of the substrate cassette. The anode connector may electrically engage with the anode contact pin of the substrate cassette.

(19) The electrophoretic system may include a system housing including the power supply, a cassette tray that extends from the system housing and is configured to receive the substrate cassette thereon, and a cassette cover that extends from the system housing above the cassette tray and includes the cathode assembly. The plurality of electrodes may project from the cassette cover toward the plurality of chambers of the substrate cassette. The cassette cover may include an anode connector configured to electrically engage with the contact pin of the substrate cassette.

(20) Particular implementations of the present disclosure described herein provide a method for analyte migration. The method may include loading a substrate comprising capture probes with a substrate cassette into an electrophoresis instrument, the substrate cassette including a plurality of buffer chambers, the substrate including a plurality of regions each comprising one or more capture probes and a biological sample containing an analyte; arranging a cathode to place a plurality of electrodes within the plurality of buffer chambers, respectively; electrically connecting a power supply to the substrate; electrically connecting the power supply to the cathode; applying a first voltage between the substrate and the cathode; detecting an output parameter in response to the first voltage; determining whether the output parameter meets a threshold value; and based on the output parameter meeting the threshold value, applying a second voltage to generate electric fields between the plurality of regions of the substrate and the plurality of electrodes through the plurality of buffer chambers to cause the analytes in the biological samples to move toward the capture probes on the substrate.

(21) In some implementations, the method described herein can optionally include one or more of the following features. The output parameter may be an impedance. The threshold value may be indicative of a propriety of at least one of an electrical connection of the power supply to the substrate, an electrical connection of the power supply to the cathode, or an electrical property of the buffers. The method may include, based on the output parameter not meeting the threshold value, generating a notification to inform an electrical connection is improper; and ceasing to apply the second voltage to generate the electric fields. The threshold value may be a predetermined range of value. The method may include, prior to loading the substrate with the substrate cassette, placing the biological samples in contact with the capture probes on the substrate; arranging the substrate cassette onto the substrate to align a plurality of apertures of the substrate cassette with a plurality of regions of the substrate and define the plurality of buffer chambers on the plurality of regions; and supplying buffers in the plurality of buffer chambers.

(22) All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, patent application, or item of information was specifically and individually indicated to be incorporated by reference. To the extent publications, patents, patent applications, and items of information incorporated by reference contradict the disclosure contained in the specification, the specification is intended to supersede and/or take precedence over any such contradictory material.

(23) Where values are described in terms of ranges, it should be understood that the description includes the disclosure of all possible sub-ranges within such ranges, as well as specific numerical values that fall within such ranges irrespective of whether a specific numerical value or specific sub-range is expressly stated.

(24) The term “each,” when used in reference to a collection of items, is intended to identify an individual item in the collection but does not necessarily refer to every item in the collection, unless expressly stated otherwise, or unless the context of the usage clearly indicates otherwise.

(25) Various embodiments of the features of this disclosure are described herein. However, it should be understood that such embodiments are provided merely by way of example, and numerous variations, changes, and substitutions can occur to those skilled in the art without departing from the scope of this disclosure. It should also be understood that various alternatives to the specific embodiments described herein are also within the scope of this disclosure.

DESCRIPTION OF DRAWINGS

- (1) The following drawings illustrate certain embodiments of the features and advantages of this disclosure. These embodiments are not intended to limit the scope of the appended claims in any manner. Like reference symbols in the drawings indicate like elements.
- (2) FIGS. 1A-B are schematic side and top views of an example electrophoretic system for analyte migration.
- (3) FIG. 2 is a schematic perspective view of an example substrate.
- (4) FIG. 3 illustrates an example configuration of a substrate and a sample undergoing electrophoretic process using the electrophoretic system of FIGS. 1A-B.
- (5) FIG. 4A is a schematic top view of an example substrate cassette.
- (6) FIG. 4B is a schematic top view of an example substrate for use with the substrate cassette of FIG. 4A.
- (7) FIG. 4C is a schematic top view of the substrate cassette of FIG. 4A that receives the substrate of FIG. 4B.
- (8) FIG. 4D is a schematic perspective view of an example substrate for use with the substrate cassette of FIG. 4A.
- (9) FIG. 5 illustrates an example of the substrate cassette of FIG. 4A.
- (10) FIG. 6 is a bottom cross sectional, perspective view of the substrate cassette, taken along line A-A of FIG. 5.
- (11) FIG. 7A shows a top view of an example substrate cassette in an open position.
- (12) FIG. 7B shows a side view of a latitudinal side of the substrate cassette of FIG. 7A.
- (13) FIG. 7C shows a cross sectional view of the latitudinal side of the substrate cassette of FIG. 7B.
- (14) FIG. 8 illustrates an example placement of the substrate into the substrate cassette of FIG. 7A.
- (15) FIG. 9 is a schematic top view of another example substrate cassette.
- (16) FIG. 10A is a schematic top view of yet another example substrate cassette.
- (17) FIG. 10B is a schematic top view of yet another example substrate for use with the substrate cassette of FIG. 10A.
- (18) FIG. 10C is a schematic top view of the substrate cassette of FIG. 10A that receives the substrate of FIG. 10B.
- (19) FIG. 10D is a schematic cross sectional side view of the substrate cassette of FIG. 10A that receives the substrate of FIG. 10B.
- (20) FIG. 11A is a schematic top view of yet another example substrate cassette.
- (21) FIG. 11B is a schematic top view of yet another example substrate for use with the substrate cassette of FIG. 11A.
- (22) FIG. 11C is a schematic top view of the substrate cassette of FIG. 11A that receives the substrate of FIG. 11B.
- (23) FIG. 11D is a schematic cross sectional side view of the substrate cassette of FIG. 11A that receives the substrate of FIG. 11B.
- (24) FIG. 12A is a schematic top view of an example cathode assembly.
- (25) FIG. 12B is a schematic top view of an example substrate cassette that receives an example substrate.
- (26) FIG. 12C is a schematic top view of the cathode assembly of FIG. 12A that is disposed on the substrate cassette and the substrate of FIG. 12B.
- (27) FIG. 12D is a schematic cross sectional side view of the cathode assembly, the substrate cassette, and the substrate of FIG. 12C.
- (28) FIG. 12E is a schematic cross sectional side view of the cathode assembly, the substrate cassette, and the substrate of FIG. 12C, in which the cathode assembly has another example electrode.
- (29) FIG. 12F is a schematic cross sectional side view of the cathode assembly, the substrate cassette, and the substrate of FIG. 12C, in which the cathode assembly has yet another example electrode.
- (30) FIG. 12G is a schematic cross sectional side view of the cathode assembly, the substrate cassette, and the substrate of FIG. 12C, in which the cathode assembly has yet another example electrode.
- (31) FIG. 13A is a schematic top view of another example cathode assembly.
- (32) FIG. 13B is a schematic top view of an example substrate.
- (33) FIG. 13C is a schematic top view of an example substrate cassette.
- (34) FIG. 13D is a schematic top view of the cathode assembly of FIG. 13A that is disposed on the substrate cassette of FIG. 13B and the substrate of FIG. 13C.
- (35) FIG. 13E is a schematic cross sectional side view of the cathode assembly, the substrate cassette, and the substrate of FIG. 13D, which illustrates an example buffer chamber.
- (36) FIG. 13F is a schematic cross sectional side view of the cathode assembly, the substrate cassette, and the substrate of FIG. 13D, which illustrates an example anode chamber.
- (37) FIG. 14A is a schematic top view of yet another example substrate cassette that receives a substrate.
- (38) FIG. 14B is a schematic cross sectional side view of the substrate cassette and the substrate of FIG. 14A.
- (39) FIG. 14C is a schematic cross sectional side view of a variation of the substrate cassette and the substrate of FIG. 14A.
- (40) FIG. 15A is a schematic top view of yet another example substrate cassette that receives a substrate.
- (41) FIG. 15B is a schematic cross sectional side view of the substrate cassette and the substrate of FIG. 15A.
- (42) FIG. 16A is a schematic top view of an example electrical connection configuration among a substrate cassette, a substrate, and a control system.
- (43) FIG. 16B is a schematic top view of another example electrical connection configuration among a substrate cassette, a substrate, and a control system.
- (44) FIG. 16C is a schematic cross sectional side view of part of the electrical connection configuration of FIG. 16B.
- (45) FIG. 17A is a schematic perspective view of an example instrument for electrophoresis.
- (46) FIG. 17B is a schematic cross sectional side view of an example cassette cover of the instrument that receives an example set of substrate cassette and substrate.
- (47) FIG. 17C is a schematic cross sectional side view of another example cassette cover of the instrument that receives an example set of substrate cassette and substrate.
- (48) FIG. 18 is a flowchart of an example process for analyte migration.
- (49) FIG. 19 is a block diagram of computing devices that may be used to implement the systems and methods described in this document, as either a client or as a server or plurality of servers.
- (50) FIG. 20A shows a schematic of an example analytical workflow in which electrophoretic migration of analytes is performed after permeabilization.
- (51) FIG. 20B shows a schematic of an example analytical workflow in which electrophoretic migration of analytes and permeabilization are performed simultaneously.
- (52) FIG. 21A shows an example perpendicular, single slide configuration for use during electrophoresis.
- (53) FIG. 21B shows an example parallel, single slide configuration for use during electrophoresis.
- (54) FIG. 21C shows an example multi-slide configuration for use during electrophoresis.
- (55) FIG. 22A-B are schematics illustrating expanded (FIG. 22A) and side views (FIG. 22B) of an electrophoretic transfer system configured to direct transcript analytes toward a spatially-barcoded capture probe array.
- (56) FIG. 23 is a schematic illustrating an exemplary workflow protocol utilizing an electrophoretic transfer system.

DETAILED DESCRIPTION

(57) As described herein, electrophoresis can be used to migrate analytes from a sample toward a substrate that includes capture probes for spatial transcriptomics applications in order to enhance sensitivity and/or resolution. The present disclosure provides various implementations of electrophoretic systems and instruments to improve electrophoresis and ease of use. Electrophoretic systems and instruments described herein provide different designs for electrode geometry and/or placement, and further provide different approaches for electrical interfacing between components. For example, various configurations of electrophoretic systems and instruments described herein utilize different electrode designs that can result in uniform electromigration of analytes. Further, the electrophoretic systems described herein consider that, where a sample tissue is very thin, relatively small angles of the electric field has no or little effect on drift of analyte migration. Moreover, various implementations of electrophoretic systems and instrumentations optimize the size and/or location (x, y, z directions) of electrodes (e.g., cathode) relative to a sample, thereby increasing uniformity of electric field and reducing or minimizing drift of analyte migration. In addition, the electrophoresis systems and instruments described herein provide a solution to conveniently check for electrical connections in the systems and instruments, and alert users of potential improper electrical connections, prior to performing electrophoresis.

(58) FIGS. 1A-B are schematic side and top views of an example electrophoretic system **100** for analyte migration. In some implementations, the system **100** is configured similarly to electrophoretic systems described herein, for example with reference to FIGS. 20A-B and 21A-C. The system **100** can be used to actively cause analytes described herein (e.g., nucleic acids, proteins, charged molecules, etc.) in a biological sample to migrate to capture probes on a substrate. In some implementations, the system **100** can include a substrate **102** as a first electrode (e.g., anode), a second electrode **104** (e.g., cathode), and a spacer **110** disposed between the substrate **102** and the second electrode **104**.

(59) Referring also to FIG. 2, the substrate **102** is configured to place a biological sample **112** that contains one or more analytes **314** (not shown in FIG. 2; see FIG. 3). By way of example, the biological sample **112** can be one or more cells or a tissue sample including one or more cells. The substrate **102** can include one or more regions **116** for placing or containing a sample **112** thereon. In some implementations, the substrate **102** can further comprise one or more capture probes **118** on a substrate region **116**. The capture probes **118** can be placed on the substrate region **116** in a variety of ways described herein. For example, the capture probes **118** can be attached (e.g., directly or indirectly, reversibly or non-reversibly) to a region on an array. Alternatively or in addition, the capture probes **118** can be immobilized on the substrate region **116** of the substrate **102**. The sample **112** can be placed and manipulated on the substrate **102** in various ways described herein. In some implementations, the biological sample can be placed in contact with the capture probes on each region. Alternatively, the biological samples do not make contact with the capture probes. For example, an intermediate layer (e.g., gels) can be disposed between the substrate and the sample, which is compatible with electrophoresis.

(60) In some implementations, the substrate **102** is configured to be used as a first electrode in the system **100**. For example, the substrate **102** can be used as an anode. In another example, the substrate **102** can be used as a cathode. The substrate **102** can be configured to be conductive at least in the substrate region **116**. In some implementations, the substrate **102** can be configured as a conductive substrate described herein. For example, the substrate **102** can include one or more conductive materials that permit the substrate **102** to function as an electrode (e.g., the anode). Examples of such a conductive material include tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), fluorine doped tin oxide (FTO), or any combination thereof. Alternatively or in addition, other materials may be used to provide desired conductivity to the substrate **102**. In some implementations, the substrate **102** can be coated with the conductive material. For example, the substrate **102** can include a conductive coating **109** (FIG. 1A) on the surface of the substrate **102**, and a sample **112** is provided on the coating **109** of the substrate **102**. In other implementations, the substrate **102** can be made partially or entirely of a conductive material. For example, a sample (e.g., tissue section) can be directly placed on a conductive material (e.g., graphite, silicon, other semiconductive material, etc.). The substrate **102** can be made of glass or other transparent material for imaging purposes.

(61) Although the substrate **102** is illustrated to include a single substrate region **116** in FIG. 2, other implementations of the substrate **102** can include a plurality of substrate regions (e.g., two substrate regions in FIGS. 1A-B, eight substrate regions in FIG. 4, etc.) configured to place multiple samples thereon, respectively. Each of such multiple substrate regions can be configured similarly to the first substrate region **116** described herein.

(62) Referring to FIGS. 1A-B, where the substrate **102** is used as the anode, the second electrode **104** is configured as a cathode. In this example, therefore, the second electrode **104** can also be referred to as the cathode **104**. As described herein, the cathode **104** can be provided in various configurations.

(63) In some implementations, the substrate **102** and the cathode **104** can be arranged within a container (e.g., substrate cassettes described herein) that defines a buffer chamber **122** between the substrate **102** and the cathode **104**. The buffer chamber **122** is configured to contain a buffer **124**. In some implementations, the substrate **102** and the cathode **104** can be fully immersed into the buffer **124**. In alternative implementations, either or both of the substrate **102** and the cathode **104** can be partially inserted into the buffer **124** located in the container.

(64) The buffer **124** can be of various types. In some implementations, the buffer **124** includes a permeabilization reagent. In some embodiments, the buffer **124** is contained in the buffer chamber **122** throughout the electrophoretic process. The permeabilization reagent can permeabilize the sample before and/or during electrophoresis. In addition or alternatively, the sample can be permeabilized using other methods described herein, independently or in conjunction with permeabilization using the permeabilization reagent.

(65) The spacer **110** can be disposed between the substrate **102** and the cathode **104** to separate the substrate **102** and the cathode **104** by a distance D. The spacer **110** can include a non-conductive material, such as plastic, glass, porcelain, rubber, silicone, etc. The distance D can be determined to provide a desired level of spatial resolution based on several factors, such as the strength and/or duration of an electric field generated between the substrate **102** and the cathode **104**, and other parameters described herein. The spacer **110** can define at least part of the buffer chamber **122** between the substrate **102** and the cathode **104**.

(66) The control system **130** can generate an electric field ($-E$) between the substrate **102** and the cathode **104**. The control system **130** can include a controller **132** configured to apply a voltage between the substrate **102** and the cathode **104** using a power supply **134**. The power supply **134** can include a high voltage power supply. The controller **132** can be electrically connected to the substrate **102** and the cathode **104**, for example using electrical wires. The control system **130** can include a user interface **136** configured to receive a user input of starting or stopping the electrophoretic process. The user interface **136** can include various types of input devices, such as a graphic user interface, physical or virtual buttons, switches, keypads, keyboard, etc., configured to receive a user input for adjusting operating parameters of the system **100** or for accessing other information (e.g., instructions) associated with the system **100**. Examples of operating parameters can include, but are not limited to, voltage applied, duration of voltage application, etc. In some implementations, the input devices can be used to select a subset of substrate regions **116** on the sample **112** so that the subset of substrate regions **116** are electrically activated to generate an electric field between the subset of substrate regions **116** and the cathode **104**. In addition, the user interface **136** can include an output device, such as a display, lamps, etc., configured to output the operating parameters of the system **100** or other information associated with the system **100**.

(67) FIG. 3 illustrates an example configuration of a substrate and a sample undergoing electrophoresis using the electrophoretic system **100**. The configuration and process illustrated in FIG. 3 may be similar to the electrophoretic processes described herein, for example with reference to FIGS. 21A-C. Referring to FIG. 3, the application of an electric field ($-E$) causes the analytes **314** (e.g., negatively charged analytes) to move towards the capture probes **318** (e.g., positively charged analytes) in the direction of the arrows shown. In some implementations, the analytes **314** include a protein or a nucleic acid. In some embodiments, the analytes **314** are negatively charged proteins or nucleic acids. In some embodiments, the analytes **314** include a positively charged protein or a nucleic acid. In some embodiments, the analytes **314** includes a negatively charged transcript. For example, the analytes **314** include a polyA transcript. In some implementations, a detergent or other reagent can be added to change the charge of the analytes. For example, SDS as used in SDS-PAGE can coat the proteins and substantially provide a uniform negative charge on the proteins. Other implementations can be used to change the charge of molecule with a covalent attachment of another charged molecule. Yet another option is to

change the pH of the solution.

(68) In some embodiments, the capture probes **318** are configured as a coating that is in contact with the substrate **302** (e.g., a coating provided on the surface of the substrate **302**). In some embodiments, the capture probes **318** can include a feature array, or can be replaced by a feature array. In some implementations, the feature array can be the same as the substrate region described herein. In alternative embodiments, the feature array can be different from the substrate region described herein. In some embodiments, the analytes **314** move towards the capture probes **318** for a distance (h). In some embodiments, the buffer **324** (e.g., permeabilization reagent) can be in contact with the sample **312**, the substrate **302**, the cathodes **304**, or any combination thereof. The buffer **324** can include any of the permeabilization reagents disclosed including but not limited to a dried permeabilization reagent, a permeabilization buffer, a buffer without a permeabilization reagent, a permeabilization gel, and a permeabilization solution.

(69) Referring to FIGS. **4-16**, various configurations of an electrophoretic system **100** are described. In general, as described herein, the system **100** can include a substrate, a substrate cassette, a cathode assembly (e.g., FIGS. **12-13**), and a control system. The substrate can be configured similarly to the substrate **102** or other substrate described herein. The substrate cassette can be configured to be disposed on the substrate and define buffer chambers (e.g., the buffer chambers **122**) on the substrate. The cathode assembly is configured to implement the cathode **104** described herein. For example, the cathode assembly can include a plurality of electrodes configured to position within the plurality of buffer chambers of the substrate cassette, respectively. The control system can be configured similarly to the control system **130**. For example, the control system includes a power supply electrically connected with the substrate and the cathode assembly and can supply power to generate electric field between the substrate (e.g., respective substrate regions thereof) and the cathode assembly (e.g., the plurality of electrodes thereof) such that the analytes in biological samples migrate toward capture probes on the substrate.

(70) Referring to FIGS. **4A-D**, an example of a substrate cassette **404** is described. The substrate cassette **404** includes a body **410** and a plurality of apertures **412**. The body **410** can be configured to mount onto the substrate **402**, or receive the substrate **402**. For example, the body **410** is configured to define a cavity that can receive at least a portion of the substrate **402**. The plurality of apertures **412** are configured and arranged in the body **410** such that, when the substrate cassette **404** is disposed on, or receives, the substrate **402**, the plurality of apertures **412** are aligned with a plurality of substrate regions **416** (FIG. **4B**) of the substrate **402** and define a plurality of buffer chambers on the plurality of substrate regions **416** of the substrate **402**.

(71) Referring also to FIG. **4D**, the substrate **402** can be configured similarly to the substrate **102**. The substrate **402** is configured to place a plurality of biological samples **413** that contains one or more analytes. By way of example, each biological sample **413** can be one or more cells or a tissue sample including one or more cells. The substrate **402** can include a plurality of substrate regions **416** for placing or containing the samples **413** thereon. In some implementations, the substrate **402** can further comprise one or more capture probes **418** on each substrate region **416**. The capture probes **418** can be placed on the substrate region **416** in a variety of ways described herein. For example, the capture probes **418** can be directly attached (e.g., reversibly or non-reversibly) to a feature on an array. In another example, the capture probes **418** can be indirectly attached (e.g., reversibly or non-reversibly) to a feature on an array. Alternatively or in addition, the capture probes **418** can be immobilized on the substrate region **416** of the substrate **402**. The sample **413** can be prepared on the substrate **402** in various ways described herein.

(72) In some implementations, the substrate **402** is configured to be used as a first electrode in the electrophoretic system **100**. For example, the substrate **402** can be used as an anode. In another example, the substrate **402** can be used as a cathode. The substrate **402** can be configured as a conductive substrate described herein. For example, the substrate **402** can include one or more conductive materials that permit the substrate **402** to function as an electrode (e.g., the anode). Examples of such a conductive material include tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), fluorine doped tin oxide (FTO), and any combination thereof. Alternatively or in addition, other materials may be used to provide desired conductivity to the substrate **402**. In some implementations, the substrate **402** can be coated with the conductive material. For example, the substrate **402** can include a conductive coating on the surface of the substrate or each substrate region of the substrate, and the sample **413** is provided on the coating of the substrate **402** or each substrate region thereof.

(73) Referring back to FIG. **4B**, the substrate **402** includes a first electrode contact **420** for electrical connection to the control system **408** (e.g., a power supply thereof). For example, the first electrode contact **420** can be electrically connected with a first wire **422** (FIG. **4C**) extending from the control system **408**. The first electrode contact **420** is electrically connected to the plurality of substrate regions **416**.

(74) The substrate cassette **404** further includes a connection interface **414** configured to expose the first electrode contact **420** of the substrate **402** for electrical connection to the control system **408**. For example, the substrate cassette **404** can include a slit **417** as the connection interface **414**. The slit **417** can be defined at the body **410**. As illustrated in FIG. **4C**, where the substrate **402** is received in the substrate cassette **404**, the slit **417** of the substrate cassette **404** can permit the substrate **402** to partially extend out from the body **410** such that the first electrode contact **420** of the substrate **402** is positioned outside the body **410**. The control system **408** can be connected to the substrate **402** via the first electrode contact **420** exposed from the substrate cassette **404**. For example, the first wire **422** extending from the control system **408** can be electrically connected to the first electrode contact **420** outside the substrate cassette **404**. The first wire **422** can include a connecting pin or clip **424** that can be removably attached to the first electrode contact **420** of the substrate **402**. Other types of fastening mechanisms can be used to connect the first wire **422** to the first electrode contact **420** of the substrate **402**.

(75) Referring to FIGS. **5-8**, various examples of a substrate cassette **204** are illustrated. A substrate cassette **204** can also be referred to as a substrate holder.

(76) FIG. **5** illustrates an example of the substrate cassette **504**. This embodiment of the substrate holder **504** can advantageously provide a single-piece component that can be arranged in an open configuration or closed configuration, when desired. In particular, FIG. **5** shows a top surface **593** of the substrate holder **504** in a closed position. The substrate holder **504** includes the plurality of apertures **512**. For example, the substrate holder **504** includes the body **510** having a main side **503** (e.g., a main surface or face), opposite longitudinal sides **505**, and opposite latitudinal sides **507**. The longitudinal sides **505** and the latitudinal sides **507** can constitute a lateral side (e.g., a lateral surface or face) extending from a periphery of the main side **503**. The plurality of apertures **512** can be provided at the main side **503** of the body **510**.

(77) The substrate holder **504** can include a substrate loading mechanism for loading and holding the substrate. For example, the substrate loading mechanism can include a first tab **550a** and a second tab **550b**. The first tab **550a** and the second tab **550b** can protrude from one of the longitudinal sides **505** of the substrate holder **504**. In some embodiments, any type of fastener or engagement feature that allows releasable engagement can be used instead of the first and second tabs **550a** and **550b**, such as, for example, screws and press fit type connectors. In some embodiments, the substrate holder **504** includes 5 tabs or less (e.g., 4 tabs or less, 3 tabs or less, 2 tabs or less, or 1 tab). In some embodiments, the substrate holder **504** is a single molded unit. Any suitable plastic or polymer can be used as a suitable molding material.

(78) FIG. **6** is a bottom cross sectional, perspective view of the substrate holder **504**, taken along line A-A of FIG. **5**. The substrate holder **504** includes a gasket **654** and defines a cavity **652** configured to receive the substrate **502**. In some embodiments, the substrate holder **504** is a single molded unit that includes the gasket **654**. That is, in some embodiments, the substrate holder **504** and the gasket **654** are one part. In some embodiments, the substrate holder **504** is overmolded with the gasket **654**. For example, the substrate holder **504** is a first injection molded plastic part with a second part (e.g., a pliable material) molded onto it to create the gasket **654**. In some embodiments, the pliable material is an elastomer. In some embodiments, the pliable material is silicone rubber. In some embodiments, the gasket **654** is a separate part that is not molded with the substrate holder **504**.

(79) Referring to FIGS. **5-6**, the substrate holder **504** includes the slit **516** defined at the lateral face of the body **510**. As described herein, the slit **516** is used as an example of the connection interface **514** for exposing the first electrode contact **420** of the substrate **502**. In some implementations, the

slit **516** is defined at one of the latitudinal sides **507**, so that the first electrode contact **620** of the substrate **502** is positioned outside the body **510**. In other implementations, the slit **516** can be defined at a different side of the body **510**, such as any suitable one of the main side **503**, the longitudinal sides **505** and the latitudinal sides **507**.

(80) FIG. 7A shows a top view of a substrate holder **704** in an open position. The opening and closing mechanism of the substrate holder **704** is a hinged mechanism. A bottom component **762** of the substrate holder **704** can be hinged to a top component **764** of the substrate holder **704** via a hinge **760**. In some embodiments, the hinge **760** can be a living hinge. In some embodiments, the substrate holder **704** includes 10 hinges or less (e.g., 9 hinges or less, 8 hinges or less, 7 hinges or less, 6 hinges or less, 5 hinges or less, 3 hinges or less, 3 hinges or less, 2 hinges or less, or 1 hinge). Non-limiting examples of hinges that the substrate holder **704** can include, include a straight or flat living hinge, a butterfly living hinge, a child safe hinge, a double living hinge, and a triple living hinge.

(81) The substrate holder **704** further includes one or more engagement features, such as a first notch **758a** and a second notch **758b**. The first and second notches **758a** and **758b** can engage the first and second tabs **750a** and **750b**, respectively, when pressed together. The first and second notches **758a** and **758b** can protrude from a longitudinal side **705** of the top component **764** of the substrate holder **704**. In some embodiments, the substrate holder **704** includes three, four, five, six, seven, eight, nine, ten or more notches. In some embodiments, the notches protrude from a latitudinal side **707** of the substrate holder **704**. In some embodiments, the first and second notches **758a** and **758b** are rigid and do not flex when engaging the first and second tabs **750a** and **750b**, respectively. In some embodiments, the first and second notches **758a** and **758b**, respectively, can be flexible.

(82) FIG. 7B shows a side view of a latitudinal side **707**. The first and second notches **758a** and **758b** can project upward and include a notch ledge **766** that engages a tab ledge **768**, as shown in FIG. 7C. Alternatively, in some embodiments, the substrate holder **704** includes a snap fit locking mechanism for releasably receiving and releasably securing the substrate. Non-limiting examples of other types of fasteners to be used in locking mechanisms of the substrate holder **704** include a catch, a projection, a male connector, and a female connector.

(83) FIGS. 7A and 8 illustrate the placement of the substrate **702** into the substrate holder **704**. In some embodiments, the substrate **702** can be “loaded” onto an inner rim or an inner edge of the bottom component **762** of the substrate holder **704**, while a portion of the substrate **702** including the first electrode contact **720** is positioned exterior of the substrate holder **704**. Once loaded, the top component **764** is closed by pressing the first notch **758a** and the second notch **758b** against the first and second tabs **750a** and **750b**, respectively, thereby forming a tight seal with the substrate **702**. In some embodiments, the substrate does not have to be tilted under tabs **750** or any other tabs. In some embodiments, the substrate holder **704** includes one or more tabs to help load the slide onto the inner rim or inner edge of the bottom component **762** of substrate holder **704**. When the substrate **702** is loaded in the substrate holder **703**, the first electrode contact **720** of the substrate **702** is exposed from the substrate holder **704** through the slit **716**.

(84) In other implementations, the hinge **760** can be provided at one of the latitudinal sides **707**. An example of this hinge arrangement is described in PCT/US20/79843, titled IMAGING SUPPORT DEVICES, filed Apr. 24, 2020, the disclosure of which is incorporated herein by reference in its entirety.

(85) Referring to FIG. 9, another example of the substrate cassette **904** can be configured similarly to the substrate cassettes **204**, **504**, and **704** described in FIGS. 4-8, except for the connection interface **214**. Unlike the substrate cassette **204**, **504**, and **704** of FIGS. 3 and 5-8, in this example, the substrate cassette **904** can be configured to fully receive the substrate **902**. Further, instead of a slit, the substrate cassette **904** includes a contact opening **900** as the connection interface **914**. The contact opening **900** is defined at the body **910** and configured to expose the first electrode contact **920** of the substrate **902**.

(86) The contact opening **900** can be defined at the main side **903** of the body **910**. In some implementations, the contact opening **900** is defined at the main side **903** adjacent the lateral side of the body **910**. As illustrated in FIG. 9, for example, the contact opening **900** is defined at the main side **903** adjacent one of the latitudinal sides **907**. The contact opening **900** can be defined at both the main side **903** and the latitudinal side **907**, so as to extend from the main side **903** to the latitudinal side **907** of the body **910**.

(87) Referring to FIGS. 10A-D, yet another example of a substrate cassette **1004** is described, which includes the body **1010** and the plurality of apertures **1012** as described herein. The substrate cassette **1004** in this example is similarly configured to the substrate cassette described in FIGS. 4-8, except for the connection interface **1014**. Unlike the substrate cassette **204**, **504**, and **704** of FIGS. 3 and 5-8, in this example, the substrate cassette **1004** can be configured to fully receive the substrate **1002**.

(88) In this example, as illustrated in FIG. 10B, the substrate **1002** includes a substrate contact pin **1015** that provides the first electrode contact **1020**. The substrate contact pin **1015** can extend from a surface of the substrate **1002** that includes a plurality of substrate regions **1026**. The substrate contact pin **1015** is electrically connected to the substrate **1002**. In embodiments where the substrate **1002** includes the conductive coating **1009**, the substrate contact pin **1015** can be electrically connected to the conductive coating **1009**, as illustrated in FIG. 10D.

(89) The substrate cassette **1004** further includes a contact hole **1010** as the connection interface **1014**. The contact hole **1000** can be defined at the body **1010** and configured to expose the substrate contact pin **1015** of the substrate **1002**. The contact hole **1000** can be defined at the main side **1003** of the body **1010**. The contact hole **1000** can be positioned to align with the substrate contact pin **1015** when the substrate **1002** is received in or held by the substrate cassette **1004**.

(90) Referring to FIGS. 11A-D, a fourth example of the substrate cassette **1104** is described, which includes the body **1110** and the plurality of apertures **1112** as described herein. The substrate cassette **1104** in this example is similarly configured to the substrate cassette described in FIGS. 4-8, except for the connection interface **1114**. Unlike the substrate cassette **1104**, **504**, and **704** of FIGS. 3 and 5-8, in this example, the substrate cassette **1104** can be configured to fully receive the substrate **1102**.

(91) The substrate cassette **1104** includes a cassette contact pin **1130** as the connection interface **1114**. The cassette contact pin **1130** can extend through the body **1110** (e.g., the main side **1103** of the body **1110**) such that a first end **1132** of the cassette contact pin **1130** is placed exterior of the body **1110**, and an opposite, second end **1134** of the cassette contact pin **1130** is placed interior of the body **1110**. The cassette contact pin **1130** is configured to electrically connect to the substrate **1102** received in the substrate cassette **1104**. In embodiments where the substrate **1102** includes the conductive coating **1109**, the cassette contact pin **1130** can be electrically connected to the conductive coating **1109**. For example, the second end **1134** of the cassette contact pin **1130** is configured to electrically connect to the conductive coating **1109**, as illustrated in FIG. 11D.

(92) In some implementations, the system **100** can include a substrate contact bracket **1136** configured to engage with the substrate **1102** and electrically connected to the substrate **1102**. For example, the substrate contact bracket **1136** can be engaged with a portion of the substrate **1102** and electrically contact the conductive coating **1109** of the substrate **1102**, as shown in FIG. 11D. The cassette contact pin **1130** can be configured to electrically connect to the substrate contact bracket **1136** when the substrate **1102** is received in the substrate cassette **1104**. For example, the second end **1134** of the cassette contact pin **1130** is configured to contact with the substrate contact bracket **1136** based on the substrate **1102** held in the substrate cassette **1104**. Thus, the cassette contact pin **1130** can be electrically connected to the substrate **1102** and further provide the connection interface **1114** at the exterior of the substrate cassette **1104**. The first wire **1122** extending from the control system **1108** can simply be connected to the cassette contact pin **1130** (e.g., the first end **1132** thereof) so that the control system **1108** can be electrically connected to the substrate **1102** that can function as one (e.g., anode) of the electrodes in electrophoresis.

(93) Referring to FIGS. 12A-D, an example of the cathode assembly **1206** is described. The cathode assembly **1206** can include a cathode body **1250** configured to be positioned at least partially on the substrate cassette **1204**. The cathode assembly **1206** can include a plurality of electrodes **1252** that project from the cathode body **1250**. The plurality of electrodes **1252** are configured to position within the plurality of apertures **1212** of the substrate cassette **1204**, respectively, when the cathode body **1250** is positioned on the substrate cassette **1204**. Thus, the electrodes **1252** is disposed

within the respective buffer chambers that are defined by the apertures **1212** on the substrate regions **1216** of the substrate **1202**. Each of the electrodes **1252** functions as a cathode in each of the buffer chambers.

(94) The cathode assembly **1206** can include a cathode contact **1254** configured for electrical connection to the control system **1208**. For example, a second wire **1226** extending from the control system **1208** is electrically connected to the cathode contact **1254**. The second wire **1226** can include a connecting pin or clip **1228** that can be removably attached to the cathode contact **1254** of the cathode assembly **1206**. Other types of fastening mechanisms can be used to connect the second wire **1226** to the cathode contact **1254** of the cathode assembly **1206**. The cathode contact **1254** is electrically connected to the plurality of electrodes **1252**. In some implementations, as shown in FIG. **12D**, the cathode assembly **1206** includes a conductive layer **1258** that is electrically connected to both the cathode contact **1254** and the plurality of electrodes **1252**. The cathode contact **1254** can be exposed at the cathode body **1250** so as to be easily accessible exterior of the cathode body **1250**.

(95) As shown in FIG. **12C**, the cathode assembly **1206** is disposed on the substrate **1204** that receives or holds the substrate **1202**. The cathode assembly **1206** is oriented and arranged such that the plurality of electrodes **1252** are inserted into the respective apertures **1212** of the substrate cassette **1204** and thus disposed on the respective substrate regions **1216** of the substrate **1202**. The cathode assembly **1206** can be used with substrates and substrate cassettes of various configurations. In the illustrated example of FIG. **12C**, the cathode assembly **1206** is used with the substrate **1202** and the substrate cassette **1204** described in FIGS. **4-8**. In other examples, the cathode assembly **1206** can be used with the substrates **1202** and the substrate cassettes described in FIGS. **9-11**.

(96) The electrode **1252** can be of various shapes. For example, the electrode **1252** can be configured as a conductive wire **1255** (FIG. **12D**), a conductive rod **1256** (FIG. **12E**), or an array of conductive wires **1257** (FIG. **12F**). Alternatively, as shown in FIG. **12G**, the electrode **1252** can include a conductive extension **1261** and a conductive plate **1263** at a distal end (i.e., a free end) of the conductive extension **1261**. The conductive extension **1261** can be of various shapes, such as the conductive wire **1255**, the conductive rod **1256**, the conductive wire array **1257**, or other suitable shapes. The conductive plate **1263** can be of various shapes, such as a circular plate **1263A**, a ring **1263B**, a square plate **1263C**, or other suitable shapes.

(97) Referring to FIGS. **13A-E**, other examples of the substrate **1302**, the substrate cassette **1304**, and the cathode assembly **1306** are described. Referring to FIG. **13B**, the substrate **1302** in this example is configured similarly to the substrate **1302** described herein. In this example, the substrate **1302** further includes an electrode region **1303** that functions as the first electrode contact **320**. The electrode region **1302** is configured to electrophoretically connect to the control system **1308** so that the substrate **1302** can be used as an electrode (e.g., anode) in the system **100**. In embodiments where the substrate **1302** includes the conductive coating **1309**, the electrode region **1303** is electrically connected to the conductive coating **1309**. For example, the electrode region **1303** can be a hole that exposes a portion of the conductive coating **1309** of the substrate **1302** to a second buffer **1307**, as illustrated in FIG. **13F**.

(98) Referring to FIG. **13C**, the substrate cassette **1304** in this example is configured similarly to the substrate cassette **1304** described herein. The substrate cassette **1304** can include an electrode aperture **1301**, as well as the plurality of apertures **1312** described herein. The electrode aperture **1301** is configured to be aligned with the electrode region **1303** of the substrate **1302** when the substrate cassette **1304** receives the substrate **1302**. As illustrated in FIG. **13C**, the substrate cassette **1304** is configured to fully receive the substrate **1302**. Further, as illustrated in FIGS. **13E-F**, the plurality of apertures **1312** and the electrode aperture **1301** of the substrate cassette **1304** are arranged on the plurality of substrate regions **1316** and the electrode region **1303** of the substrate **1302**, respectively. As described herein, the plurality of apertures **1312** (either independently or in cooperation with the gasket **1354** as shown in FIG. **13E**) can define a plurality of first buffer chambers (e.g., the buffer chambers **122**) on the plurality of substrate regions **1316**, which are configured to receive first buffers **1305** (e.g., the buffer **124**) for analyte migration under electrophoresis. In addition, the electrode aperture **1301** (either independently or in cooperation with the gasket **1354** as shown in FIG. **13F**) can define a second buffer chamber on the electrode region **1303**, which is configured to receive a second buffer **1307**. In some implementations, the second buffer **1307** is different from the first buffer **1305**. The second buffer **1307** can have an electrolyte strength that is greater than the first buffer **1305**. For example, the second buffer **1307** can be selected to have ionic strength that ranges from 1 to 1300 mmol/L. One example of the second buffer **1307** is a sodium chloride solution. Other examples of the second buffer **1307** include sodium hydroxide, potassium chloride, or any buffer with a high concentration of ions, which can include the first buffer **1305** at a higher concentration. Further examples of the first and second buffers can be found in Reijenga, et al., Buffer Capacity, Ionic Strength and Heat Dissipation in Capillary Electrophoresis, Journal of Chromatography A, 744 (1996) 1347-153, the disclosure of which is incorporated herein by reference in its entirety.

(99) Referring to FIG. **13A**, the cathode assembly **1306** includes the cathode body **1350** configured to be positioned at least partially on the substrate cassette **1304**. The cathode assembly **1306** can include a plurality of electrodes **1352** that project from the cathode body **1350**. The plurality of electrodes **1352** are configured to position within the plurality of apertures **1312** of the substrate cassette **1304**, respectively, when the cathode body **1350** is positioned on the substrate cassette **1304**. Thus, the electrodes **1352** is disposed within the respective buffer chambers that are defined by the apertures **1312** of the substrate cassette **1304** on the substrate regions **316** of the substrate **1302**. Each of the electrodes **1352** functions as a cathode in each of the buffer chambers.

(100) The cathode assembly **1306** can include a second electrode contact **1354** configured for electrical connection to the control system **1308**. The second electrode contact **1354** can be used for a cathode contact. For example, a second wire **1326** extending from the control system **1308** is electrically connected to the cathode contact **1354**. The second wire **326** can include a connecting pin or clip **1328** that can be removably attached to the cathode contact **1354** of the cathode assembly **1306**. Other types of fastening mechanisms can be used to connect the second wire **1326** to the cathode contact **1354** of the cathode assembly **1306**. The cathode contact **1354** is electrically connected to the plurality of electrodes **1352**. In some implementations, as shown in FIG. **13E**, the cathode assembly **1306** includes a conductive layer **1358** that is electrically connected to both the cathode contact **1354** and the plurality of electrodes **1352**. The cathode contact **1354** can be exposed at the cathode body **1350** so as to be easily accessible exterior of the cathode body **1350**.

(101) In this example, the cathode assembly **1306** further includes a second electrode **1310** that projects from the cathode body **1350**. As illustrated in FIG. **13F**, the second electrode **1310** is configured to position within the electrode aperture **1301** of the substrate cassette **1304** when the cathode body **1350** is positioned on the substrate cassette **1304**. Thus, the second electrode **1310** is disposed within the second buffer chamber that is defined by the electrode aperture **1301** of the substrate cassette **1304** (independently or in cooperation with the gasket **354**) on the electrode region **1303** of the substrate **1302**. The second electrode **1310** can function as part of the anode in electrophoresis.

(102) The cathode assembly **1306** can include an anode contact **1313** configured for electrical connection to the control system **1308**. For example, the first wire **322** extending from the control system **1308** is electrically connected to the anode contact **1313**. The first wire **1322** can include a connecting pin or clip **1328** that can be removably attached to the anode contact **1313** of the cathode assembly **1306**. Other types of fastening mechanisms can be used to connect the first wire **1322** to the anode contact **1313** of the cathode assembly **1306**. The anode contact **1313** is electrically connected to the second electrode **1310**. In some implementations, as shown in FIG. **13F**, the cathode assembly **1306** includes a conductive layer **1314** that is electrically connected to both the anode contact **1313** and the second electrode **1310**. As illustrated in FIG. **13D**, the anode contact **1313** can be exposed at the cathode body **1350** so as to be easily accessible exterior of the cathode body **1350**. As illustrated in FIG. **13F**, with the control system **1308** being electrically connected to the anode contact **1313** of the cathode assembly **1306**, the control system **1308** is electrophoretically connected to the electrode region **1303** (i.e., the first electrode contact **320**) of the substrate **1302** via the second electrode **1310** that is positioned within the second chamber containing the second buffer **1307** on the electrode region **1303** of the substrate **1302**.

(103) Referring to FIGS. **14A-C**, yet another example of a substrate cassette **1404** is described. As described herein, the substrate cassette **1404** includes the body **1410** and the plurality of apertures **1412**. The substrate cassette **1404** in this example is configured similarly to the substrate

cassette **1404** described in FIGS. **4-8**. In addition, the substrate cassette **1404** is configured to incorporate at least part of the cathode assembly **1406** described herein. For example, as illustrated in FIG. **14B**, the substrate cassette **1404** includes a plurality of electrodes **1403** that function as the electrodes **1452** described herein. The plurality of electrodes **1403** are configured to position within the buffer chambers **1405** (similar to the buffer chambers **122** described herein) defined at least by the apertures **1412** of the substrate cassette **1404**. The electrode **1403** can be of various configurations, similarly to the electrodes **1452**.

(104) Referring to FIG. **14B**, the substrate cassette **1404** can include a plurality of cathode contact pins **1406** that extend through a wall of the cassette body **1410** and electrically connect to the plurality of electrodes **1403**. The plurality of cathode contact pins **1406** are electrically connected to the second electrode contact or cathode contact **1454** that is exposed at the cassette body **1410**. Alternatively, one of the cathode contact pins **1456** can be exposed exterior of the cassette body **1410** and used as the cathode contact **1454**. As described herein, the cathode contact **1454** is used to connect to the control system **1408**, such as via the second wire **1426** extending from the control system **1408**.

(105) In this example, the substrate cassette **1404** is used with the substrate **202** described in FIGS. **4-8**. In other examples, the substrate cassette **1404** can be used with the substrates **902** described in FIGS. **9-13**.

(106) The electrode **1403** can be of various configurations, similarly to the electrodes **1452**. For example, the electrode **1403** can be configured as a conductive wire (similar to the conductive wire **1255** in FIG. **12D**), a conductive rod (similar to the conductive rod **1256** in FIG. **12E**), or an array of conductive wires (similar to the conductive wire array **1257** in FIG. **12F**). Alternatively, the electrode **1403** can be configured similarly to the electrode **1252** shown in FIG. **12G**. (e.g., extensions with various tips such as a circular plate, a ring, a square plate, or other suitable shapes). Referring to FIG. **14C**, each buffer chamber **1405** can be provided with two or more electrodes **1403**.

(107) Referring to FIGS. **15A-B**, yet another example of a substrate cassette **1504** is described. The substrate cassette **1504** in this example is configured similarly to the substrate cassette **1504** of FIGS. **14A-C**. In addition, the substrate cassette **1504** includes the connection interface **1514** shown in FIGS. **11A-D**. In particular, the substrate cassette **1504** in this example further includes the cassette contact pin **1530** as the connection interface **1514**. The cassette contact pin **1530** can extend through the body **1510** such that the first end **1532** of the cassette contact pin **1530** is placed exterior of the body **1510**, and the opposite, second end **1534** of the cassette contact pin **1530** is placed interior of the body **1510**. The cassette contact pin **1530** is configured to electrically connect to the substrate **1502** received in the substrate cassette **1504**. In embodiments where the substrate **1502** includes the conductive coating **1509**, the cassette contact pin **1530** can be electrically connected to the conductive coating **1509**. For example, the second end **1534** of the cassette contact pin **1530** is configured to electrically connect to the conductive coating **1509**, as illustrated in FIG. **15B**.

(108) In some implementations, the system **1500** can include the substrate contact bracket **1536** configured to engage with the substrate **1502** and electrically connected to the substrate **1502**. For example, the substrate contact bracket **1536** can be engaged with a portion of the substrate **1502** and electrically contact the conductive coating **1509** of the substrate **1502**, as shown in FIG. **15B**. The cassette contact pin **1530** can be configured to electrically connect to the substrate contact bracket **1536** when the substrate **1502** is received in the substrate cassette **1504**. For example, the second end **1534** of the cassette contact pin **1530** is configured to contact with the substrate contact bracket **1536** based on the substrate **1502** held in the substrate cassette **1504**. Thus, the cassette contact pin **1530** can be electrically connected to the substrate **1502** and further provide the connection interface **1514** at the exterior of the substrate cassette **1504**.

(109) Accordingly, the control system **1508** can be easily connected to the system **1500**. For example, the first wire **1522** extending from the control system **1508** can simply be connected to the cassette contact pin **1530** (e.g., the first end **1532** thereof) so that the control system **1508** can be electrically connected to the substrate **1502** that can function as one (e.g., anode) of the electrodes in electrophoresis. Further, the second wire **1526** extending from the control system **1508** can simply be connected to the cathode contact **1554** so that the control system **1508** can be electrically connected to the electrodes **602** (e.g., cathodes) in the electrophoresis system.

(110) Referring to FIGS. **16A-C**, example connection schemes are described. Referring to FIG. **16A**, the substrate cassette **1604** and the substrate **1602** are configured similarly to the substrate cassette **1404** and the substrate **1402** shown in FIGS. **14A-C**. In addition, the substrate cassette **1604** includes an anode contact pin **1630** extending from the cassette body **1610**. Further, a conductive wire **1632** is provided to electrically connect the anode contact pin **1630** to the first electrode contact **1620** of the substrate **1602**. For example, the conductive wire **1632** includes a first end **1634** and an opposite second end **1635**. The first end **1634** can be electrically connected to the anode contact pin **1630**, and the second end **1635** can be electrically connected to the first electrode contact **1620** that is positioned outside the body **1610**. Connecting pins, clips, or other suitable fastening mechanisms can be provided at the first and second ends **1634**, **1635** to connect the conductive wire **1632** to the anode contact pin **1630** and the first electrode contact **1620**, respectively.

(111) Referring to FIG. **16B**, the substrate cassette **1604** and the substrate **1602** are configured similarly to the substrate cassette **1404** and the substrate **1402** shown in FIGS. **14A-C**. In addition, the substrate cassette **1604** includes an anode cassette body **1650**. The anode cassette body **1650** can define a cavity configured to receive a portion of the substrate **1602**. For example, the anode cassette body **1650** can be configured to receive at least the first electrode contact **1620** of the substrate **1602** that is positioned exterior of the cassette body **1610** of the substrate cassette **1604**.

(112) The anode cassette body **1650** can include a connection interface **1614** that is similar to the connection interface **1114** described in FIGS. **11A-D**. In particular, as shown in FIG. **16C**, the anode cassette body **1650** includes a cassette contact pin **1630** as the connection interface **1614**. The cassette contact pin **1630** can extend through anode cassette body **1650** such that a first end **1631** of the cassette contact pin **1630** is placed exterior of anode cassette body **1650**, and an opposite, second end **1634** of the cassette contact pin **1630** is placed interior of anode cassette body **1650**. The cassette contact pin **1630** is configured to electrically connect to the substrate **1602** received in the anode cassette body **1650**. In embodiments where the substrate **1602** includes a conductive coating **1609**, the cassette contact pin **1630** can be electrically connected to the conductive coating **1609**. For example, the second end **1634** of the cassette contact pin **1630** is configured to electrically connect to the conductive coating **1609**, as illustrated in FIG. **16C**.

(113) In some implementations, the system **100** can include a substrate contact bracket **1636** configured to engage with the substrate **1602** and electrically connected to the substrate **1602**. For example, the substrate contact bracket **1636** can be engaged with a portion of the substrate **1602** and electrically contact the conductive coating **1609** of the substrate **1602**, as shown in FIG. **16C**. The cassette contact pin **1630** can be configured to electrically connect to the substrate contact bracket **1636** when the substrate **1602** is received in the anode cassette body **1650**. For example, the second end **1634** of the cassette contact pin **1630** is configured to contact with the substrate contact bracket **1636** based on the substrate **1602** held in the anode cassette body **1650**. Thus, the cassette contact pin **1630** can be electrically connected to the substrate **1602** and further provide the connection interface **1614** at the exterior of the anode cassette body **1650**. The first wire **1622** extending from the control system **1608** can simply be connected to the cassette contact pin **1630** (e.g., the first end **1631** thereof) so that the control system **1608** can be electrically connected to the substrate **1602** that can function as one (e.g., anode) of the electrodes in electrophoresis.

(114) Referring to FIGS. **17A-C**, an example instrument **1700** for electrophoresis is described. The instrument **1700** can be used with one or more components of the electrophoretic system **100** described herein. In some implementations, the instrument **1700** is configured to automate at least part of the electrophoretic process that uses the substrate **1702**, the substrate cassette **1704**, the cathode assembly **1707**, and the control system **1708**.

(115) Referring to FIG. **17A**, the instrument **1700** has an instrument housing **1703**, a cassette tray **1704**, and a cassette cover **1706**. The instrument housing **1703** is configured to receive at least one or all of the components of the electrophoretic system **100**. For example, the instrument housing **1703** is configured to include the control system **1708**. In some implementations, the instrument housing **1703** can include the controller **132** and the power supply **134**. The power supply **134** can be connected to an electrical outlet external to the instrument housing **1703**. Alternatively or in addition, the power supply **134** can include a battery that is received in the instrument housing **1703**. The instrument housing **1703** can further include the user interface **136** of the control system **108**. The user interface **136** can be placed in a location (e.g., adjacent the cassette tray **1705**)

suitable for a user to control the instrument **1700**.

(116) The cassette tray **1705** is configured to receive the substrate cassette **1704**. For example, the substrate cassette **1704**, which receives the substrate **1702**, can be placed on the cassette tray **1705**. In some implementations, the cassette tray **1705** extends from the instrument housing **1703** so that the substrate cassette **1704** can be conveniently placed on the cassette tray **1705**. In some implementations, the cassette tray **1705** is retractable into the instrument housing **1703** so that electrophoresis can be performed within the instrument housing **1703**. Once the electrophoresis is completed, the cassette tray **1705** is extended from the instrument housing **1703** so that the substrate cassette **1704** can be removed from the cassette tray **1705**. Alternatively, the cassette tray **1705** can remain exterior of the instrument housing **1703** before, during, and after electrophoresis is performed.

(117) The cassette tray **1705** can be configured to receive various implementations of the substrate cassette **1704** described herein. Referring to FIG. **17B**, in one example, the substrate cassette **1704** described in FIGS. **15A-B** is placed on the cassette tray **1705**. In this example, the cassette cover **1706** is configured to be located above the cassette tray **1705** and provide electrical connections between the substrate cassette **1704** and the control system **1708**. Referring to FIG. **17B**, in embodiments where the substrate cassette **1704** described in FIGS. **15A-B** is placed on the cassette tray **1705**, the cassette cover **1706** can include a cathode connector **1710** and an anode connector **1711**. The cathode connector **1710** is configured to electrically engage with the cathode contact **1754** (or an end of the cathode contact pin **606**) of the substrate cassette **1704**. The cathode connector **1710** is further electrically connected to the control system **1708**. For example, the cathode connector **1710** is electrically connected to the second wire **1726** extending from the control system **1708**. The anode connector **1711** is configured to electrically engage with the anode contact (i.e., the first end **1732** of the cassette contact pin **1730**) of the substrate cassette **1704**. The anode connector **1711** is further electrically connected to the control system **1708**. For example, the anode connector **1711** is electrically connected to the first wire **1722** extending from the control system **1708**.

(118) In some implementations, the cassette cover **1706** can be movable between a raised position and a lowered position. In the raised position, the cassette cover **1706** is positioned at a first predetermined distance from the cassette tray **1705** so that the substrate cassette **1704** can be placed on the cassette tray **1705** without interference with the cassette cover **1706**. Once the substrate cassette **1704** is placed on the cassette tray **1705**, the cassette cover **1706** can be moved to the lowered position in which the cassette cover **1706** is positioned closer to the cassette tray **1705** and positioned at a second predetermined distance (shorter than the first predetermined distance) from the cassette tray **1705**. In the lowered position, the cathode connector **1710** and the anode connector **1711** are engaged with the cathode contact **1754** and the anode contact (i.e., the first end **1732** of the cassette contact pin **1730**) of the substrate cassette **1704**, respectively. The cathode connector **1710** and the anode connector **1711** can be positioned so as to be aligned with the cathode contact **1754** and the anode contact (i.e., the first end **1732** of the cassette contact pin **1730**) of the substrate cassette **1704** when the cassette cover **1706** is positioned in the lowered position relative to the substrate cassette **1704**. In some implementations, the cassette cover **1706** can be manually operated between the raised position and the lowered position. Alternatively or in addition, the cassette cover **1706** is automatically actuated using an actuator (e.g., a motor) that is controlled by a controller (e.g., the control system **1708**). For example, a user can place the substrate cassette **1704** on the cassette tray **1705** and select a menu via a user interface (e.g., the user interface **136** (FIG. **1**)) to lower the cassette cover **1706**, and the control system **1708** operates the actuator to move the cassette cover **1706** so that the cassette cover **1706** engages with the substrate cassette **1704** and makes necessary electrical connection with the substrate cassette **1704** as described above.

(119) Referring to FIG. **17C**, in another example, the substrate cassette **1704** described in FIGS. **11A-D** is placed on the cassette tray **1705**. In this example, the cassette cover **1706** is configured to be located above the cassette tray **1705** and provide at least part of the cathode assembly **1707**. For example, the cassette cover **1706** can include a plurality of electrodes **1720** that project from the cassette cover **1706** and configured to extend into the plurality of buffer chambers **1723** (defined at least by the plurality of apertures **1712** of the substrate cassette **1704**). The cassette cover **1706** can further include a cathode connector **1724** that electrically connects the plurality of electrodes **1720** and is further configured to electrically connect the control system **1708** via, for example, the second wire **1726** extending from the control system **1708**.

(120) Further, the cassette cover **1706** can include an anode connector **1726**. The anode connector **1727** is configured to electrically engage with the anode contact (i.e., the first end **1732** of the cassette contact pin **1730**) of the substrate cassette **1704**. The anode connector **1727** is further electrically connected to the control system **1708**. For example, the anode connector **1726** is electrically connected to the first wire **1722** extending from the control system **1708**.

(121) As described above, the cassette cover **1706** can be movable between the raised position and the lowered position. Once the substrate cassette **1704** is placed on the cassette tray **1705** with the cassette cover **1706** being in the raised position, the cassette cover **1706** can be moved to the lowered position. In the lowered position, the plurality of electrodes **1720** are positioned within the buffer chambers **1723**, respectively. Further, in the lowered position, the anode connector **1711** are engaged with the anode contact (i.e., the first end **1732** of the cassette contact pin **1730**) of the substrate cassette **1704**.

(122) FIG. **18** is a flowchart of an example process **1800** for analyte migration. The process **1800** can be performed using the electrophoretic system **100** (including the instrument **1700**) described herein. The process **1800** can be used to check for electrical connections in the system **100** prior to execution of electrophoresis.

(123) In some implementations, the process **1800** can include loading a substrate (e.g., the substrate **1002**) to an electrophoresis instrument (e.g., the instrument **1700**) (**1802**). For example, as described herein, the substrate **1002** comprises capture probes on the substrate regions **1016**, and one or more biological samples can be placed on the substrate **1002** so that the biological samples contact with the capture probes. Alternatively, the biological samples do not make contact with the capture probes. For example, an intermediate layer (e.g., gels) can be disposed between the substrate and the sample, which is compatible with electrophoresis. A substrate cassette (e.g., the substrate cassette **1004**) can be arranged on the substrate **1002** so that the plurality of apertures **1012** of the substrate cassette **1004** are aligned with the substrate regions **1016** of the substrate **1002** and define a plurality of buffer chambers (e.g., the buffer chambers **122**, **1723**) on the substrate regions **1016** of the substrate **1002**. Then, buffers (e.g., the buffers **124**) are supplied in the buffer chambers.

(124) The process **1800** can include arranging a cathode relative to the substrate (**1804**). For example, the cathode can include a plurality of electrodes (e.g., the electrodes **1252**, **1720**) that can be placed within the plurality of chambers, respectively.

(125) The process **1800** can include electrically connecting a power supply to the substrate and the cathode (**1806**). As described herein, the power supply can be included in and controlled by the control system **1008**. The power supply (or the control system) can be electrically connected to the substrate (as the anode) and the cathode (e.g., the plurality of electrodes) in various ways described herein, such as using various connection interfaces implemented in the substrate **1002**, the substrate cassette **1004**, the cathode assembly **1006**, the cassette cover **1706**, etc.

(126) The process **1800** can include applying a test voltage between the substrate (i.e., the anode) and the cathode (**1808**). The test voltage can be lower than a voltage necessary for executing electrophoresis in the instrument **1700**. The test voltage can be applied to ensure that electrical connections are made in the electrophoretic system **100** (including the instrument **1700**) before electrophoresis is performed. Such electrical connections can include at least one of an electrical connection of the power supply to the substrate, an electrical connection of the power supply to the cathode, or an electrical property of the buffers.

(127) The process **1800** can include detecting an output parameter in response to the test voltage (**1810**). In some implementations, the output parameter can be an impedance. For example, the test voltage can allow a current to flow through the system **100** (including the instrument **1700**), and the process **1800** can measure the current in the system **100**, and calculate an impedance based on the measured current and the test voltage being applied. In other implementations, other parameters in the system **100** can be used.

(128) The process **1800** can include determining whether the output parameter meets a threshold value (**1812**). The threshold value can be a value that is indicative of proper electrical connection in the system **100** (including the instrument **1700**). For example, the threshold value can be an

impedance of the system **100** when proper electrical connections are made in the system **100**, such as the electrical connection of the power supply to the substrate, the electrical connection of the power supply to the cathode, the electrical property of the buffers, etc. The threshold value can be a single value. Alternatively, the threshold value can be a range of value.

(129) The process **1800** can include, upon determining that the output parameter meets the threshold value, applying a voltage for analyte migration (**1814**). For example, when the output parameter meets the threshold value, the electrical connections in the system can be considered to be proper. Therefore, an actual voltage for electrophoresis can be applied to generate electric fields between the substrate (e.g., the plurality of substrate regions of the substrate) and the cathode (e.g., the plurality of electrodes) through one or a plurality of buffer chambers. The electrical fields can cause the analytes in the biological samples to migrate toward the capture probes on the substrate.

(130) The process **1800** can include, upon determining that the output parameter does not meet the threshold value, ceasing to apply the voltage for analyte migration (**1816**). For example, when the output parameter does not meet the threshold value, the electrical connections in the system can be considered to be improper. Thus, the instrument **1700** can stop applying the actual voltage if it has already started it, or does not proceed with applying the actual voltage if it hasn't started yet.

(131) The process **1800** can include, upon determining that the output parameter does not meet the threshold value, generating a notification of a potential electrical connection issue (**1818**). The notification can be output to the instrument **1700** or other devices that the user can access. For example, the notification can be output via the user interface **136**. The notification can include information that alerts users of an improper electrical connection in the system **100** (including the instrument **1700**). The notification can be output before advancing to subsequent steps for electrophoresis (e.g., applying the actual voltage for electrophoresis), so that the user can decide to take any action to investigate and/or remediate such improper connection. For example, based on the notification, the user can examine physical connections between the power supply and the substrate or between the power supply and the cathode, and check for the buffers. The user can then adjust the physical connections, or replace/refill the buffers, to resolve any issue. Alternatively, the notification can be output during or after performing such subsequent steps for electrophoresis (e.g., applying the actual voltage for electrophoresis). The user can consider such possibility of connection issues when analyzing the analytes captured by the electrophoresis.

(132) FIG. **19** is a block diagram of computing devices **1900** that may be used to implement the systems and methods described in this document, as either a client or as a server or plurality of servers. Computing device **1900** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. Computing device **1950** is intended to represent various forms of mobile devices, such as personal digital assistants, cellular telephones, smartphones, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be examples only, and are not meant to limit implementations described and/or claimed in this document.

(133) Computing device **1900** includes a processor **1902**, memory **1904**, a storage device **1906**, a high-speed interface **1908** connecting to memory **1904** and high-speed expansion ports **1910**, and a low speed interface **1912** connecting to low speed bus **1914** and storage device **1906**. Each of the components **1902**, **1904**, **1906**, **1908**, **1910**, and **1912**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **1902** can process instructions for execution within the computing device **1900**, including instructions stored in the memory **1904** or on the storage device **1906** to display graphical information for a graphic user interface (GUI) on an external input/output device, such as display **1916** coupled to high-speed interface **1908**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **1900** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(134) The memory **1904** stores information within the computing device **1900**. In one implementation, the memory **1904** is a volatile memory unit or units. In another implementation, the memory **1904** is a non-volatile memory unit or units. The memory **1904** may also be another form of computer-readable medium, such as a magnetic or optical disk.

(135) The storage device **1906** is capable of providing mass storage for the computing device **1900**. In one implementation, the storage device **1906** may be or contain a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. A computer program product can be tangibly embodied in an information carrier. The computer program product may also contain instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **1904**, the storage device **1906**, or memory on processor **1902**.

(136) The high-speed controller **1908** manages bandwidth-intensive operations for the computing device **1900**, while the low speed controller **1912** manages lower bandwidth-intensive operations. Such allocation of functions is an example only. In one implementation, the high-speed controller **1908** is coupled to memory **1904**, display **1916** (e.g., through a graphics processor or accelerator), and to high-speed expansion ports **1910**, which may accept various expansion cards (not shown). In the implementation, low-speed controller **1912** is coupled to storage device **1906** and low-speed expansion port **1914**. The low-speed expansion port, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet) may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(137) The computing device **1900** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **1920**, or multiple times in a group of such servers. It may also be implemented as part of a rack server system **1924**. In addition, it may be implemented in a personal computer such as a laptop computer **1922**. Alternatively, components from computing device **1900** may be combined with other components in a mobile device (not shown). Each of such devices may contain one or more of computing device **1900**, and an entire system may be made up of multiple computing devices **1900** communicating with each other.

(138) Additionally computing device **1900** can include Universal Serial Bus (USB) flash drives. The USB flash drives may store operating systems and other applications. The USB flash drives can include input/output components, such as a wireless transmitter or USB connector that may be inserted into a USB port of another computing device.

(139) Various implementations of the systems and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(140) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” “computer-readable medium” refers to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

(141) To provide for interaction with a user, the systems and techniques described here can be implemented on a computer having a display device (e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor) for displaying information to the user and a keyboard and a pointing device (e.g., a mouse or a trackball) by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback (e.g., visual feedback, auditory feedback, or tactile

feedback); and input from the user can be received in any form, including acoustic, speech, or tactile input.

(142) The systems and techniques described herein can be implemented in a computing system that includes a back end component (e.g., as a data server), or that includes a middleware component (e.g., an application server), or that includes a front end component (e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the systems and techniques described here), or any combination of such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication (e.g., a communication network). Examples of communication networks include a local area network (“LAN”), a wide area network (“WAN”), peer-to-peer networks (having ad-hoc or static members), grid computing infrastructures, and the Internet.

(143) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

Active Capture Methods

(144) In some of the methods described herein, an analyte in a biological sample (e.g., in a cell or tissue section) can be transported (e.g., passively or actively) to a capture probe (e.g., a capture probe affixed to a substrate (e.g., a substrate or bead)).

(145) For example, analytes can be transported to a capture probe (e.g., an immobilized capture probe) using an electric field (e.g., using electrophoresis), pressure, fluid flow, gravity, temperature, and/or a magnetic field. For example, analytes can be transported through, e.g., a gel (e.g., hydrogel), a fluid, or a permeabilized cell, to a capture probe (e.g., an immobilized capture probe) using a pressure gradient, a chemical concentration gradient, a temperature gradient, and/or a pH gradient. For example, analytes can be transported through a gel (e.g., hydrogel), a fluid, or a permeabilized cell, to a capture probe (e.g., an immobilized capture probe).

(146) In some examples, an electrophoretic field can be applied to analytes to facilitate migration of analytes towards a capture probe. In some examples, a sample containing analytes contacts a substrate having capture probes fixed on the substrate (e.g., a slide, cover slip, or bead), and an electric current is applied to promote the directional migration of charged analytes towards capture probes on a substrate. An electrophoresis assembly (e.g., an electrophoretic chamber), where a biological sample is in contact with a cathode and capture probes (e.g., capture probes fixed on a substrate), and where the capture probes are in contact with the biological sample and an anode, can be used to apply the current.

(147) In some embodiments, methods utilizing an active capture method can employ a conductive substrate (e.g., any of the conductive substrates described herein). In some embodiments, a conductive substrate includes paper, a hydrogel film, or a glass slide having a conductive coating. In some embodiments, a conductive substrate (e.g., any of the conductive substrates described herein) includes one or more capture probes.

(148) FIGS. 20A and 20B show different example analytical workflows of active capture methods using an electric field (e.g., using electrophoresis). In some examples, a biological sample **2002** (e.g., a tissue sample) can be in contact with a first substrate **2004**. In some embodiments, first substrate **2004** can have one or more coatings (e.g., any of the conductive substrates described herein) on its surface. Non-limiting examples of coatings include, nucleic acids (e.g., RNA) and conductive oxides (e.g., indium tin oxide). In some embodiments, first substrate **2004** can have a functionalization chemistry on its surface. In the examples shown in FIGS. 20A and 20B, first substrate **2004** is overlaid with a first coating **2006**, and first coating **2006** (e.g., a conductive coating) is further overlaid with a second coating **2008**. In some embodiments, first coating **2006** is an indium tin oxide (ITO) coating. In some embodiments, second coating **2008** is a lawn of capture probes (e.g., any of the capture probes described herein). In some embodiments, a substrate can include an ITO coating. In some embodiments, a substrate can include capture probes or capture probes attached to features on the substrate.

(149) Biological sample **2002** and second coating **2008** (e.g., a lawn of capture probes) can be in contact with a permeabilization solution **2010**. Non-limiting examples of permeabilization solutions includes enzymes (e.g., proteinase K, pepsin, and collagenase), detergents (e.g., sodium dodecyl sulfate (SDS), polyethylene glycol tert-octylphenyl ether, polysorbate 80, and polysorbate 20), ribonuclease inhibitors, buffers optimized for electrophoresis, buffers optimized for permeabilization, buffers optimized for hybridization, or combinations thereof. Permeabilization reagents can also include but are not limited to a dried permeabilization reagent, a permeabilization buffer, a buffer without a permeabilization reagent, a permeabilization gel, and a permeabilization solution. In some examples, biological samples (e.g., tissue samples) can be permeabilized first and then be subjected to electrophoresis.

(150) FIG. 20A shows an example analytical workflow including a first step **2012** in which biological sample **2002** can be permeabilized prior to subjecting the sample **2002** to electrophoresis. Any of the permeabilization methods disclosed herein can be used during first step **2012**. Biological sample **2002** includes an analyte **2014**. In some embodiments, the analyte **2014** is a negatively charged analyte. First substrate **2004** can include a capture probe **2016** that is fixed or attached to the first substrate **2004** or attached to features (e.g., beads **2018**) on the substrate. In some embodiments, capture probe **2016** can include any of the capture probes disclosed herein. In some embodiments, capture probes are directly or indirectly attached to the first substrate **2004**. In some embodiments, the capture probe **2016** is positively charged.

(151) In step **2020**, after permeabilization of biological sample **2002** concludes, the sample **2002** can be subjected to electrophoresis. During electrophoresis, the biological sample **2002** is subjected to an electric field that can be generated by sandwiching biological sample **2002** between the first substrate **2004** and a second substrate **2022**, connecting each substrate to a cathode and an anode, respectively, and running an electric current through the substrates. The application of the electric field “-E” causes the analyte **2004** (e.g., a negatively charged analyte) to migrate towards the substrate **2004** and capture probe **2016** (e.g., a positively charged capture probe) in the direction of the arrows shown in FIG. 20A. In some embodiments, the analyte **2014** migrates towards the capture probe **2016** for a distance “h.” In some embodiments, the analyte **2014** migrates towards a capture probe **2016** through one or more permeabilized cells within the permeabilized biological sample (e.g., from an original location in a permeabilized cell to a final location in or proximal to the capture probe **2016**). Second substrate **2022** can include the first coating **2006** (e.g., a conductive coating), thereby allowing electric field “-E” to be generated.

(152) In some embodiments, the analyte **2014** is a protein or a nucleic acid. In some embodiments, the analyte **2014** is a negatively charged protein or a nucleic acid. In some embodiments, the analyte **2014** is a positively charged protein or a nucleic acid. In some embodiments, the capture probe **2016** is a protein or a nucleic acid. In some embodiments, the capture probe **2016** is a positively charged protein or a nucleic acid. In some embodiments, the capture probe **2016** is a negatively charged protein or a nucleic acid. In some embodiments, the analyte **2014** is a negatively charged transcript. In some embodiments, the analyte **2014** is a poly(A) transcript. In some embodiments, the capture probe **2016** is attached to a feature in a feature array. In some embodiments, permeabilization reagent **2010** can be in contact with sample **2002**, first substrate **2004** second substrate **2022**, or any combination thereof.

(153) FIG. 20B shows an example analytical workflow in which biological sample **2002** can be permeabilized and subjected to electrophoresis simultaneously. In some embodiments, simultaneous permeabilization and electrophoresis of biological sample **2002** can decrease the total duration of the analytical workflow translating into a more efficient workflow.

(154) In some embodiments, the permeabilization reaction is conducted at a chilled temperature (e.g., about 4° C.). In some embodiments, conducting the permeabilization reaction at a chilled temperature controls the enzyme activity of the permeabilization reaction. In some embodiments, the permeabilization reaction is conducted at a chilled temperature in order to minimize drift and/or diffusion of the analyte **2014** from an original location (e.g., a location in a cell of the biological sample **2002**) until a user is ready to initiate the permeabilization reaction. In some embodiments, the permeabilization reaction is conducted at a warm temperature (e.g., a temperature ranging from about 15° C. to about 37° C. or more) in order to initiate and/or increase the rate of the permeabilization reaction. In some embodiments, once electrophoresis is applied and/or the permeabilization reaction is heated, the permeabilization reaction allows for analyte migration from an original location (e.g., a location in a cell of the biological sample **2002**) to the capture probe **2016** on the first substrate **2004**.

(155) Referring generally to FIGS. 21A-C, example substrate configurations for use in the active migration of analytes from a first location to a second location via electrophoresis are shown. FIG. 21A shows an example substrate configuration for use in electrophoresis in which the first substrate **2104** and the second substrate **2122** are aligned at about 90 degrees with respect to each other. In this example, the first substrate **2104** including biological sample **2102** is placed beneath second substrate **2122**. Both the first substrate **2104** and the second substrate **2122** can be connected to electrical wires **2124** that direct an electric current from a power supply to the substrates, thereby generating an electric field between the substrates. FIG. 21B shows an additional example substrate configuration for use during electrophoresis in which the first substrate **2104** and the second substrate **2122** are parallel with respect to each other. In this example, the first substrate **2104** including biological sample **2102** is also placed beneath second substrate **2122**.

(156) FIG. 21C shows yet an additional example substrate configuration for use in electrophoresis in which the second substrate **2122** and a third substrate **2126** are aligned at about 90 degrees with respect to the first substrate **2104**. Thus, in this example, a first biological sample **2102a** and a second biological sample **2102b** can be subjected to electrophoresis simultaneously. In some embodiments, 3, 4, 5, 6, 7, 8, 9, 10, or more biological samples can be placed on a same substrate and be subjected to electrophoresis simultaneously. In some embodiments, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more top substrates can be placed above a same bottom substrate containing one or more samples in order to simultaneously subject the one or more samples to electrophoresis. In some embodiments, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more top substrates can be perpendicularly placed (e.g., at about 90 degrees) above a same bottom substrate containing one or more samples in order to simultaneously subject the one or more samples to electrophoresis. In some embodiments, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more top substrates can be placed at a particular angle (e.g., at about 10, 20, 30, 40, 45, 50, 60, 70, or 80 degrees) with respect to a bottom substrate containing one or more samples in order to simultaneously subject the one or more samples to electrophoresis. In some embodiments, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more top substrates can be placed in a parallel orientation above a same bottom substrate containing one or more biological samples in order to simultaneously subject the one or more samples to electrophoresis. In some embodiments, a configuration of top substrates can be arranged above a same bottom substrate containing one or more biological samples in order to simultaneously subject the one or more samples to electrophoresis. In some embodiments, a first configuration of top substrates can be arranged above a second array of bottom substrates containing one or more biological samples in order to simultaneously subject the one or more biological samples to electrophoresis. In some embodiments, simultaneously subjecting two or more biological samples on a same substrate to electrophoresis can provide the advantage of a more effective workflow. In some embodiments, one or more of the top substrates can contain the biological sample. (157) In some embodiments, methods utilizing an active capture method can include one or more solutions between the biological sample and the substrate (e.g., a substrate including capture probes). In some embodiments, the one or more solutions between the biological sample and the substrate including capture probes can include a permeabilization buffer (e.g., any of the permeabilization buffers described herein). In some embodiments, the one or more solutions between the biological sample and the substrate including capture probes can include an electrophoresis buffer.

(158) In some embodiments, actively capturing analytes can include one or more porous materials between the biological sample and the substrate including capture probes. In some embodiments, the one or more porous materials between the biological sample and substrate including capture probes can include a paper or a blotting membrane. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can include a gel containing one or more solutions. For example, in a non-limiting way, the gel can be a SDS-PAGE gel. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can contain a permeabilization buffer. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can contain an electrophoresis buffer. In some embodiments, actively capturing analytes can include one or more solutions and one or more porous materials between the biological sample and the substrate including capture probes.

(159) In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes (e.g., an array) can act as a filter to separate analytes (e.g., analytes of interest) from other molecules or analytes present in the biological sample. In some embodiments, the analytes (e.g., analytes of interest) are RNA transcripts. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can act as a filter to separate RNA transcripts from other molecules (e.g., analytes) such as proteins, lipids and/or other nucleic acids. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can act as a filter to separate the analytes and other molecules based on physicochemical properties. For example, in a non-limiting way, analytes can be separated on properties such as charge, size (e.g., length, radius of gyration, effective diameters, etc.), hydrophobicity, hydrophilicity, molecular binding (e.g., immunoaffinity), and combinations thereof. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can separate the analytes from other molecules to reduce non-specific binding near the capture probes and therefore improve binding between the analytes and the capture probes, thus improving subsequent assay performance.

(160) In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can act as molecular sieving matrices for electrophoretic analyte separation. For example, in a non-limiting way, separation of analytes can occur based on physicochemical properties such as charge, size (e.g., length, radius of gyration, and effective diameters, etc.), electrophoretic mobility, zeta potential, isoelectric point, hydrophobicity, hydrophilicity, molecular binding (e.g., immunoaffinity), and combinations thereof. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can be of a uniform pore size. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can have discontinuities in pore sizes, as generally used in different gel electrophoresis schemes. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can have gradients in pore sizes. For example, the one or more porous materials (e.g., a hydrogel) can have a gradient of pore sizes such that the gradient separates the analytes as the analytes migrate to the substrate including capture probes (e.g., an array).

(161) In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can separate the analytes based on length. For example, shorter analytes will have a higher electrophoretic mobility, and therefore migrate faster towards the capture probes relative to longer analytes in an electrophoretic setup. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes separate the analytes based on length, such that only shorter analytes can migrate through the one or more porous materials to reach the capture probes, while longer analytes cannot reach the capture probes.

(162) In some embodiments, specific subsets of analytes (e.g., a subset of transcripts) can be captured by applying an electrophoretic field for a certain amount of time. In some embodiments, specific subsets of analytes (e.g., a subset of transcripts) can be captured by selecting different porous materials (e.g., porous materials with different compositions) between the biological sample and the substrate including capture probes. In some embodiments, specific subsets of analytes (e.g., a subset of transcripts) can be captured by applying an electrophoretic field for a certain amount of time and selecting different porous materials between the biological sample and the substrate including capture probes (e.g., an array).

(163) In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can have discontinuities in pore sizes that can cause an increase in the concentration of the migrating analytes (e.g., “stacking”). For example, the one or more porous materials (e.g., a hydrogel) between the biological sample and the substrate including capture probes can have discontinuities in pore sizes that can cause an increase in the concentration of the analytes near the capture probes resulting in favorable binding kinetics and increased sensitivity. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can have discontinuities in pore sizes that enhance the separation between migrating analytes of different sizes and/or lengths. In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can include a first porous material and a second

porous material, with the first porous material having a larger pore size than the second porous material. In some embodiments, the first porous material is located on the surface, or near the surface, of the biological sample. In some embodiments, the second porous material (e.g., second porous material with a smaller pore size than the first porous material) can be placed on the surface, or near the surface, of the first porous material. In some embodiments, as analytes migrate (e.g., migrate via electrophoresis) from the biological sample through the first porous material and the second porous material sequentially, the migrating analytes can collect (e.g., “stack”) at the interface between the first porous material and the second porous material.

(164) In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can include a gradient in pore sizes for continuous stacking as analytes migrate through decreasing pore sizes (e.g., decreasing pore diameter). In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can include a gradient in pore sizes such that the pores decrease in diameter as the analytes migrate from the biological sample to the substrate including capture probes. In some embodiments, the pore size gradient can increase the resolution among analytes of different sizes. In some embodiments, the pore size gradient can increase the concentration of the analytes near the capture probes. In some embodiments, the pore size gradient can continuously reduce the speed at which the analytes migrate and collect (e.g., “stack”) as the analytes migrate through the gradient of decreasing pore sizes (e.g., decreasing pore diameter).

(165) In some embodiments, the one or more porous materials between the biological sample and the substrate including capture probes can include a gradient gel for continuous stacking as analytes migrate through decreasing pore sizes (e.g., decreasing pore diameter) of the gradient gel. In some embodiments, the gradient gel can have pores with a decreasing diameter as the analytes migrate toward the capture probes. In some embodiments, the gradient gel can increase the separation resolution among analytes of different sizes. In some embodiments, the gradient gel can increase the concentration of analytes near the capture probes. In some embodiments, the gradient gel can continuously reduce the speed at which analytes migrate and collect (e.g., “stack”) as the analytes migrate through the gradient gel of decreasing pore sizes (e.g., decreasing in diameter).

(166) In some embodiments, a biological sample can be placed in a first substrate holder (e.g., a substrate holder described herein). In some embodiments, a spatially-barcoded capture probe array (e.g., capture probes, barcoded array) can be placed on a second substrate holder (e.g., a substrate holder described herein). In some embodiments, a biological sample can be placed in a first substrate holder that also contains capture probes. In some embodiments, the first substrate holder, the second substrate holder, or both can be conductive (e.g. any of the conductive substrates described herein). In some embodiments, the first substrate holder including the biological sample, the second substrate holder including capture probes, or both, can be contacted with permeabilization reagents (e.g., a permeabilization buffer) and analytes can be migrated from the biological sample toward the barcoded array using an electric field.

(167) In some embodiments, electrophoresis can be applied to a biological sample on a barcoded array while in contact with a permeabilization buffer. In some embodiments, electrophoresis can be applied to a biological sample on a barcoded array while in contact with an electrophoresis buffer (e.g. a buffer that lacks permeabilization reagents). In some embodiments, the permeabilization buffer can be replaced with an electrophoresis buffer after a desired amount of time. In some embodiments, electrophoresis can be applied simultaneously with the permeabilization buffer or electrophoresis buffer. In some embodiments, electrophoresis can be applied after a desired amount of time of contact between the biological sample and the permeabilization buffer or electrophoresis buffer.

(168) In some embodiments, the biological sample can be placed on a substrate (e.g., a porous membrane, a hydrogel, paper, etc.). In some embodiments, the biological sample placed on the substrate can have a gap (e.g., a space) between the substrate and the substrate holder (e.g., conductive substrate holder). In some embodiments, the barcoded array can be placed on a substrate (e.g., a porous membrane, a hydrogel, paper, etc.). In some embodiments, the barcoded array can have a gap between the substrate and substrate holder (e.g., conductive substrate holder). In some embodiments, the barcoded array can be placed in direct proximity to the biological sample or at a desired distance from the biological sample. In some embodiments, a buffer reservoir can be used between the substrate holder (e.g., conductive substrate holder) and the barcoded array, the biological sample, or both. This setup allows the analytes to be migrated to a barcoded array while not in proximity with the electrodes (e.g. conductive substrate holder), thus resulting in more stable electrophoresis.

(169) In some embodiments, a combination of at least two buffers with different ionic compositions can be used to differentially migrate analytes based on their ionic mobility (e.g., isotachopheresis (ITP)). For example, using two or more buffers with different ionic compositions can increase the concentration of analytes prior to contact with a barcoded array. Isotachopheresis includes at least two buffers that contain a common counter-ion (e.g., ions that have different charge sign than the analytes) and different co-ions (e.g., ions that have the same charge sign as the analytes) (Smejkal P, et al., Microfluidic isotachopheresis: A review, Electrophoresis, 34.11 1493-1509, (2013) which is incorporated herein by reference in its entirety). In some embodiments, one buffer can contain a co-ion with a higher ionic mobility (e.g. speed at which they travel through solution in an electric field) than the analytes (e.g., the “leading” buffer). In some embodiments, a second buffer can contain a co-ion with a lower ionic mobility than the analytes (e.g., the “trailing” buffer). In some embodiments, a third buffer can contain a co-ion with an ionic mobility that is between the electric mobility of the analytes. In some embodiments, a biological sample can be placed on a first substrate holder (e.g., a conductive substrate holder) and the barcoded array can be placed on a second substrate holder (e.g., a second conductive substrate holder) and contacted with a permeabilization buffer and the analytes can be migrated away from the biological sample and toward the barcoded array using an electric field. As the electric field is applied to the biological sample the analytes can be concentrated in the buffer as they are migrated toward the capture probes. In some embodiments, isotachopheresis can be used with gel-based separations (e.g., any of the gel-based separations described herein).

(170) In some embodiments, a permeabilization buffer can be applied to a region of interest (e.g., region of interest as described herein) in a biological sample. In some embodiments, permeabilization reagents (e.g. a hydrogel containing permeabilization reagents) can be applied to a region of interest in a biological sample. For example, a region of interest can be a region that is smaller in area relative to the overall area of the biological sample. In some embodiments, the permeabilization buffer or permeabilization reagents can be contacted with the biological sample and a substrate including capture probes (e.g., an array). In some embodiments, the biological sample can have more than one region of interest (e.g. two, three). In some embodiments, the biological sample, the substrate including capture probes, or both, can be placed in a conductive substrate holder. In some embodiments, analytes can be released from the region(s) of interest and migrated from the biological sample toward the capture probes with an electric field.

(171) In some embodiments, electrophoretic transfer of analytes can be performed while retaining the relative spatial locations of analytes in a biological sample while minimizing passive diffusion of an analyte away from its location in a biological sample. In some embodiments, an analyte captured by a capture probe (e.g., capture probes on a substrate) retains the spatial location of the analyte present in the biological sample from which it was obtained (e.g., the spatial location of the analyte that is captured by a capture probe on a substrate when the analyte is actively migrated to the capture probe by electrophoretic transfer can be more precise or representative of the spatial location of the analyte in the biological sample than when the analyte is not actively migrated to the capture probe). In some embodiments, electrophoretic transport and binding process is described by the Damkohler number (Da), which is a ratio of reaction and mass transport rates. The fraction of analytes bound and the shape of the biological sample will depend on the parameters in the Da. There parameters include electromigration velocity $U_{sub,e}$ (depending on analyte electrophoretic mobility $\mu_{sub,e}$ and electric field strength E), density of capture probes (e.g., barcoded oligonucleotides) $p_{sub,0}$, the binding rate between probes (e.g., barcoded oligonucleotides) and analytes $k_{sub,on}$, and capture area thickness L.

$$(172) Da \sim \frac{k_{on} p_0 L}{U_{sub,e} E}$$

Fast migration (e.g., electromigration) can reduce assay time and can minimize molecular diffusion of analytes.

(173) In some embodiments, the spatial alignment of the analytes in the sample. As such, an analyte captured by the capture probes (e.g., capture probes on a substrate) retains the spatial information of the cell or the biological sample from which it was obtained. Applying an electrophoretic field to analytes can also result in an increase in temperature (e.g., heat). In some embodiments, the increased temperature (e.g., heat) can facilitate the migration of the analytes towards a capture probe.

(174) In some examples, a spatially-addressable microelectrode array is used for spatially-constrained capture of at least one charged analyte of interest by a capture probe. For example, a spatially-addressable microelectrode array can allow for discrete (e.g., localized) application of an electric field rather than a uniform electric field. The spatially-addressable microelectrode array can be independently addressable. In some embodiments, the electric field can be applied to one or more regions of interest in a biological sample. The electrodes may be adjacent to each other or distant from each other. The microelectrode array can be configured to include a high density of discrete sites having a small area for applying an electric field to promote the migration of charged analyte(s) of interest. For example, electrophoretic capture can be performed on a region of interest using a spatially-addressable microelectrode array.

(175) A high density of discrete sites on a microelectrode array can be used. The surface can include any suitable density of discrete sites (e.g., a density suitable for processing the sample on the conductive substrate in a given amount of time). In one embodiment, the surface has a density of discrete sites greater than or equal to about 500 sites per 1 mm.^{sup.2}. In some embodiments, the surface has a density of discrete sites of about 100, about 200, about 300, about 400, about 500, about 600, about 700, about 800, about 900, about 1,000, about 2,000, about 3,000, about 4,000, about 5,000, about 6,000, about 7,000, about 8,000, about 9,000, about 10,000, about 20,000, about 40,000, about 60,000, about 80,000, about 100,000, or about 500,000 sites per 1 mm.^{sup.2}. In some embodiments, the surface has a density of discrete sites of at least about 200, at least about 300, at least about 400, at least about 500, at least about 600, at least about 700, at least about 800, at least about 900, at least about 1,000, at least about 2,000, at least about 3,000, at least about 4,000, at least about 5,000, at least about 6,000, at least about 7,000, at least about 8,000, at least about 9,000, at least about 10,000, at least about 20,000, at least about 40,000, at least about 60,000, at least about 80,000, at least about 100,000, or at least about 500,000 sites per 1 mm.^{sup.2}.

(176) Schematics illustrating an electrophoretic transfer system configured to direct nucleic acid analytes (e.g., mRNA transcripts) toward a spatially-barcoded capture probe array are shown in FIG. 22A and FIG. 22B. In this exemplary configuration of an electrophoretic system, a sample 2202 is sandwiched between the cathode 2201 and the spatially-barcoded capture probe array 2204, 2205, and the spatially-barcoded capture probe array 2204, 2205 is sandwiched between the sample 2202 and the anode 2203, such that the sample 2202 is in contact with the spatially-barcoded capture probes 2207. When an electric field is applied to the electrophoretic transfer system, negatively charged nucleic acid analytes 2206 will be pulled toward the positively charged anode 2203 and into the spatially-barcoded array 2204, 2205 containing the spatially-barcoded capture probes 2207. The spatially-barcoded capture probes 2207 interact with the nucleic acid analytes (e.g., mRNA transcripts hybridize to spatially-barcoded nucleic acid capture probes forming DNA/RNA hybrids) 2207, making the analyte capture more efficient. The electrophoretic system set-up may change depending on the target analyte. For example, proteins may be positive, negative, neutral, or polar depending on the protein as well as other factors (e.g., isoelectric point, solubility, etc.). The skilled practitioner has the knowledge and experience to arrange the electrophoretic transfer system to facilitate capture of a particular target analyte.

(177) FIG. 23 is an illustration showing an exemplary workflow protocol utilizing an electrophoretic transfer system. In the example, Panel A depicts a flexible spatially-barcoded feature array being contacted with a sample. The sample can be a flexible array, wherein the array is immobilized on a hydrogel, membrane, or other flexible substrate. Panel B depicts contact of the array with the sample and imaging of the array-sample assembly. The image of the sample/array assembly can be used to verify sample placement, choose a region of interest, or any other reason for imaging a sample on an array as described herein. Panel C depicts application of an electric field using an electrophoretic transfer system to aid in efficient capture of a target analyte. Here, negatively charged mRNA target analytes migrate toward the positively charged anode. Panel D depicts application of reverse transcription reagents and first strand cDNA synthesis of the captured target analytes. Panel E depicts array removal and preparation for library construction (Panel F) and next-generation sequencing (Panel G).

(178) Spatial analysis methodologies and compositions described herein can provide a vast amount of analyte and/or expression data for a variety of analytes within a biological sample at high spatial resolution, while retaining native spatial context. Spatial analysis methods and compositions can include, e.g., the use of a capture probe including a spatial barcode (e.g., a nucleic acid sequence that provides information as to the location or position of an analyte within a cell or a tissue sample (e.g., mammalian cell or a mammalian tissue sample) and a capture domain that is capable of binding to an analyte (e.g., a protein and/or a nucleic acid) produced by and/or present in a cell. Spatial analysis methods and compositions can also include the use of a capture probe having a capture domain that captures an intermediate agent for indirect detection of an analyte. For example, the intermediate agent can include a nucleic acid sequence (e.g., a barcode) associated with the intermediate agent. Detection of the intermediate agent is therefore indicative of the analyte in the cell or tissue sample.

(179) Non-limiting aspects of spatial analysis methodologies and compositions are described in U.S. Pat. Nos. 10,774,374, 10,724,078, 10,480,022, 10,059,990, 10,041,949, 10,002,316, 9,879,313, 9,783,841, 9,727,810, 9,593,365, 8,951,726, 8,604,182, 7,709,198, U.S. Patent Application Publication Nos. 2020/239946, 2020/080136, 2020/0277663, 2020/024641, 2019/330617, 2019/264268, 2020/256867, 2020/224244, 2019/194709, 2019/161796, 2019/085383, 2019/055594, 2018/216161, 2018/051322, 2018/0245142, 2017/241911, 2017/089811, 2017/067096, 2017/029875, 2017/0016053, 2016/108458, 2015/000854, 2013/171621, WO 2018/091676, WO 2020/176788, Rodrigues et al., Science 363(6434):1463-1467, 2019; Lee et al., Nat. Protoc. 10(3):442-458, 2015; Trejo et al., PLoS ONE 14(2):e0212031, 2019; Chen et al., Science 348(6233):aaa6090, 2015; Gao et al., BMC Biol. 15:50, 2017; and Gupta et al., Nature Biotechnol. 36:1197-1202, 2018; the Visium Spatial Gene Expression Reagent Kits User Guide (e.g., Rev C, dated June 2020), and/or the Visium Spatial Tissue Optimization Reagent Kits User Guide (e.g., Rev C, dated July 2020), both of which are available at the 10× Genomics Support Documentation website, and can be used herein in any combination. Further non-limiting aspects of spatial analysis methodologies and compositions are described herein.

(180) Some general terminology that may be used in this disclosure can be found in Section (I)(b) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Typically, a “barcode” is a label, or identifier, that conveys or is capable of conveying information (e.g., information about an analyte in a sample, a bead, and/or a capture probe). A barcode can be part of an analyte, or independent of an analyte. A barcode can be attached to an analyte. A particular barcode can be unique relative to other barcodes. For the purpose of this disclosure, an “analyte” can include any biological substance, structure, moiety, or component to be analyzed. The term “target” can similarly refer to an analyte of interest.

(181) Analytes can be broadly classified into one of two groups: nucleic acid analytes, and non-nucleic acid analytes. Examples of non-nucleic acid analytes include, but are not limited to, lipids, carbohydrates, peptides, proteins, glycoproteins (N-linked or O-linked), lipoproteins, phosphoproteins, specific phosphorylated or acetylated variants of proteins, amidation variants of proteins, hydroxylation variants of proteins, methylation variants of proteins, ubiquitylation variants of proteins, sulfation variants of proteins, viral proteins (e.g., viral capsid, viral envelope, viral coat, viral accessory, viral glycoproteins, viral spike, etc.), extracellular and intracellular proteins, antibodies, and antigen binding fragments. In some embodiments, the analyte(s) can be localized to subcellular location(s), including, for example, organelles, e.g., mitochondria, Golgi apparatus, endoplasmic reticulum, chloroplasts, endocytic vesicles, exocytic vesicles, vacuoles, lysosomes, etc. In some embodiments, analyte(s) can be peptides or proteins, including without limitation antibodies and enzymes. Additional examples of analytes can be found in Section (I)(c) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. In some embodiments, an analyte can be detected indirectly, such as through detection of an intermediate agent, for example, a ligation product or an analyte capture agent (e.g., an oligonucleotide-conjugated antibody), such as those described herein.

(182) A “biological sample” is typically obtained from the subject for analysis using any of a variety of techniques including, but not limited to, biopsy, surgery, and laser capture microscopy (LCM), and generally includes cells and/or other biological material from the subject. In some

embodiments, a biological sample, a tissue section. In some embodiments, a biological sample can be a fixed and/or stained biological sample (e.g., a fixed and/or stained tissue section). Non-limiting examples of stains include histological stains (e.g., hematoxylin and/or eosin) and immunological stains (e.g., fluorescent stains). In some embodiments, a biological sample (e.g., a fixed and/or stained biological sample) can be imaged. Biological samples are also described in Section (I)(d) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(183) In some embodiments, a biological sample is permeabilized with one or more permeabilization reagents. For example, permeabilization of a biological sample can facilitate analyte capture. Exemplary permeabilization agents and conditions are described in Section (I)(d)(ii)(13) or the Exemplary Embodiments Section of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(184) Array-based spatial analysis methods involve the transfer of one or more analytes from a biological sample to an array of features on a substrate, where each feature is associated with a unique spatial location on the array. Subsequent analysis of the transferred analytes includes determining the identity of the analytes and the spatial location of the analytes within the biological sample. The spatial location of an analyte within the biological sample is determined based on the feature to which the analyte is bound (e.g., directly or indirectly) on the array, and the feature's relative spatial location within the array.

(185) A “capture probe” refers to any molecule capable of capturing (directly or indirectly) and/or labelling an analyte (e.g., an analyte of interest) in a biological sample. In some embodiments, the capture probe is a nucleic acid or a polypeptide. In some embodiments, the capture probe includes a barcode (e.g., a spatial barcode and/or a unique molecular identifier (UMI)) and a capture domain. In some embodiments, a capture probe can include a cleavage domain and/or a functional domain (e.g., a primer-binding site, such as for next-generation sequencing (NGS)). See, e.g., Section (II)(b) (e.g., subsections (i)–(vi)) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Generation of capture probes can be achieved by any appropriate method, including those described in Section (II)(d)(ii) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(186) In some embodiments, more than one analyte type (e.g., nucleic acids and proteins) from a biological sample can be detected (e.g., simultaneously or sequentially) using any appropriate multiplexing technique, such as those described in Section (IV) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(187) In some embodiments, detection of one or more analytes (e.g., protein analytes) can be performed using one or more analyte capture agents. As used herein, an “analyte capture agent” refers to an agent that interacts with an analyte (e.g., an analyte in a biological sample) and with a capture probe (e.g., a capture probe attached to a substrate or a feature) to identify the analyte. In some embodiments, the analyte capture agent includes: (i) an analyte binding moiety (e.g., that binds to an analyte), for example, an antibody or antigen-binding fragment thereof; (ii) analyte binding moiety barcode; and (iii) an analyte capture sequence. As used herein, the term “analyte binding moiety barcode” refers to a barcode that is associated with or otherwise identifies the analyte binding moiety. As used herein, the term “analyte capture sequence” refers to a region or moiety configured to hybridize to, bind to, couple to, or otherwise interact with a capture domain of a capture probe. In some cases, an analyte binding moiety barcode (or portion thereof) may be able to be removed (e.g., cleaved) from the analyte capture agent. Additional description of analyte capture agents can be found in Section (II)(b)(ix) of WO 2020/176788 and/or Section (II)(b)(viii) U.S. Patent Application Publication No. 2020/0277663.

(188) There are at least two methods to associate a spatial barcode with one or more neighboring cells, such that the spatial barcode identifies the one or more cells, and/or contents of the one or more cells, as associated with a particular spatial location. One method is to promote analytes or analyte proxies (e.g., intermediate agents) out of a cell and towards a spatially-barcoded array (e.g., including spatially-barcoded capture probes). Another method is to cleave spatially-barcoded capture probes from an array and promote the spatially-barcoded capture probes towards and/or into or onto the biological sample.

(189) In some cases, capture probes may be configured to prime, replicate, and consequently yield optionally barcoded extension products from a template (e.g., a DNA or RNA template, such as an analyte or an intermediate agent (e.g., a ligation product or an analyte capture agent), or a portion thereof), or derivatives thereof (see, e.g., Section (II)(b)(vii) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663 regarding extended capture probes). In some cases, capture probes may be configured to form ligation products with a template (e.g., a DNA or RNA template, such as an analyte or an intermediate agent, or portion thereof), thereby creating ligation products that serve as proxies for a template.

(190) As used herein, an “extended capture probe” refers to a capture probe having additional nucleotides added to the terminus (e.g., 3' or 5' end) of the capture probe thereby extending the overall length of the capture probe. For example, an “extended 3' end” indicates additional nucleotides were added to the most 3' nucleotide of the capture probe to extend the length of the capture probe, for example, by polymerization reactions used to extend nucleic acid molecules including templated polymerization catalyzed by a polymerase (e.g., a DNA polymerase or a reverse transcriptase). In some embodiments, extending the capture probe includes adding to a 3' end of a capture probe a nucleic acid sequence that is complementary to a nucleic acid sequence of an analyte or intermediate agent specifically bound to the capture domain of the capture probe. In some embodiments, the capture probe is extended using reverse transcription. In some embodiments, the capture probe is extended using one or more DNA polymerases. The extended capture probes include the sequence of the capture probe and the sequence of the spatial barcode of the capture probe.

(191) In some embodiments, extended capture probes are amplified (e.g., in bulk solution or on the array) to yield quantities that are sufficient for downstream analysis, e.g., via DNA sequencing. In some embodiments, extended capture probes (e.g., DNA molecules) act as templates for an amplification reaction (e.g., a polymerase chain reaction).

(192) Additional variants of spatial analysis methods, including in some embodiments, an imaging step, are described in Section (II)(a) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Analysis of captured analytes (and/or intermediate agents or portions thereof), for example, including sample removal, extension of capture probes, sequencing (e.g., of a cleaved extended capture probe and/or a cDNA molecule complementary to an extended capture probe), sequencing on the array (e.g., using, for example, in situ hybridization or in situ ligation approaches), temporal analysis, and/or proximity capture, is described in Section (II)(g) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Some quality control measures are described in Section (II)(h) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(193) Spatial information can provide information of biological and/or medical importance. For example, the methods and compositions described herein can allow for: identification of one or more biomarkers (e.g., diagnostic, prognostic, and/or for determination of efficacy of a treatment) of a disease or disorder; identification of a candidate drug target for treatment of a disease or disorder; identification (e.g., diagnosis) of a subject as having a disease or disorder; identification of stage and/or prognosis of a disease or disorder in a subject; identification of a subject as having an increased likelihood of developing a disease or disorder; monitoring of progression of a disease or disorder in a subject; determination of efficacy of a treatment of a disease or disorder in a subject; identification of a patient subpopulation for which a treatment is effective for a disease or disorder; modification of a treatment of a subject with a disease or disorder; selection of a subject for participation in a clinical trial; and/or selection of a treatment for a subject with a disease or disorder.

(194) Spatial information can provide information of biological importance. For example, the methods and compositions described herein can allow for: identification of transcriptome and/or proteome expression profiles (e.g., in healthy and/or diseased tissue); identification of multiple analyte types in close proximity (e.g., nearest neighbor analysis); determination of up- and/or down-regulated genes and/or proteins in diseased tissue; characterization of tumor microenvironments; characterization of tumor immune responses; characterization of cells types and their co-localization in tissue; and identification of genetic variants within tissues (e.g., based on gene and/or protein expression profiles associated with specific disease or disorder biomarkers).

(195) Typically, for spatial array-based methods, a substrate functions as a support for direct or indirect attachment of capture probes to features of the array. A “feature” is an entity that acts as a support or repository for various molecular entities used in spatial analysis. In some embodiments,

some or all of the features in an array are functionalized for analyte capture. Exemplary substrates are described in Section (II)(c) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. Exemplary features and geometric attributes of an array can be found in Sections (II)(d)(i), (II)(d)(iii), and (II)(d)(iv) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(196) Generally, analytes and/or intermediate agents (or portions thereof) can be captured when contacting a biological sample with a substrate including capture probes (e.g., a substrate with capture probes embedded, spotted, printed, fabricated on the substrate, or a substrate with features (e.g., beads, wells) comprising capture probes). As used herein, “contact,” “contacted,” and/or “contacting,” a biological sample with a substrate refers to any contact (e.g., direct or indirect) such that capture probes can interact (e.g., bind covalently or non-covalently (e.g., hybridize)) with analytes from the biological sample. Capture can be achieved actively (e.g., using electrophoresis) or passively (e.g., using diffusion). Analyte capture is further described in Section (II)(e) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(197) In some cases, spatial analysis can be performed by attaching and/or introducing a molecule (e.g., a peptide, a lipid, or a nucleic acid molecule) having a barcode (e.g., a spatial barcode) to a biological sample (e.g., to a cell in a biological sample). In some embodiments, a plurality of molecules (e.g., a plurality of nucleic acid molecules) having a plurality of barcodes (e.g., a plurality of spatial barcodes) are introduced to a biological sample (e.g., to a plurality of cells in a biological sample) for use in spatial analysis. In some embodiments, after attaching and/or introducing a molecule having a barcode to a biological sample, the biological sample can be physically separated (e.g., dissociated) into single cells or cell groups for analysis. Some such methods of spatial analysis are described in Section (III) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663.

(198) In some cases, spatial analysis can be performed by detecting multiple oligonucleotides that hybridize to an analyte. In some instances, for example, spatial analysis can be performed using RNA-templated ligation (RTL). Methods of RTL have been described previously. See, e.g., Credle et al., *Nucleic Acids Res.* 2017 Aug. 21; 45(14):e128. Typically, RTL includes hybridization of two oligonucleotides to adjacent sequences on an analyte (e.g., an RNA molecule, such as an mRNA molecule). In some instances, the oligonucleotides are DNA molecules. In some instances, one of the oligonucleotides includes at least two ribonucleic acid bases at the 3' end and/or the other oligonucleotide includes a phosphorylated nucleotide at the 5' end. In some instances, one of the two oligonucleotides includes a capture domain (e.g., a poly(A) sequence, a non-homopolymeric sequence). After hybridization to the analyte, a ligase (e.g., SplintR ligase) ligates the two oligonucleotides together, creating a ligation product. In some instances, the two oligonucleotides hybridize to sequences that are not adjacent to one another. For example, hybridization of the two oligonucleotides creates a gap between the hybridized oligonucleotides. In some instances, a polymerase (e.g., a DNA polymerase) can extend one of the oligonucleotides prior to ligation. After ligation, the ligation product is released from the analyte. In some instances, the ligation product is released using an endonuclease (e.g., RNase H). The released ligation product can then be captured by capture probes (e.g., instead of direct capture of an analyte) on an array, optionally amplified, and sequenced, thus determining the location and optionally the abundance of the analyte in the biological sample.

(199) During analysis of spatial information, sequence information for a spatial barcode associated with an analyte is obtained, and the sequence information can be used to provide information about the spatial distribution of the analyte in the biological sample. Various methods can be used to obtain the spatial information. In some embodiments, specific capture probes and the analytes they capture are associated with specific locations in an array of features on a substrate. For example, specific spatial barcodes can be associated with specific array locations prior to array fabrication, and the sequences of the spatial barcodes can be stored (e.g., in a database) along with specific array location information, so that each spatial barcode uniquely maps to a particular array location.

(200) Alternatively, specific spatial barcodes can be deposited at predetermined locations in an array of features during fabrication such that at each location, only one type of spatial barcode is present so that spatial barcodes are uniquely associated with a single feature of the array. Where necessary, the arrays can be decoded using any of the methods described herein so that spatial barcodes are uniquely associated with array feature locations, and this mapping can be stored as described above.

(201) When sequence information is obtained for capture probes and/or analytes during analysis of spatial information, the locations of the capture probes and/or analytes can be determined by referring to the stored information that uniquely associates each spatial barcode with an array feature location. In this manner, specific capture probes and captured analytes are associated with specific locations in the array of features. Each array feature location represents a position relative to a coordinate reference point (e.g., an array location, a fiducial marker) for the array. Accordingly, each feature location has an “address” or location in the coordinate space of the array.

(202) Some exemplary spatial analysis workflows are described in the Exemplary Embodiments section of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. See, for example, the Exemplary embodiment starting with “In some non-limiting examples of the workflows described herein, the sample can be immersed . . .” of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663. See also, e.g., the Visium Spatial Gene Expression Reagent Kits User Guide (e.g., Rev C, dated June 2020), and/or the Visium Spatial Tissue Optimization Reagent Kits User Guide (e.g., Rev C, dated July 2020).

(203) In some embodiments, spatial analysis can be performed using dedicated hardware and/or software, such as any of the systems described in Sections (II)(e)(ii) and/or (V) of WO 2020/176788 and/or U.S. Patent Application Publication No. 2020/0277663, or any of one or more of the devices or methods described in Sections Control Slide for Imaging, Methods of Using Control Slides and Substrates for, Systems of Using Control Slides and Substrates for Imaging, and/or Sample and Array Alignment Devices and Methods, Informational labels of WO 2020/123320.

(204) Suitable systems for performing spatial analysis can include components such as a chamber (e.g., a flow cell or sealable, fluid-tight chamber) for containing a biological sample. The biological sample can be mounted for example, in a biological sample holder. One or more fluid chambers can be connected to the chamber and/or the sample holder via fluid conduits, and fluids can be delivered into the chamber and/or sample holder via fluidic pumps, vacuum sources, or other devices coupled to the fluid conduits that create a pressure gradient to drive fluid flow. One or more valves can also be connected to fluid conduits to regulate the flow of reagents from reservoirs to the chamber and/or sample holder.

(205) The systems can optionally include a control unit that includes one or more electronic processors, an input interface, an output interface (such as a display), and a storage unit (e.g., a solid state storage medium such as, but not limited to, a magnetic, optical, or other solid state, persistent, writeable and/or re-writable storage medium). The control unit can optionally be connected to one or more remote devices via a network. The control unit (and components thereof) can generally perform any of the steps and functions described herein. Where the system is connected to a remote device, the remote device (or devices) can perform any of the steps or features described herein. The systems can optionally include one or more detectors (e.g., CCD, CMOS) used to capture images. The systems can also optionally include one or more light sources (e.g., LED-based, diode-based, lasers) for illuminating a sample, a substrate with features, analytes from a biological sample captured on a substrate, and various control and calibration media.

(206) The systems can optionally include software instructions encoded and/or implemented in one or more of tangible storage media and hardware components such as application specific integrated circuits. The software instructions, when executed by a control unit (and in particular, an electronic processor) or an integrated circuit, can cause the control unit, integrated circuit, or other component executing the software instructions to perform any of the method steps or functions described herein.

(207) In some cases, the systems described herein can detect (e.g., register an image) the biological sample on the array. Exemplary methods to detect the biological sample on an array are described in PCT Application No. 2020/061064 and/or U.S. patent application Ser. No. 16/951,854.

(208) Prior to transferring analytes from the biological sample to the array of features on the substrate, the biological sample can be aligned with the array. Alignment of a biological sample and an array of features including capture probes can facilitate spatial analysis, which can be used to detect differences in analyte presence and/or level within different positions in the biological sample, for example, to generate a three-dimensional map of

the analyte presence and/or level. Exemplary methods to generate a two- and/or three-dimensional map of the analyte presence and/or level are described in PCT Application No. 2020/053655 and spatial analysis methods are generally described in WO 2020/061108 and/or U.S. patent application Ser. No. 16/951,864.

(209) In some cases, a map of analyte presence and/or level can be aligned to an image of a biological sample using one or more fiducial markers, e.g., objects placed in the field of view of an imaging system which appear in the image produced, as described in the Substrate Attributes Section, Control Slide for Imaging Section of WO 2020/123320, PCT Application No. 2020/061066, and/or U.S. patent application Ser. No. 16/951,843. Fiducial markers can be used as a point of reference or measurement scale for alignment (e.g., to align a sample and an array, to align two substrates, to determine a location of a sample or array on a substrate relative to a fiducial marker) and/or for quantitative measurements of sizes and/or distances.

(210) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular implementations of particular inventions. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(211) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(212) Thus, particular implementations of the subject matter have been described. Other implementations are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous.

Claims

1. An electrophoretic system for analyte migration, the system a substrate including a plurality of regions, each region configured to be electrically conductive, each region configured to comprise one or more capture probes and a biological sample comprising an analyte, wherein the substrate includes a first electrode contact, the first electrode contact being electrically connected to at least one of the plurality of regions, the substrate including a conductive material, the conductive material including at least one of tin oxide (TO), indium tin oxide (ITO), a transparent conductive oxide (TCO), aluminum doped zinc oxide (AZO), fluorine doped tin oxide (FTO), or any combination thereof; a substrate cassette configured to be disposed at the substrate and including a plurality of apertures corresponding to the plurality of regions of the substrate, the plurality of apertures configured to define a plurality of buffer chambers comprising the plurality of regions of the substrate, wherein the substrate cassette comprises a connection interface electrically coupled to the first electrode contact of the substrate; a cathode assembly including a plurality of electrodes positioned within the plurality of buffer chambers of the substrate cassette, respectively, wherein the cathode assembly includes a second electrode contact, the second electrode contact being electrically connected to at least one of the plurality of electrodes; and a power supply electrically connected with the first electrode contact of the substrate at the connection interface of the substrate cassette, and electrically connected with the second electrode contact of the cathode assembly, the power supply configured to generate an electric field between the plurality of regions and the plurality of electrodes, respectively, such that the analyte in the biological sample moves toward the one or more capture probes on the substrate.
2. The electrophoretic system of claim 1, wherein the plurality of buffer chambers are configured to receive a buffer.
3. The electrophoretic system of claim 1, further comprising: a light configured to illuminate the plurality of regions.
4. The electrophoretic system of claim 1, further comprising: a first contact pin electrically attached to the substrate and providing the first electrode contact; wherein the substrate cassette includes: a body defining a cavity configured to receive the substrate, wherein the connection interface of the substrate cassette includes a contact hole defined at the body and configured to expose the first contact pin therethrough; wherein the body includes a main face and a lateral face extending from a periphery of the main face; wherein the plurality of the apertures are defined at the main face of the body; and wherein the contact hole is defined at the main face.
5. The electrophoretic system of claim 1, wherein the substrate cassette includes: a body defining a cavity configured to receive the substrate; and a contact pin extending through a wall of the body and having a first end and a second end opposite to the first end, the first end arranged exterior of the body, and the second end arranged interior of the body and configured to electrically contact with the first electrode contact of the substrate.
6. The electrophoretic system of claim 5, further comprising: a contact bracket that electrically engages with the substrate and provides the first electrode contact of the substrate, wherein the second end of the contact pin is configured to electrically contact with the contact bracket.
7. The electrophoretic system of claim 5, further comprising: a system housing including the power supply; a cassette tray that extends from the system housing and is configured to receive the substrate cassette thereon; and a cassette cover that extends from the system housing above the cassette tray and includes the cathode assembly, wherein the plurality of electrodes project from the cassette cover toward the plurality of buffer chambers of the substrate cassette, the cassette cover further including: an anode connector configured to electrically engage with the contact pin of the substrate cassette.
8. The electrophoretic system of claim 1, wherein the substrate cassette includes a second aperture corresponding to the first electrode contact of the substrate and configured to define a second buffer chamber on the first electrode contact of the substrate, wherein the plurality of buffer chambers are configured to contain a first buffer, and wherein the second buffer chamber is configured to contain a second buffer different from the first buffer; wherein the second buffer has an electrolyte strength greater than the first buffer; wherein the cathode assembly includes: a cathode body configured to position at least partially on the substrate cassette, wherein the plurality of electrodes projecting from the cathode body and configured to position within the plurality of buffer chambers, respectively; and a second electrode projecting from the cathode body and configured to position within the second buffer chamber.
9. The electrophoretic system of claim 8, wherein the cathode body includes: an anode contact that is electrically connected to the second electrode, wherein the power supply is electrically connected to the anode contact of the cathode body so that the power supply is electrically connected with the first electrode contact of the substrate via the second electrode that positions within the second buffer chamber containing the second buffer on the first electrode contact of the substrate.
10. The electrophoretic system of claim 1, wherein the substrate cassette includes: a cassette body defining a cavity configured to receive the substrate, wherein the cassette body includes the cathode assembly such that the plurality of electrodes are configured to position within the plurality of buffer chambers; and a plurality of cathode contact pins extending through a wall of the cassette body and electrically contacting with the second electrode contact such that a cathode contact pin of the cathode contact pins is electrically connected to the at least one of the plurality of electrodes; wherein each of the plurality of electrodes includes at least one of a conductive wire, a conductive rod, or an array of conductive wires; and wherein

each of the plurality of electrodes includes a conductive plate at a distal end of the at least one of the conductive wire, the conductive rod, or the array of conductive wires.

11. The electrophoretic system of claim 10, wherein the cassette body includes: an anode contact pin extending through the wall of the cassette body and having a first end and a second end opposite to the first end, the first end arranged exterior of the cassette body and the second end arranged interior of the cassette body and configured to electrically contact with the first electrode contact of the substrate.

12. The electrophoretic system of claim 11, further comprising: a contact bracket that electrically engages with the substrate and provides the first electrode contact of the substrate, wherein the second end of the anode contact pin is configured to electrically contact with the contact bracket.

13. The electrophoretic system of claim 1, wherein the substrate cassette includes: a body defining a cavity configured to partially receive the substrate, wherein the connection interface of the substrate cassette includes a slit defined at the body and configured for the substrate to partially extend out from the body such that the first electrode contact of the substrate is positioned outside the body; wherein the body includes the cathode assembly such that the plurality of electrodes are configured to position within the plurality of buffer chambers, wherein the substrate cassette further includes: a cathode contact pin extending through a wall of the body and electrically contacting with the second electrode contact such that the cathode contact pin is electrically connected to the at least one of the plurality of electrodes; an anode contact pin extending from the body; and a conductive wire having a first end and an opposite second end, the first end electrically connected to the anode contact pin, and the second end electrically connected to the first electrode contact that is positioned outside the body.

14. The electrophoretic system of claim 1, wherein the substrate cassette includes a body defining a cavity configured to partially receive the substrate, wherein the connection interface of the substrate cassette includes a slit defined at the body and configured for the substrate to partially extend out from the body such that the first electrode contact of the substrate is positioned outside the body; and the electrophoretic system further comprising: an anode cassette body defining a cavity configured to receive the first electrode contact of the substrate that is positioned outside the body of the substrate cassette; and an anode contact pin extending through a wall of the anode cassette body and having a first end and a second end opposite to the first end, the first end arranged exterior of the anode cassette body and the second end arranged interior of the anode cassette body and configured to electrically contact with the first electrode contact of the substrate.

15. A method for analyte migration, the method comprising: loading a substrate comprising capture probes with a substrate cassette into an electrophoresis instrument, the substrate cassette including a plurality of buffer chambers, the substrate including a plurality of regions each comprising one or more capture probes and a biological sample containing analytes; arranging a cathode to place a plurality of electrodes within the plurality of buffer chambers, respectively; electrically connecting a power supply to the substrate; electrically connecting the power supply to the cathode; applying a first voltage between the substrate and the cathode; detecting an output parameter in response to the first voltage; determining whether the output parameter meets a threshold value; and based on the output parameter meeting the threshold value, applying a second voltage to generate electric fields between the plurality of regions of the substrate and the plurality of electrodes through the plurality of buffer chambers to cause the analytes in the biological sample to move toward the capture probes on the substrate.

16. The method of claim 15, wherein the output parameter is an impedance.

17. The method of claim 15, wherein the threshold value is indicative of a property of at least one of an electrical connection of the power supply to the substrate, an electrical connection of the power supply to the cathode, or an electrical property of one or more buffers within the plurality of buffer chambers.

18. The method of claim 15, further comprising: based on the output parameter not meeting the threshold value, generating a notification that an electrical connection is improper; and ceasing to apply the second voltage to generate the electric fields.

19. The method of claim 15, wherein the threshold value is a predetermined range of values.

20. The method of claim 15, further comprising: prior to loading the substrate with the substrate cassette, placing the biological sample in contact with the capture probes on the substrate; arranging the substrate cassette onto the substrate to align a plurality of apertures of the substrate cassette with the plurality of regions of the substrate and define the plurality of buffer chambers on the plurality of regions; and supplying buffers in the plurality of buffer chambers.
